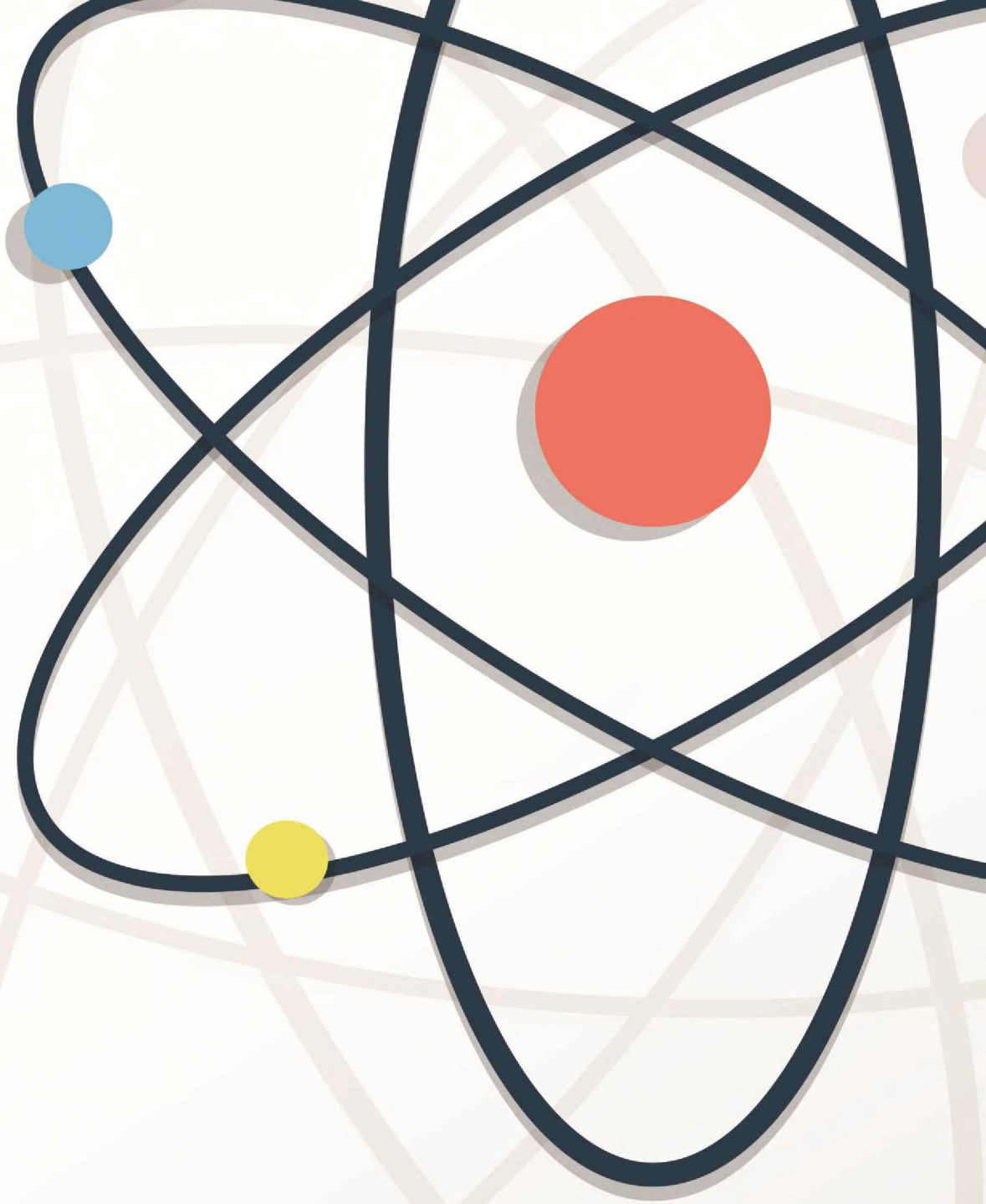


Modern Physics

for IIT-JEE

NEETIN AGRAWAL

Modern Physics for IIT-JEE

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Modern Physics for IIT-JEE

Neetin Agrawal



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This book is dedicated to
my beloved family

About The Author

Mr Neetin

Agrawal is a renowned faculty for IIT-JEE Physics for last 10 years. He has been teaching in some of the most popular coaching institutes of the country. He had mentored many students in top 100 AIR for IIT-JEE and other competitive exams.

He had done his B.Tech from IIT Madras. He has been a scientist and has several inventions on his name. Some of them are:

http://www.google.com/patents/US8203383
http://www.ee.iitm.ac.in/~nagendra/papers/isc09-tappedlcfil-pap.pdf

About The Book

This book includes theory, solved examples and exercise for chapters of Modern Physics. This book will help students preparing for Board exams after class 12th or equivalent. This book will be a complete knowledge house for Modern Physics for students preparing for IIT-JEE and other similar competitive exams. Best of luck to the students using this book!

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01 PHOTO ELECTRIC EFFECT

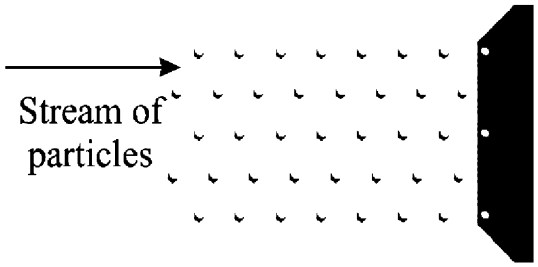
1.1 Introduction

In the history of physics, we had some great scientists like Newton, Maxwell, Faraday, Galileo, etc. They did a lot of prominent research and unfolded the secrets of science to mankind. By the end of 1890, scientists around the world started feeling that they had discovered a lot of physics and now they can explain most of the physical phenomena. But after1890, some new experimental results were obtained by scientists. Scientists were not able to explain these results at all with the help of previous theories. Soon they realized that a lot of physics was yet to be discovered. So many theories came up after 1890. Here, in modern physics, we will study about some of those theories.
Now, to understand modern physics, first we will recall some basic differences between particles and waves.

1.2 Particle and Wave

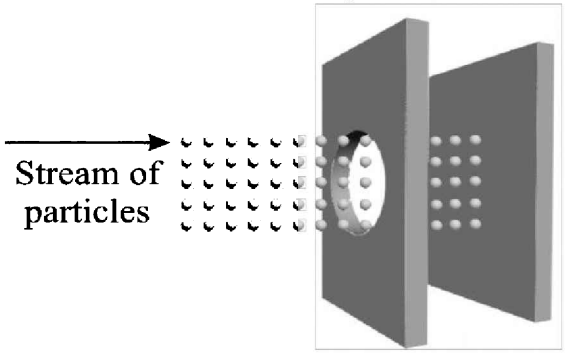
By now you must know about particle and wave. Some simple examples of particles are electron, ball, bullet, etc. and examples of waves are sound waves, string wave, etc. Now we will discuss some important differences between them, which will be useful for us in this chapter ahead.

(i) Nature of Energy Flow - In case of particles, flow of energy is discrete. Say for example, a stream of particles is moving towards a wall as shown in the figure below.

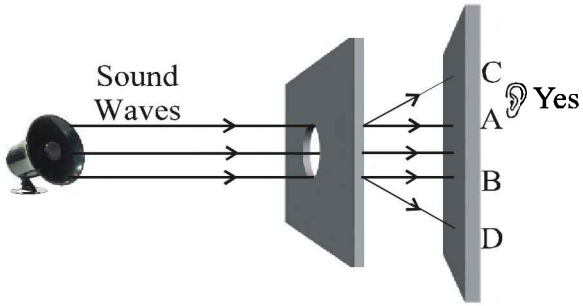


Here each particle has some amount of kinetic energy. These particles hit the wall one by one and transfer their K.E. to the wall. Each time a particle hits the wall, some amount of K.E. is transferred to the wall. It means that the energy of a stream of particles will reach the wall in small amounts or in a discrete way (i.e. not a continuous flow of energy).
While in case of wave, flow of energy is continuous. Like, when you listen music or talk to some person, you hear a continuous sound. It means that energy comes continuously through sound waves.

(ii) Diffraction- In case of particles, diffraction does not occur. For example, when a stream of particles passes through a hole, it does not bend at the edge of the hole as shown below.



Here we can see that particles hit the wall only on the area which is just in front of the hole (i.e. no diffraction).
While in case of waves, diffraction occurs. For example, when sound wave passes through a hole, it bends or spreads at the edge of the hole as shown below.



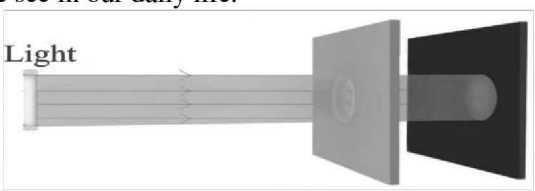
From above figure, we can see that due to diffraction of sound wave, a person can hear sound beyond the region AB as well. This means that waves show diffraction, but particles do not. Diffraction can crudely be thought of as bending at the edges.

1.3 Photo Electric Effect

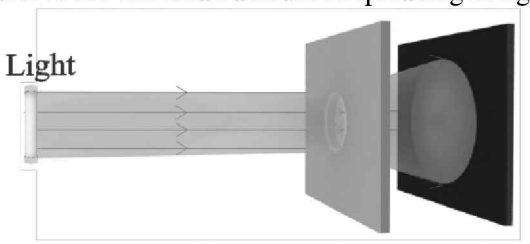
The word photo means 'light'. Mankind has always been curious to understand the concept of light. In the past, many theories were given about light by some great scientists like Newton, Young, Maxwell, etc. They all tried to find out the nature of light. The main question about the nature of light was, "is light a stream of particles or is light a wave?" Here we will talk about some prominent theories and experiments related to light.

(i) Corpuscles Theory-This was the first meaningful theory about light and was given by Newton in early 18th century. On the basis of some experimental results, Newton proposed that light is composed of tiny particles. He named these particles as ***Corpuscles.***

(ii) Young' s Theory-By 1800, Thomas Young performed experiments (like famous Young's double slit experiment) and observed that light shows interference. As interference of light can only be explained considering light a wave, so Young concluded that ***'light is a wave'***.
When Newton proposed light to be a particle, he had observed that when light passes through an opening as in figure below, only the area in front of the opening becomes bright. This is a very common experience that we see in our daily life.



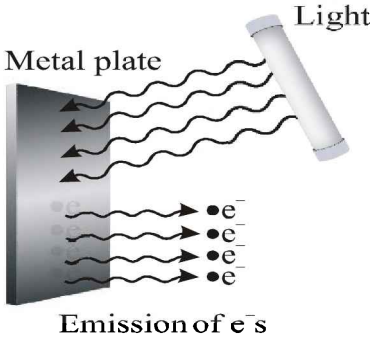
This implies that light is not showing diffraction and thus Newton's idea of light being particle was justified.
But later Young argued that light does spread around the edge of the opening, but the spreading is too little to be observed by naked eye. When seen with good microscopes, it was found that Young was right, which meant that light shows diffraction and is thus a wave. Around Newton's time, such good microscopes were not available to see this small amount of spreading of light.



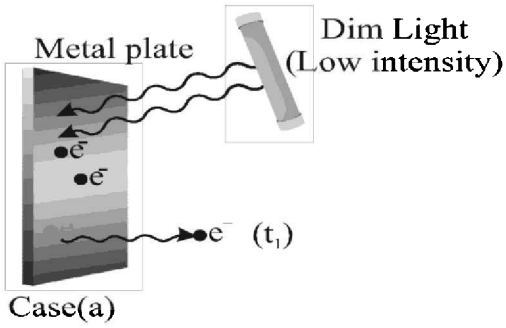
Around 1860, Maxwell mathematically proved that light is a wave. So, by that time, the wave nature of light was fully accepted. But towards the end of the 19th century, debate on the nature of light was triggered once again. Scientists found it hard to explain newly discovered phenomena of light called **photo emission.**

1.4 Photo Emission

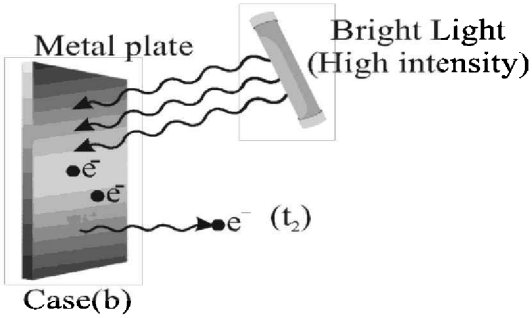
It was discovered, when electromagnetic waves fall on the surface of metals, electrons are emitted from the metal surface. This phenomenon was named as **photo-emission**.



Scientists explained photo emission as follows:
Every metal contains free electrons. These free electrons are free enough to move randomly inside the metal but not free enough to come out of the metal. When light falls on a metal surface, electrons gain energy from light. After some time, when electrons accumulate the required energy to come out from metal, emission of electrons takes place. Although this explanation seems to be right, but scientists were not able to explain many experimental results (related to photo-emission) with the help of this theory. Let's see some of such results.
(i) Say there are two metal plates of the same metal as shown in below figure. Now here we will discuss about the emission time of electrons from metal plates in two different cases. Say electrons are emitted from these two metal plates at time 't_1' and 't_2'.



Case(a)



Case(b)

In case (a), as dim (low intensity) light is falling on metal plate so the energy of electrons will increase slowly. While in case (b) as bright (high intensity) light is falling on metal plate so the energy of electrons will increase at a faster rate. So, in case (b), electrons will quickly accumulate the required energy to come out of the metal. Thus electrons in case (b) should come out in lesser time than the electrons in case (a).

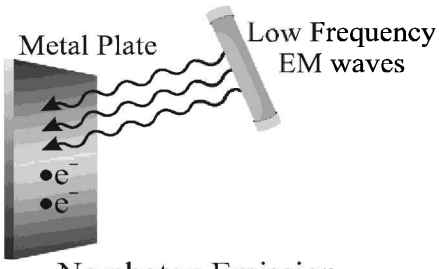
i.e., $t_2 < t_1$ (Theoretical result)

Scientists were also expecting the same result. But experimentally, it was found that in both cases electrons are emitted at the same time. Also, the emission time of electrons in both cases was extremely small.

i.e., $t_1 = t_2$ (Experimental result)

This disagreement between theory and experiment perplexed scientists.

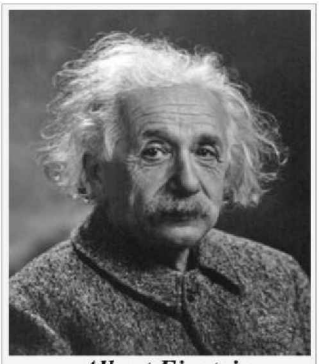
(ii) One more contradictory result: emission of electrons does not take place when low frequency EM waves fall on a metal surface.



No photon Emission

As per scientist's theory, EM waves bring energy to electrons irrespective of their frequency. This means that electrons should have been emitted at all frequencies. So this observation also remained unexplained.

Such contradictions between theory and experiment proved that the understanding of light as per that time had flaws. In March 1905, Albert Einstein published a revolutionary theory that explained all the experimental results. Today this theory is known as **'Photoelectric effect'**. Einstein was awarded the Nobel Prize in 1921 for this great work.



Albert Einstein
(1879 - 1955)
Awarded Nobel in 1921

1.4.1 Einstein's Explanation of Photoemission (Photoelectric Effect)

Einstein gave this theory to explain the contradictory results of photoemission. Here, through some points we will understand that how Einstein explained the phenomenon of photo emission.

(i) Einstein proposed that light is composed of tiny particles called ***'photons'***. Photons are special kind of particles as some properties of photons are same as normal particles but some properties are different from normal particles. Photons travel with the speed of light (c) and rest mass of a photon is zero. Each photon carries a certain amount of energy and this energy depends on the frequency of light. It is given as-

$$\mathbf{E = h\nu}$$

Here **'h'** is plank constant and **'ν'** is frequency of light. Einstein's theory is also known as the ***photon theory of light.***



Quiz-1

Will the energy of photons for red and blue colour light be same? If not, then for which colour energy of photons will be more?



Sol. We know that frequency of blue light is more than red light so the energy of photons will be more For blue light

(i) Einstein further explains that when a photon falls on a metal surface, it collides with an electron in the metal. The electron absorbs the photon and its energy increases by **'hν'** (energy of the photon).

(ii) In a collision, the photon is either fully absorbed or not absorbed at all. It means that there is no partial absorption. Say for example, if a photon with energy 'hν' collides with an electron then either energy of electron will increase by 'hν' or energy of electron will remain same (i.e. electron doesn't absorb the photon).

Generally, in a metal, the number of free electrons is much larger than the number of photons falling on the surface of metal. So, it is very less probable that an electron collides with more than

one photon at a time. So, here we assume that one electron absorbs only one photon at a time.

(iii) Work Function - It is defined as *the minimum energy to be given to an electron to eject it fromthe metal.* It is denoted by symbol 'ϕ'. Work function is a property of material. It means that the value of work function will be different for different materials.



Quiz-2

Work function of a metal is 5 eV. Check in below options, for which value of energy of photon, photo emission will happen.

(A) E = 2 eV (B) E = 4.9 eV (C) E = 5.1 eV (D) E = 6 eV

Sol. Here work function of metal is 5 eV. It means that minimum energy required to eject an electron from the metal is 5 eV. So, if an electron receives energy equal to 2 eV or 4.9 eV then it will not come out. But if received energy by electron is greater than 5 eV then it will come out from the. metal. So photo emission will take place if (c) E = 5.1 eV and (d) E = 6 eV

(iv) We have seen that energy of an electron increases by 'hν' when it collides with a photon. If this increment in energy of electron is greater than the work function of metal, then electron will come out of the metal. But if it is less than the work function of metal, then the electron will not come out. So photo emission will take place only if-

$$h\nu \geq \phi$$

$$\Rightarrow \quad \nu \geq \frac{\phi}{h}$$

Here we define a new term **'threshold frequency(ν_{th})'** and its value is given as -

$$\nu_{th} = \frac{\phi}{h}$$

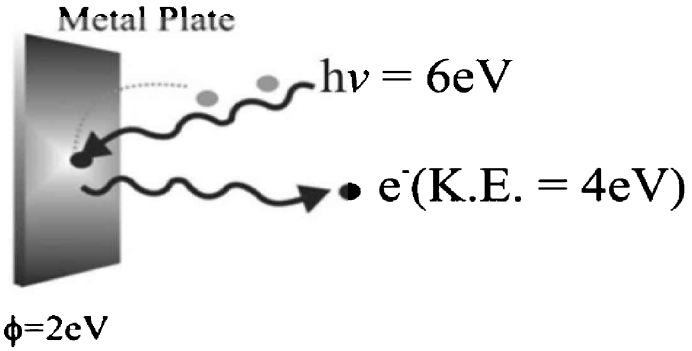
Threshold frequency is the minimum frequency of a photon that can eject an electron from the metal.

(v) Possibilities in Photo Emission when $h\nu > \phi$ - We know that when an electron collides with a photon, energy of the electron increases by 'hν'. The electron loses some amount of this increased energy (equal to ϕ) to come out from the metal. Now the remaining or excess energy of the electron (i.e. $h\nu - \phi$) appears as kinetic energy of the photon. Mathematically, we can write this as –

$$KE = h\nu - \phi$$

For example, say the work function of a metal is 2 eV and energy of a photon is 6 eV. Here when an electron collides with a photon, its energy increases by 6 eV. Electron spends 2 eV energy to come out from metal and remaining energy of electron (4eV) appears as its kinetic energy as shown below:

$$KE. = h\nu - \phi$$
$$= 6 - 2 = 4eV$$



In the above description,we have assumed that the electron does no collision with any other inside metal while coming out of the metal. This happens very rarely. Practically, in most of the cases when an electron tries to come out of metal, it does collisions with other electrons or lattice inside metal. So, the electron loses some amount of its increased energy in collisions. In this situation, ejection of an electron depends on its remaining energy after collisions. In below points, we will discuss two possibilities in such cases.

(a) Energy of electron remains greater than 'ϕ' after collisions: - In this situation, the electron will be able to come out of the metal. But electron loses some amount of its energy in collisions inside metal. So, kinetic energy of emitted electron will be less than the case of no collision.

i.e. K.E. < $h\nu - \phi$

For example, say work function of a metal is 2eV, energy of a photon is 6 eV and energy loss in collisions is 1.5 eV. Here K.E. of electron will be–

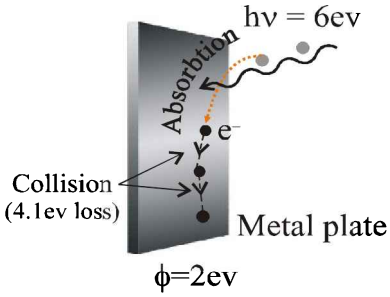
K.E. = $(h\nu - \phi) - 1.5 = (6 - 2) - 1.5 = 2.5$ eV

(b) Energy of electron becomes less than 'ϕ' after collisions: - In this situation, the electron will not have the required energy to come out from the metal after collisions. So, it will not be able to come out from the metal.

For example, say the work function of a metal is 2eV, the energy of a photon is 6 eV and energy loss in collisions is 4.1 eV. So, here remaining energy of the electron after collisions will be-

E = 6 – 4.1 = 1.9 eV

As E < ϕ, electron will not come out from the metal.



No Photon Emission

Finally, we can say that there are three possible cases when an electron absorbs a photon of energy $h\nu > \phi$.

- Electron comes out from metal with kinetic energy-

  K.E. = $h\nu - \phi$ ... (1)

- Electron comes out from metal with kinetic energy-

  K.E. < $h\nu - \phi$ ... (2)

- Electron does not come out.

On combining equation (1) and (2), we can write kinetic energy of all the ejected electrons as-

$$\mathbf{K.E} \leq \mathbf{h\nu - \phi} \quad \text{... (3)}$$

Here we can see that maximum kinetic energy of an ejected electron can be -

$$\mathbf{KE_{max} = h\nu - \phi}$$

Above equation is called ***Einstein's equation of photo- electric effect.***

(vi) Photon Efficiency- When light falls on a metal, many electrons absorb photons but only a few out of them come out of the metal. Therefore the numbers of emitted electrons are always less than the number of absorbed photons. To show this ratio of emitted electrons and absorbed photons, we use the term photon efficiency. It is denoted by 'η' and defined as -

$$\eta = \frac{\text{Number of emitted electrons}}{\text{Number of absorbed photons}}$$

For example, say number of absorbed photons is 1000 and photon efficiency (η) is 0.2%. Then, number of emitted electrons (N) -

$$N = 1000 \times \frac{0.2}{100} = 2 \text{ electrons}$$

In this example, although 1000 electrons absorb photons, only two out of them come out of the metal. Typical η is less than 1%. This happens because most of the electrons lose their energy in collisions after absorbing photons.

Illustration 1

Light of wavelength 500 nm falls on a metal plate which have work function equal to 2eV.

(i) Find energy of each photon.

Sol. We know

$$E = h\nu = h\left(\frac{c}{\lambda}\right) = \frac{\left(4.14\times10^{-15}\right)\left(3\times10^{8}\right)}{\left(500\times10^{-9}\right)} = 2.5 \text{ eV}$$

So energy of a photon will be 2.5 eV.

(ii) Will photo emission happen?

Sol. As $h\nu > \phi$, photo-emission will happen.

(iii) Find maximum and minimum KE of ejected electrons?

Sol. We know $KE_{max} = h\nu - \phi = 2.5 - 1 = 1.5$ eV

$KE_{min} = 0$

1.4.2 Explanation of Contradictory Results By Einstein's Photon Theory -

Previously, we have seen that experimental observations of photo-emission could not be explained on old theories of light. Here we shall see how Einstein's idea easily proves the experimental results.

(i) No photo emission of low frequency electromagnetic waves -

We have already seen that photo-emission happens only if $\nu \geq \frac{\phi}{h}$. This means that for frequencies lower than threshold frequency, no photo-emission happens.

(ii) When light of different intensities falls on metal, emission of electrons take place at the same time in all cases -

According to Einstein, light is composed of photons. Here, for dim light, the number of photons will be less and for bright light, the number of photons will be more.

We know that in the case of particles, flow of energy is discrete. So when an electron absorbs a photon, its energy increases by '$h\nu$'. But here increment in energy of electron is sudden. It means that as soon as an electron absorbs a photon, its energy will increase by '$h\nu$'. So, in both cases, electrons

will have the energy of photons at the same time. If this energy 'hν' is greater than 'ϕ', the emission of electrons will take place at the same time in both the cases.

i.e. $t_1 = t_2$

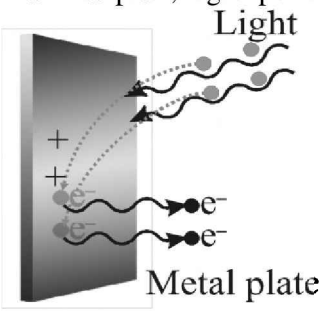
Although Einstein's photon theory explains photo-emission properly, scientists of that time were not fully convinced by his theory. Scientists had been thinking of light as a wave since long. Now it was difficult for them to suddenly digest the fact that light behaved rather as particles. So, many scientists tried to prove Einstein wrong. One of them was 'Millikan'. He did a lot of experiments to prove Einstein wrong. But as he did more and more experiments, he got more and more convinced that Einstein was actually right. Millikan was awarded the Nobel Prize in 1929 for proving 'particle nature of light'.



Robert Andrews Millikan
(1868 - 1953)
Awarded Nobel in 1923

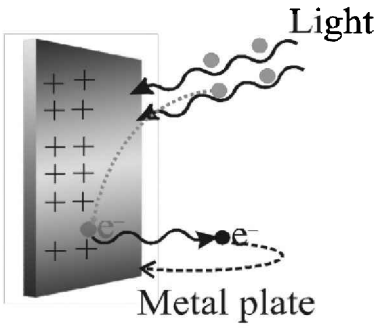
1.5 Photo Electric Cell

When photo emission takes place from a metal plate, it gets positively charged as shown below:

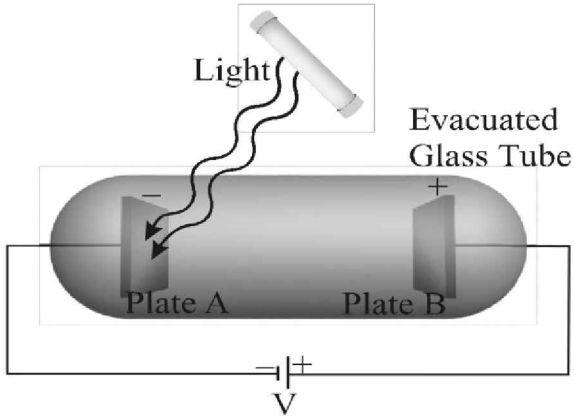


This positive charge on plate attracts the ejected electrons towards itself. Initially, when the charge on the metal plate is less, electrons with higher kinetic energy escape from the metal plate. But electrons with lower kinetic energy come back to the metal plate.

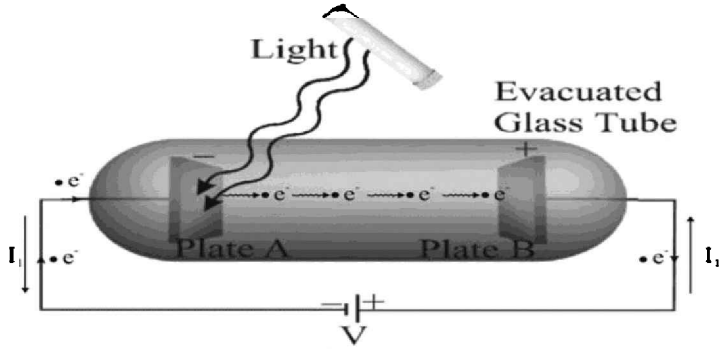
As some more electrons escape from the metal plate, charge on metal plate increases. After some time, positive charge on metal plate becomes so large that it is able to pull back all ejected electrons. So, effectively no electron comes out after some time.



Here we can see that photo emission will stop after some time. So, for continuous photo emission from a metal, we use a device named ***'photo-electric cell (PEC)'***. The basic structure of PEC is shown in below figure:



In PEC, two metal plates (A and B) are used which are kept in an evacuated glass tube. A battery is connected across these two plates. When light falls on plate A, electrons are emitted from it. These emitted electrons travel towards plate B due to the electric field generated by a battery. They hit plate B and again reach to plate A through the wire. As emitted electrons move in a loop, electric current flows in the circuit as shown below.



Here, we can see that as ejected electrons come back to plate A**,** positive charge does not develop on the plate. Thus continuous photo emission can occur from plate A**.** Here evacuated glass tube is used so that there will be no collision between air molecules and ejected electrons and electrons can move smoothly inside tube. Now, we will see some terms related to PEC.

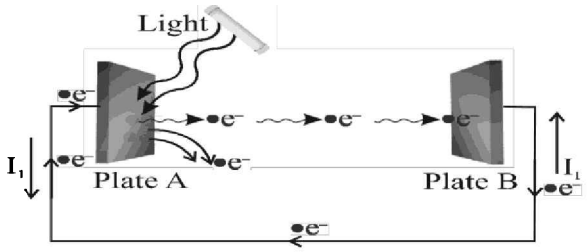
(i) Saturation Current

Here we will discuss how current in photo-electric cell changes when we change the emf (V) of the battery. We will understand about it through some cases.

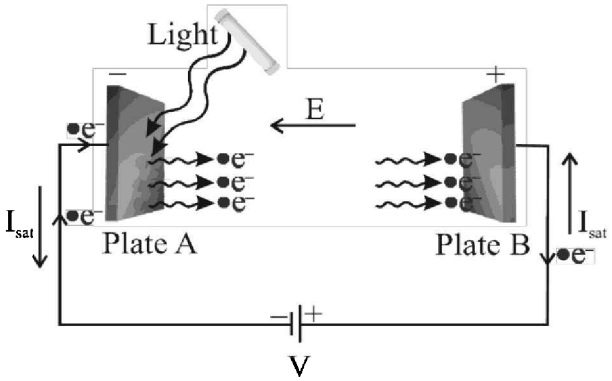
- First, we will check current in PEC for zero emf of the battery (i.e. there is no battery connected across the plates or simply the plates are connected through a wire). Here when light falls on plate A, electrons are emitted with different kinetic energies.

$$0 \le KE \le h\nu - \phi$$

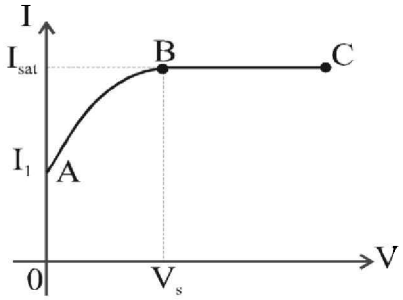
Electrons with high KE reach the other plate but electrons with very low KE fail to reach plate B. As some electrons reach plate B and travel in a loop so current flows in the circuit (say I_1) as shown below:



- Now say, we connect a battery across the plates and increase its emf. Battery will create a potential difference between the plates and thus, an electric field will be created in the region between the plates. Plate B being at higher potential, direction of electric field will be from B to A. In this case, electrostatic force acts on emitted electrons towards plate B. Due to this force, electrons with slower speed also start to reach plate B. As we further increase emf of the battery, electric field and thus force on ejected electrons increases and more electrons reach plate B. So, the value of current increases in the photoelectric cell. At a certain value of emf (say V_S), all the emitted electrons start to reach plate B and the current becomes maximum or saturated. Now, if we further increase emf of the battery beyond V_S, current does not increase. This maximum current is called **'Saturation Current (I_{sat})'.**



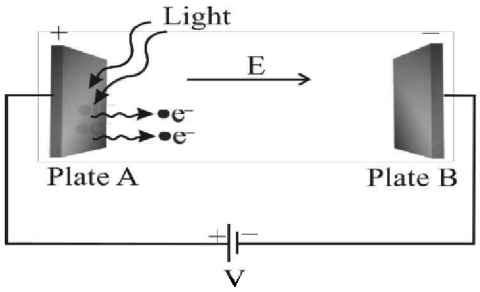
On the basis of above discussion, we can draw a graph between current and potential difference between plates as shown below.



In part **AB** - Not all electrons reach plate **B**
In part **BC** - All electrons reach plate **B**

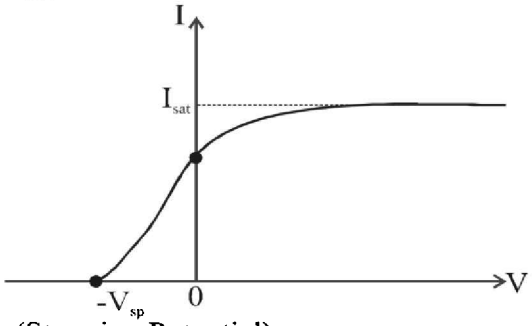
(ii) Stopping Potential

Till now we have seen the changes in current when we increase '**V**' positively. Now we will see how current changes when we increase emf of the battery after reversing its polarity.



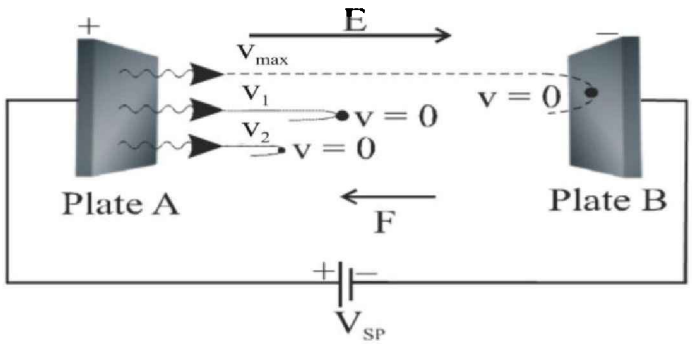
Now we can see that direction of electric field is from plate A to plate B. So repelling force will act on ejected electron towards plate A.

We have already seen that if emf of the battery is zero, then only some electrons with higher KE reach plate B. Now here if we increase emf of the battery, then repelling force on electrons will increase and very less number of electrons will reach plate B. So the magnitude of current will decrease. At a certain value of the emf, electric force becomes sufficient enough that no emitted electron reaches plate B. So current in circuit becomes zero. **This value of the emf at which current becomes zero, is called 'stopping potential (V_{SP})'.**



(Stopping Potential)

Now we will discuss about the path of electrons when stopping potential is applied across the metal plates. In this situation, electrons will move towards plate **B** due to their initial kinetic energy. But as an electric force acts on electrons in opposite directions of their motion, their speed will decrease and after some time they will reverse their direction of motion as shown below:



Here $v_2 < v_1 < v_{max}$

From above figure, we can see that electrons with higher KE travel larger distance towards plate B before reversing their direction of motion. For example, an electron with maximum KE will reverse its direction of motion just before reaching plate B, while other electrons with less KE will do the same thing earlier as shown in above figure.

Now, to find the value of stopping potential, we will apply energy conservation theorem for the electron with maximum kinetic energy.
According to the energy conservation theorem, change in magnitude of the kinetic energy of electron will be equal to change in magnitude of potential energy between the plates.

$\Rightarrow \quad |\Delta KE| = |\Delta PE|$

$\Rightarrow \quad |KE_{plate\ A} - KE_{plate B}| = |Q(V_{plate A} - V_{plate B})|$

$\Rightarrow \quad |KE_{max} - 0| = |e(V_{SP})|$

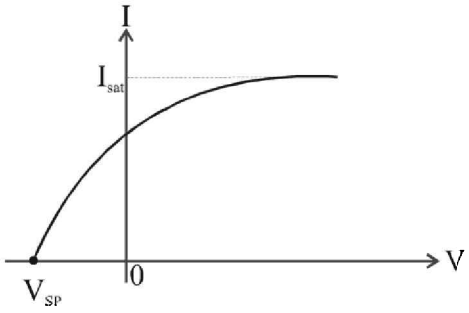
$\Rightarrow \quad KE_{max} = eV_{SP}$

$\Rightarrow \quad \mathbf{V_{SP} = \frac{KE_{max}}{e}} \qquad \Rightarrow V_{SP} = \frac{h\nu - \phi}{e}$

In this way, we can find value of stopping potential.

Illustration 2

Find out how the below graph will change if we increase the number of incoming photons in photo electric cell.



Sol. We will see how the saturation current and stopping potential change when we increase the Number of coming photons

- Change in I_{sat} - We know that if more photons will fall on plate A, more electrons will come out from A. So, saturation current will increase when we increase the number of photons.
- Change in V_{SP} – We know:

$$V_{SP} = \frac{h\nu - \phi}{e}$$

Here as 'h' and 'e' are constants, so the value of 'V_{SP}' will depend on the frequency of light (ν) and work function (ϕ). As we have not changed metal so 'ϕ' is constant here. Also, we have not changed frequency ν of photons. So, value of stopping potential will remain same when we increase number of photons.
The given graph will change as shown below on increasing the number of incoming photons from N_1 to N_2.

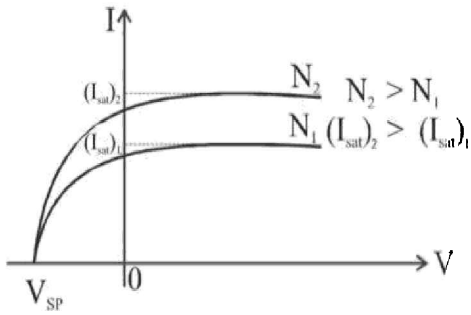
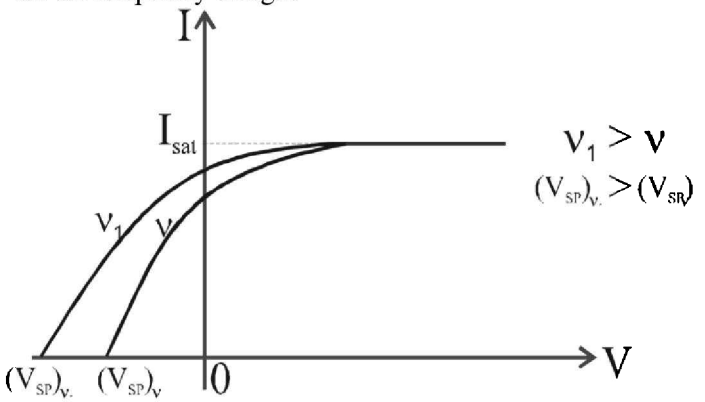


Illustration 3

How the graph in the last question changes, if we increase frequency of light.

Sol. When we increase frequency of light, energy of photons increases but number of photons remains same. Now, we will see how I_{sat} and V_{SP} change when we increase frequency of light.

- Change in I_{sat} - As same number of photons fall on plate A, so the number of emitted electrons will remain same. Thus I_{sat} remains unchanged.
- Change in V_{SP} - We know that V_{SP} depends on frequency of light (ν). V_{SP} increases when we increase the frequency of light



Key Points – (very important)

(1) When we increase number of incoming photons-
  (a) I_{sat} increases
  (b) V_{SP} remains same

(2) When we increase frequency of light-
  (a) I_{sat} remains same
  (b) V_{SP} increases.

Illustration 4

In a photo-electric cell, show graphically how stopping potential will change when we change frequency of light.

Sol. We know

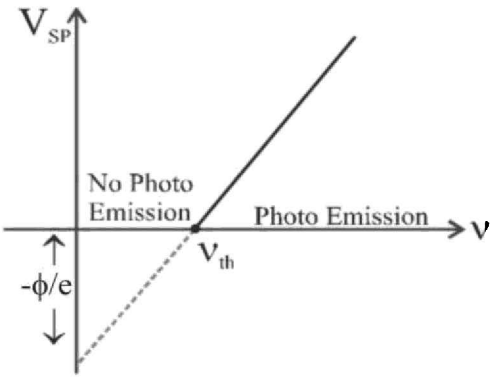
$$V_{SP} = \frac{h\nu - \phi}{e}$$

$$\Rightarrow \quad V_{SP} = \frac{h\nu}{e} - \frac{\phi}{e} \qquad \text{... (1)}$$

Here we can see that equation (1) represents a straight line. Comparing this equation with $y = mx + c$

Slope (m) $= \frac{h}{e}$ (positive)

And y intercept (c) $= \frac{-\phi}{e}$ (negative)



Note that we have removed the graph for ν less than threshold frequency as no emission happens in that case.



Quiz-3

Work function of two metals A and B are ϕ_A and ϕ_B. Draw graph between stopping potential and frequency of light for these two metals if $\phi_A > \phi_B$.

Sol. We have seen in the previous example that V_{SP} changes linearly with frequency of light according to the equation

$$V_{SP} = \frac{h}{e}\nu - \frac{\phi}{e}$$

Here $\phi_A > \phi_B$

$$\Rightarrow \quad \frac{\phi_A}{e} > \frac{\phi_B}{e}$$

$$\Rightarrow \quad (y_{intercept})_A > (y_{intercept})_B$$

So, graph between V_{SP} and ν for these two metals will be as shown in below figure.

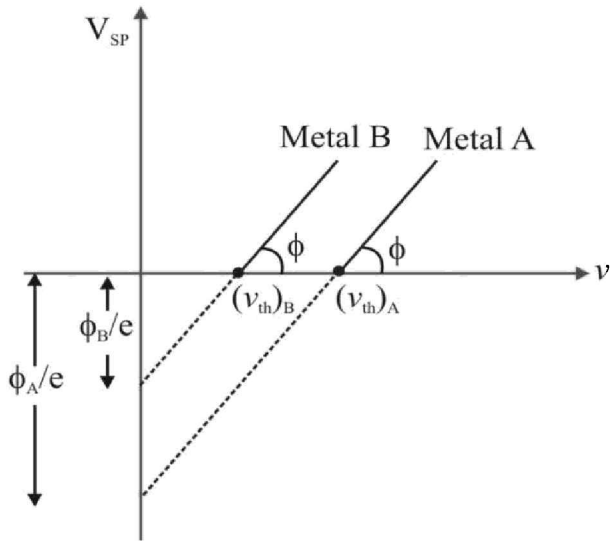
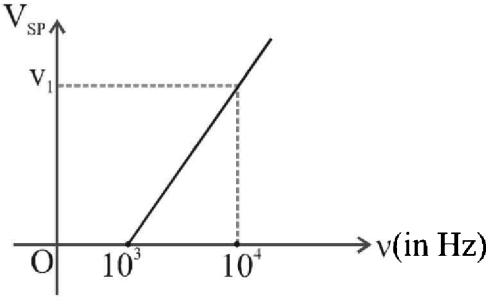


Illustration 5

The graph below shows the relation between stopping potential and frequency of light for a metal.



(i) Find out work function of metal.

Sol. From above graph we can see that -

$$V_{th} = 10^3 \text{ Hz.}$$

We know -

$$V_{th} = \frac{\phi}{h}$$

$$\Rightarrow \quad \phi = V_{th}(h)$$

$$\Rightarrow \quad \phi = (10^3)(6\cdot 63 \times 10^{-34})$$

$$\Rightarrow \quad \phi = 6\cdot 63 \times 10^{-31} \text{ J}$$

(ii) Find value of stopping potential at frequency 10^4 Hz.

Sol. We know -

$$V_{SP} = \frac{h\nu - \phi}{e}$$

So we can simply find value of V_{SP} by putting appropriate values of ν and ϕ in the above equation.

Illustration 6

In a photo-electric cell, 1000 photons/sec falls on metal plate A. If photon efficiency is 1% then find -

(i) Number of emitted electrons per second?

Sol. We know that photon efficiency is defined as -

$$\eta = \frac{\text{Number of electron emitted}}{\text{Number of photons caming}}$$

$$\Rightarrow \quad \frac{1}{100} = \frac{N}{1000}$$

$$\Rightarrow \quad N = 10$$

(ii) Find saturation current.

Sol. We know the saturation current is defined as the maximum current in PEC. Here maximum current will flow in PEC when all the emitted electrons start to reach plate B. In the first part of this problem we have seen that number of emitted electrons per second is 10.

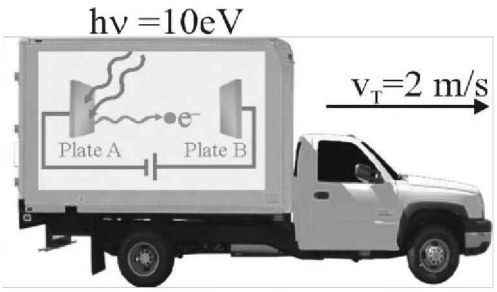
$$I = \frac{Q}{t}$$

$$\Rightarrow \qquad I = \frac{Ne}{t}$$

$$\Rightarrow \qquad I_{sat} = \frac{10\left(1\cdot 6\times 10^{-19}\right)}{1} = 1\cdot 6\times 10^{-18}\ A$$

Illustration 7

A photo-electric cell is placed in a truck which is moving forward with velocity 2 m/sec as shown in below figure.



Work function of metal plate A is 2 eV. Find out maximum velocity of ejected electrons.

Sol. We know that maximum KE of emitted electrons is -

$$KE_{max} = h\nu - \phi$$

$$\Rightarrow \qquad KE_{max} = 10 - 2 = 8\ eV$$

$$\Rightarrow \qquad \frac{1}{2}m\left(v_{max}\right)^2 = 8\left(1.6\times 10^{-19}\right)J$$

$$\Rightarrow \qquad \left(v_{max}\right)^2 = \frac{2\times 8\left[1.6\times 10^{-19}\right]}{9.1\times 10^{-31}}$$

$$\Rightarrow \qquad v_{max} = 1.68\times 10^{-6}\ m/sec$$

Here one important point is that V_{max} shows the maximum velocity of ejected electrons in the frame of plate A.

In the given problem, as no specific frame is mentioned so we will find the velocity of emitted electrons in ground frame. Here plate A is moving forward (with truck) with velocity of 2 m/sec (w.r.t. ground). So we can find the maximum velocity of electrons in ground frame by using equation of relative motion-

$$(V_{max})_{Gnd} = (V_{max})_A + V_{Truck}$$

Here $(V_{max})_A = 1\cdot 68\times 10^{-6}$ m/sec

And $V_{Truck} = 2$ m/sec

Key point –

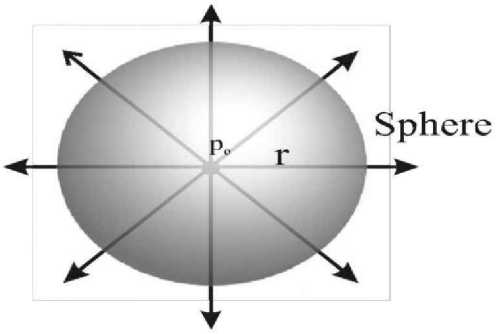
$$\frac{1}{2}m\left(V_{max}\right)^2 = h\nu - \phi$$

In above equation 'V_{max}' denotes the maximum velocity of emitted electrons w.r.t. the plate emitting electrons.

1.6 Photon Flux

Photon flux is defined as **the number of photons passing through a unit area in unit time**.

Now we will find photon flux for a point source of light of power P_0. To find this, let's draw a sphere of radius 'r' taking point source as its centre. Here point source will emit light in all directions as shown in below figure.



We know power is defined as -

$$\text{Power} = \frac{\text{energy}}{\text{time}}$$

Here the power of point source is P_0. It means that point source will emit P_0 joule energy in 1 sec or we can say P_0 joule energy will pass through the sphere (in the form of photons) in unit time.

We know that intensity is defined as energy passing through per unit area in unit time.

i.e. $$I = \frac{(\text{energy})}{(\text{area})(\text{time})}$$

$$\Rightarrow \quad I = \frac{\text{power}}{\text{area}} \qquad \Rightarrow I = \frac{P_0}{4\pi r^2}$$

Say, photon flux of point source is 'N'. It means that N photons pass through unit area of a sphere in unit time. As energy of each photon is 'hν' so 'N (hν)' amount of energy will pass through the unit area of a sphere in unit time. Thus-

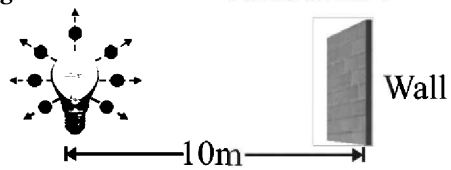
N (hν) = Intensity of light

$$\Rightarrow \quad \mathbf{N = \frac{I}{h\nu}}$$

So, in this way we can find photon flux for a light source.

Illustration 8

Light is emitted from a bulb of 100 watt. A wall is there at a distance of 10 m from the bulb as shown below. Frequency of light is 5×10^4 Hz and area of the wall is $1mm^2$.



Find the number of photons hitting the wall per second.

Sol. To find this, we can assume bulb as a point source and draw a sphere of radius 10 m taking bulb as its centre. Here intensity of light-

$$I = \frac{P}{\text{area}} = \frac{100}{4\pi(10)^2} = \frac{1}{4\pi}\,W/m^2$$

Here photon flux will represent the number of photons hitting unit area of wall in unit time. So total number of photons which will hit the wall in unit time will be -

= (photon flux) (Area of wall)

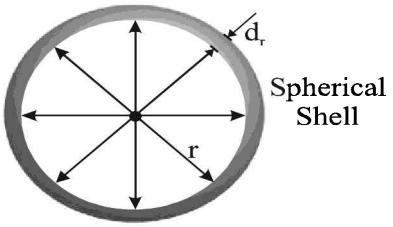
$$= \left(\frac{I}{h\nu}\right)(A)$$

Now we can find the number of photons hitting the wall per second by putting appropriate values in the above equation.

1.7 Photon Density

It is defined as **the number of photons present in a unit volume at any time.**

Now, we will find value of photon density for a point source of light (of power P) which spreads light in all directions. Let's draw a thin spherical shell of thickness 'dr' at a distance 'r' from a point source as shown in below figure.



Volume of spherical shell = (Area of sphere) (Thickness)

$\Rightarrow \quad dV = (4\pi r^2)\,(dr)$ ... (1)

Say, photon density or the number of photons in unit volume of spherical shell is 'x'. So the number of photons, present in spherical shell will be –

= (photon density) (volume of spherical shell)

= x dV

Now energy present in spherical shell will be-

dE = (Energy of one photon) (Total number of photons present in spherical shell)

$\Rightarrow \quad dE = (h\nu)\,(x\,dV)$

$\Rightarrow \quad dE = (h\nu)\,x\,(4\pi r^2\,dr)$

Divide above equation by 'dt', we get -

$$\Rightarrow \quad \frac{dE}{dt} = h\nu.x.4\pi r^2.\frac{dr}{dt} \qquad \text{... (2)}$$

Now here photons will travel a distance 'dr' in time 'dt'. So, we can write -

$$\frac{dr}{dt} = \text{Velocity of photon} = c$$

We can also see that-

$$\frac{dE}{dt} = \text{Power of point source} = P$$

So, from equation (2), we get -

$$P = h\nu.\,x.\,4\pi r^2.\,c$$

$$\Rightarrow \quad x = \left(\frac{P}{4\pi r^2}\right)\left(\frac{1}{h\nu c}\right)$$

$$\Rightarrow \quad x = \frac{I}{h\nu c}$$

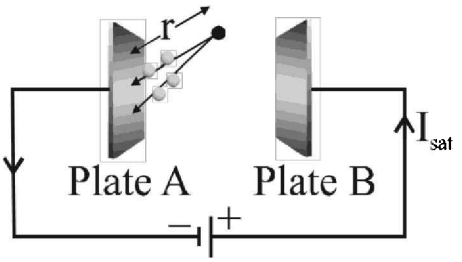
$$\Rightarrow \quad \text{Photon density} = \frac{\text{Photon flux}}{c}$$

Above equation shows the relation between photon density and photon flux.



Quiz-4

In PEC, when light source is situated at a distance 'r' from plate A, saturation current is 'I_{sat}' as shown below.



Find out value of saturation current when distance between source and plate A is doubled?

Sol. We know that -

$$\text{Photon flux} = \frac{P}{4\pi r^2}\cdot\frac{1}{h\nu}$$

Here we can see that

$$\text{Photon flux } \alpha \frac{1}{r^2}$$

It means that as the distance 'r' will increase, photon flux (i.e. number of photons falling on unit area of the plate A in unit time) will decrease. Thus smaller number of electrons will come out from the plate. As a result value of saturation current will decrease.

Mathematically, we can write this as-

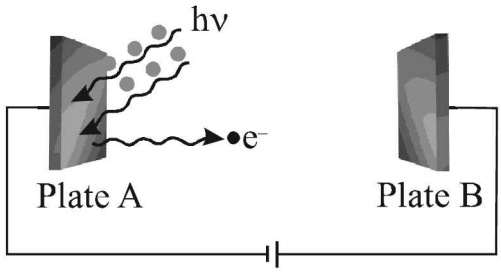
$$\text{Saturation current } \alpha \text{ Photon flux } \alpha \frac{1}{r^2}$$

$$\Rightarrow \text{ Saturation current } \alpha \frac{1}{r^2} \qquad \text{... (1)}$$

From above equation we can see that as distance 'r' will be doubled, value of saturation current will become $\frac{'I_{sat}'}{4}$. Equation (1) shows an important relation between 'I_{sat}' and 'r'. Say, if we increase distance 'r' by four times then the value of saturation current will be $\frac{I_{sat}}{(4)^2}$ or $\frac{I_{sat}}{16}$

1.8 Effect of Intensity of light in Photo-Electric Cell

Here we will see how saturation current and stopping potential, in PEC, will change when we increase the intensity of light.



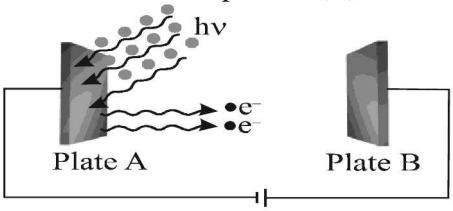
But first we will understand how we can increase intensity of light. We have seen that photo flux can be written as-

$$N = \frac{I}{h\nu}$$

$$\Rightarrow \quad I = N.\, h\nu \qquad \text{... (1)}$$

From equation (1), we can see that the intensity of light depends on photon flux (N) and frequency of photons (ν). So, we can increase the intensity of light either by increasing the photon flux (number of photons per unit area per unit time) or by increasing frequency of light. Now, we will discuss about each method one by one.

- Increasing photon flux or number of photons (N)-



If we increase the number of photons then more photons and thus more energy will fall on plate A. So, the intensity of light increases and more number of electrons come out from plate A. As a result saturation current (I_{sat}) increases in PEC.

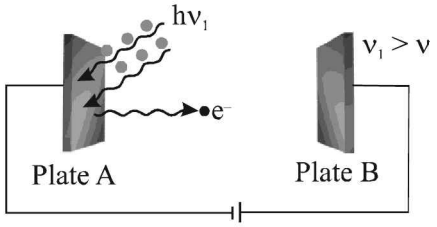
Now we know -

$$V_{SP} = \frac{h\nu - \phi}{C}$$

We can see that value of stopping potential depends on the frequency of light. Here, as we have not changed the frequency of light so stopping potential will remain same.

- Increasing frequency of light (ν)-

If we increase frequency of photons then energy of each photon increases. So, more energy falls on plate A i.e. intensity of light increases.



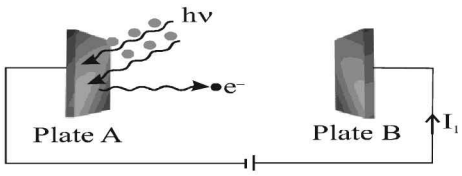
As here we haven't increased the number of photons so numbers of emitted electrons will not change and saturation current (I_{sat}) will remain same.

As here we have increased the frequency of photons (ν) so the value of stopping potential will increase. Now, we will see an example related to this discussion.



Quiz-5

Find out how saturation current will change in a photo- electric cell when –



(i) We increase frequency by two times and intensity by four times.

Sol. We know that intensity of light can be written as-

$$I = N(h\nu)$$

Here we have increased frequency only by two times, but increment in intensity is four times. It means that there must be an increment in the number of photons by two times.

As the number of photons is increased by two times so 'I_{sat}' will also increase by two times.

(ii) We increase frequency by two times and Intensity by two times.

Sol. We know- $I = N(h\nu)$

Here both frequency and intensity are doubled, it means that the photon flux (N) remains same. So, 'I_{sat}' will also remain same as well.

(iii) We increase frequency by four times and intensity by two times.

Sol. We know- $I = N(h\nu)$

Here frequency is increased by four times, but intensity has increased only by two times. It means that number of photons is decreased by two times so 'I_{sat}' will also decrease by two times.

1.9 Momentum of Photon

Generally, we associate momentum with the objects which have some mass. Earlier we have seen that mass of photon is zero, so it looks weird that how can a massless particle have momentum But Einstein
said that even photons have momentum and it is given as -

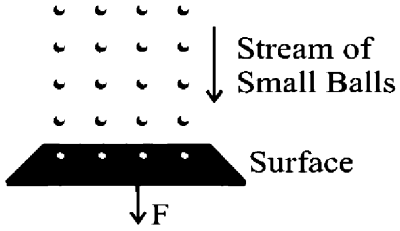
$$\text{Momentum of photon} = \frac{\text{Energy of photon}}{\text{Speed of photon}}$$

$$\Rightarrow \quad \mathbf{p = \frac{h\nu}{c}}$$

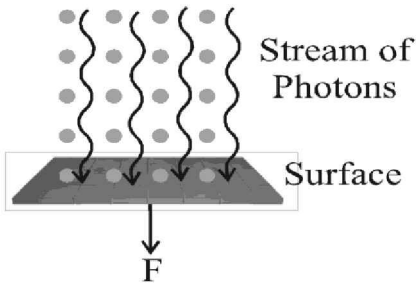
$$\Rightarrow \quad \mathbf{p = \frac{h}{\lambda}} \; ; \qquad (\nu\lambda = c)$$

1.10 Radiation Pressure

In practical life, you have seen that when some moving particles (say balls) strike a surface, they exert a force on the surface as shown below.



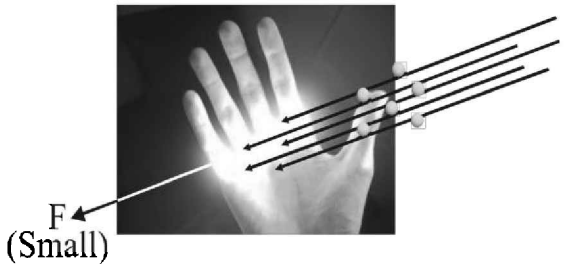
In the same way, when light falls on a surface, it also exerts a force on the surface. This is same as above case but here we have a stream of photons instead of small balls as shown below.



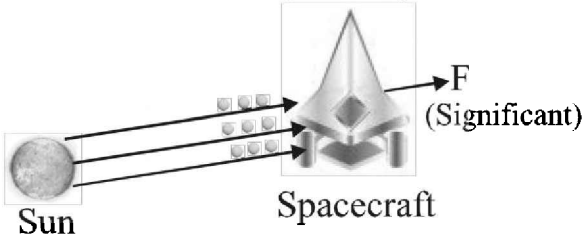
Now say, light is falling on a surface of area 'A' so pressure on surface will be -

$$P = \frac{\text{Force on surface due to light}}{\text{Area of surface}}$$

This pressure, which is exerted on a surface by photons, is called 'Radiation pressure'. Now, you can ask if light exerts force when it falls on a surface, why don't we feel it when it falls on our body as shown below. Actually light exerts a force on us but generally that force is so small that we cannot feel it.



That's why, we normally neglect radiation pressure. But, in some special cases, we cannot neglect radiation pressure. For example, in case of spacecraft, radiation pressure (due to the light of the sun) has a significant effect on the calculations for navigation in space.



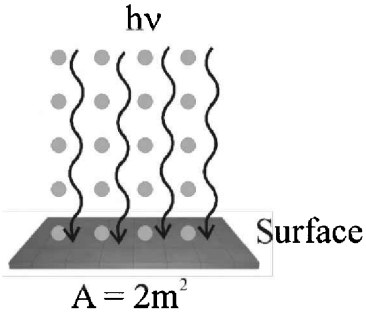
Now we will discuss about the three possible cases which can take place when light falls on a surface. These possible cases are –

- **Light is fully absorbed by the surface.**
- **Light is fully reflected by the surface.**
- **Light is partially reflected and partially absorbed by the surface.**

Now, we will understand these cases with the help of some examples.

Illustration 9

Calculate the value of pressure on a surface when light falls on it. The frequency of light is 10^4 Hz and intensity of light is 10 W/m^2. Area of surface is 2 m^2. Assume that the light is fully absorbed by the surface.



Sol. Here, light is fully absorbed by the surface. It means that all the photons are absorbed by the surface. We know -

$$\text{Photon flux} = \frac{I}{h\nu}$$

So number of photons hitting the whole area of surface in unit time will be-

$$N = \left(\frac{I}{h\nu}\right)A$$

$$\Rightarrow \quad N = \left(\frac{I}{E}\right)A$$

We have seen that momentum of one photon is given as $\frac{'E'}{C}$. So, momentum of all the photons hitting surface in unit time or initial momentum of photons will be -

p_I = N (momentum of one photon)

$$= \left(\frac{I}{E}A\right)\left(\frac{E}{C}\right) = \frac{IA}{C}$$

As photons are absorbed by surface so momentum of photons becomes zero after striking the surface.

$p_F = 0$

Now change in momentum of photons in unit time-

$$\Delta p = p_I - p_F = \frac{IA}{C}$$

We know that force is defined as rate of change of momentum. So, force on the surface due to light.

$$F = \frac{\Delta p}{\Delta t} = \frac{IA}{C}$$

Radiation pressure is defined as-

$$\text{R.P.} = \frac{F}{A}$$

$$\Rightarrow \quad \textbf{R.P.} = \frac{\mathbf{I}}{\mathbf{C}}$$

This is an important formula which shows the value of radiation pressure in case of full absorption. Here intensity of light is 10 W/m^2. So

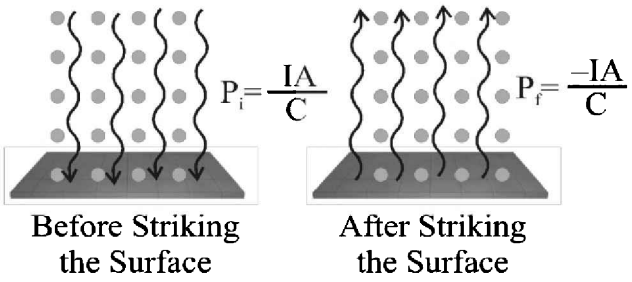
$$\text{Radiation pressure} = \frac{10}{3\times10^8} = 3\cdot 33 \times 10^{-8}\ \text{N/m}^2$$

Here we can see that value of radiation pressure is very less. That's why, in normal cases, we don't feel it.

Illustration 10

Repeat the above question if light is fully reflected by surface.

Sol. Here, light is fully reflected by the surface. It means that all the photons are reflected by the surface. Now here initial momentum of photons (coming in unit time) will be same as in case of. full absorption. But here final momentum of photons will be different (i.e. not equal to zero) as shown below



Here change in momentum of photons in 1 sec-

$$\Delta p = p_i - p_F = \frac{2IA}{C}$$

So force on surface due to light -

$$F = \frac{\Delta p}{\Delta t} = \frac{2IA}{C}$$

Now radiation pressure on surface -

$$R.P = \frac{F}{A}$$

$$\Rightarrow \quad \mathbf{R.P = \frac{2I}{C}}$$

Here intensity of light is 10 W/m^2. So radiation pressure on surface-

$$R.P = 6 \cdot 66 \times 10^{-8} \ N/m^2$$

In this way, we can calculate the value of radiation pressure in case of full reflection.

Illustration 11

Calculate the value of radiation pressure on a surface due to light. It is given that 40% of light is absorbed by the surface and 60% of light is reflected by the surface. Intensity of light is 'I'.

Sol. This is the case of partial absorption where some photons are absorbed by the surface and Some are reflected by the surface. Now, we will find radiation pressure on the surface due to absorption and reflection of light individually.

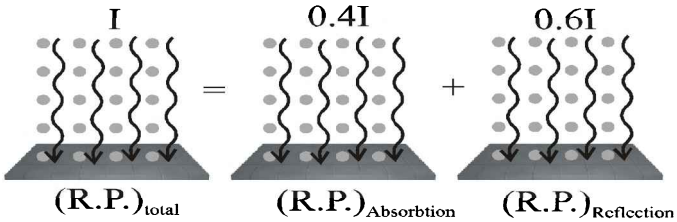
- As 40% of light (0.4I) is absorbed by surface so radiation pressure due to absorption of photons-

$$(R.P)_{absorption} = \frac{0.4I}{C}$$

- As 60% of light (0.6 I) is reflected by surface so radiation pressure due to reflection of photons-

$$(R.P)_{reflection} = \frac{2(0.6I)}{C}$$

- So, the total pressure on surface will be equal to sum of these two as shown below.



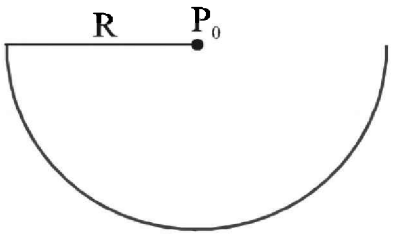
i.e. $R.P = (R.P)_{absorption} + (R.P.)_{reflection}$.

$$= \frac{0\cdot 4I}{C} + \frac{1\cdot 2I}{C} = \frac{1\cdot 6I}{C}$$

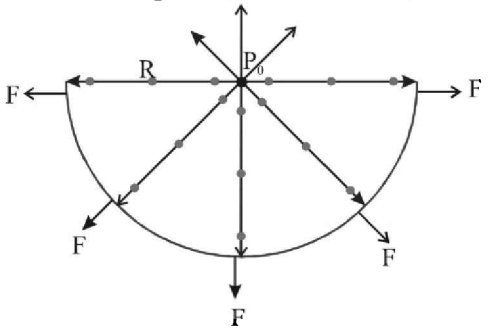
So, in this way we can find radiation pressure in case of partial reflection.

Illustration 12

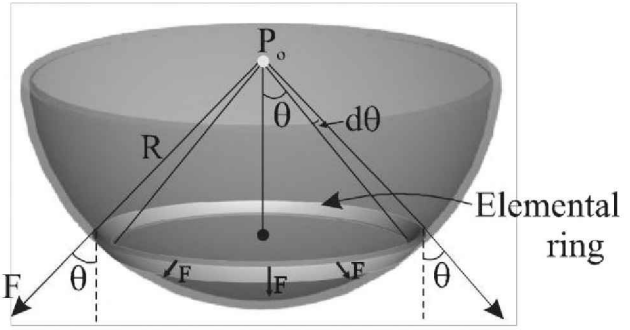
A point source is placed in a hemisphere as shown below. Find out the value of radiation pressure if light is fully absorbed by the surface of a hemisphere. Power of point source is 'P_0'



Sol. Here we can see that point source will emit light in all directions. When this light will hit the hemisphere, it will exert force on hemisphere (in all directions) as shown below.



Now, to find the resultant force on a hemisphere, let's take an elemental ring on hemisphere as shown below.



Here we can see that as light will fall on the elemental ring, it will exert force on it. As ring absorbs all the photons falling on it so the magnitude of force on ring will be-

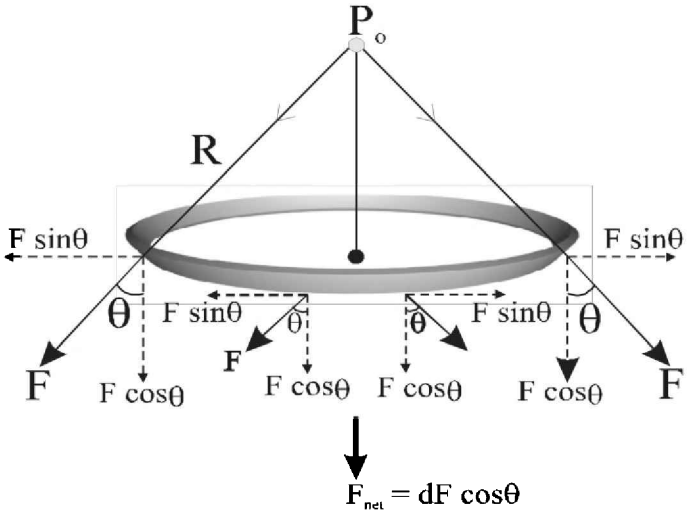
dF = (Radiation pressure on ring) (Area of ring)

$$\Rightarrow \quad dF = \left(\frac{I}{C}\right)(dA)$$

$$\Rightarrow \quad dF = \left(\frac{P_0}{4\pi R^2}\cdot\frac{1}{C}\right)dA \qquad \text{... (1)}$$

$$\left(\text{As } I = \frac{P_0}{4\pi R^2}\right)$$

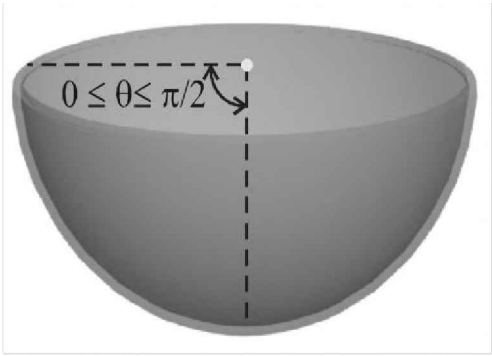
From above figure, we can see that light will exert force on the ring in different directions. But here horizontal component of forces on ring will cancel out each other so resultant force on ring will be in a downward direction as shown below.



So here, net force on elemental ring will be in downward direction.

i.e. $F_{net} = dF\cos\theta$

Now, here 'θ' will vary from 0 to $\frac{\pi}{2}$ as shown in below figure.



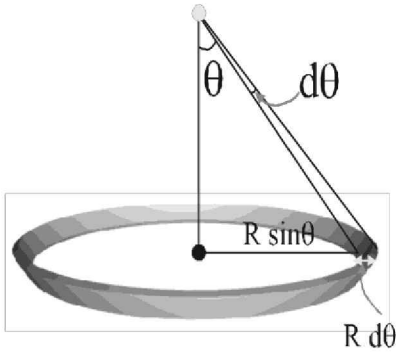
Now, to find the force on a hemisphere, we will integrate '$dF\cos\theta$' from 0 to $\frac{\pi}{2}$. So, net force on hemisphere will be-

$$F = \int_0^{\frac{\pi}{2}} dF\cos\theta$$

$$\Rightarrow \quad F = \int_0^{\frac{\pi}{2}} \frac{P_0}{4\pi R^2 C} dA\cos\theta$$

$$\Rightarrow \qquad F = \frac{P_0}{4\pi R^2 C}\int_0^{\frac{\pi}{2}} dA\cos\theta \qquad \text{... (2)}$$

Now we will find area of ring (dA).



(Thickness of ring)

dA = (perimeter of ring) (Thickness of ring)

$= (2\pi R\sin\theta)(Rd\theta)$

$= 2\pi R^2 \sin\theta d\theta$

Put value of 'dA' in equation (2) -

$$F = \frac{P_0}{2C}\int_0^{\frac{\pi}{2}} \sin\theta\cos\theta\, d\theta$$

$$\Rightarrow \quad F = \frac{P_0}{4C}\int_0^{\frac{\pi}{2}} \sin 2\theta\, d\theta$$

$$\Rightarrow \quad F = \frac{P_0}{4C}$$

So radiation pressure on hemisphere -

$$P = \frac{F}{A} = \frac{P_0}{4C}\left(\frac{1}{4\pi R^2}\right) = \frac{P_0}{16\pi R^2 C}$$

1.11 Dual Nature of Light

Previously we have seen that we cannot explain the phenomenon of photoelectric effect if we consider light as wave. But we can easily explain this phenomenon if we consider light as a stream of particles. So the photoelectric effect gives a firm experimental proof about the particle theory of light.

But if we consider light as particle, we will not be able to explain some other phenomenon like interference, diffraction and polarization. However, this phenomenon's can be explained easily by considering light as wave.

Above discussion again puts the same question 'what is light'. We can see that some experiments show that light behaves as wave, while other experiments show that it behaves as stream of particles. That's why scientists came to a conclusion that "light have dual nature". It means, sometimes light behave as wave and sometimes it behave as stream of particles.

1.12 Dual Nature of Matter

We have already studied about the dual nature of light. Now, we will study about the dual nature of matter. In 1923 De Broglie came up with a new theory. After reading Einstein's theory of photo

emission, he thought of a very natural thing- if light, which was considered to be a wave for many years, can behave as a particle then things which are normally considered to be particles might also behave as waves.



Louis de Broglie
(1892 - 1987)
Awarded Nobel in 1929

Einstein much appreciated this theory. But some other scientists of that time were confused that how the objects in our day to day life like pen, stone etc. can behave as particles. So to remove the confusion, De Broglie gave a very logical explanation about the dual nature of matter. Now we will understand that explanation.

The basic characteristic of a particle is its momentum (p) and basic characteristic of a wave is its wavelength (λ). That's why, when Einstein proposed that light is a stream of particles (i.e. photons), he also defined momentum of a photon as –

$$P = \frac{h}{\lambda}$$

In the same way, when De- Broglie proposed that particle can behave as wave, he also defined the wavelength of particles as -

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

This wavelength associated with matter is known as 'de Broglie wavelength'

Now, say for example a 0.50 kg ball is moving with a speed 1 m/sec then its wavelength will be -

$$\lambda_{ball} = \frac{h}{mv} = \frac{6.62\times10^{-34}\,\text{Js}}{(0.50\,\text{kg})(10\,\text{m/s})} = 1{\cdot}3\times10^{-33}\,\text{m}$$

Here we can see that wavelength of the ball is very small. Actually wavelength associated with the objects in day to day life is very small. That's the reason that we do not feel the wave nature of objects in our day to day life.

Now, let's talk about the wavelength of subatomic particles. Say, for example, an electron is moving with velocity 1m/sec then its wavelength will be -

$$\lambda_{electron} = \frac{h}{mv} = \frac{6{\cdot}62\times10^{-34}\,\text{J.S}}{(9.1\times10^{-31}\,\text{kg})(1\,\text{m/S})} = 0.72\ \text{mm}$$

Here we can see that wavelength of electron is significant. So, De Broglie said that we will be able to notice the wave nature of electrons and other subatomic particles.

In 1927, Division and Germer experimentally proved that electrons show interference when they are scattered from the surface of a metal crystal. As interference can only be shown by waves, wave nature of electrons was proved experimentally.

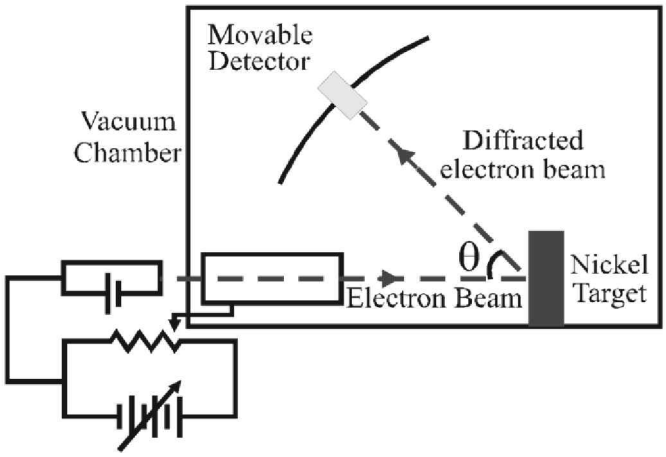


Illustration 13

Find De Broglie wavelength of a 5kg object if -

(i) Object is moving with velocity 10 m/sec.

Sol. We know $\lambda = \dfrac{h}{mv}$

$\Rightarrow \quad \lambda_{object} = \dfrac{6.62 \times 10^{-34}}{(5)(10)}$

$\Rightarrow \quad \lambda = 1.3 \times 10^{-35}\ m$

(ii) Kinetic energy of object is 40 J.

Sol. We know $K.E. = \dfrac{1}{2}mv^2$

$\Rightarrow \quad K.E. = \dfrac{m^2v^2}{2m}$

$\Rightarrow \quad K.E. = \dfrac{p^2}{2m}$ [p = mv]

$\Rightarrow \quad \mathbf{p = \sqrt{2m(K.E.)}}$

DE Broglie wavelength of object -

$$\lambda = \frac{h}{p}$$

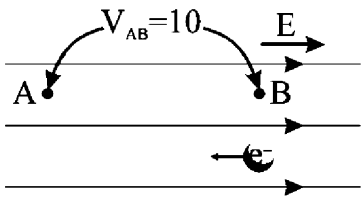
$\Rightarrow \quad \boldsymbol{\lambda = \dfrac{h}{\sqrt{2m(KE)}}}$

$\Rightarrow \quad \lambda = 3.31 \times 10^{-35}\ m$

Illustration 14

An electron is accelerated through a potential difference of 10 V. Find out its de Broglie wavelength.

Sol.. To understand this problem, we can imagine an electric field. Say the potential difference between A and B is 10V.



$V_A - V_B = 10V$

As shown above, an electron is released at point B. This electron will accelerate towards point A due to the electric force on it. Now, de Broglie wavelength of electron will be-

$$\lambda_{elect} = \frac{h}{\sqrt{2m(KE)}} \quad \text{... (1)}$$

To find the kinetic energy of electron, we will apply the law of energy conservation between point A and B. That is-

|Change in potential energy | = |Change in KE of electron|

$\Rightarrow |\Delta PE| = |\Delta KE|$

$\Rightarrow |(V_B - V_A)e| = |(KE)_B - (KE)_A|$

$\Rightarrow |(V_A - V_B)e| = |0 - (KE)_A|$

$\Rightarrow (KE)_A = 10e$

So from equation (1), we get-

$$\lambda_{elect} = \frac{h}{\sqrt{2m(10e)}}$$

From the above equation, we can find the wavelength of an electron. In general, when an electron is accelerated through a voltage difference of 'V' volt then its wavelength is given as-

$$\boldsymbol{\lambda_{elect} = \frac{h}{\sqrt{2m(Ve)}}}$$

Exercise of Photoelectric Effect

Multiple Choice Types

Only One Option Correct

Q.1 What is the energy of photon of wavelength 12400 Å?
(a) 1eV (b) 2eV (c) 5eV (d) 12 eV

Ans. **[a]**

Q.2 The work function of a substance is 4.0eV. The maximum wavelength of light that can cause photoelectron emission from this substance is approximately?
(a) 540 nm (b) 400 nm (c) 310 nm (d) 220 nm.

Ans. **[c]**

Q.3 For light of wavelength 5000 Å, the photon energy is nearly 2.5 eV. For 1 Å wavelength, the photon energy will be close to?

(a) $2.5 \div 5000\,\text{eV}$ (b) $2.5 \div (5000)^2\,\text{eV}$

(c) 2.5×5000 eV (d) $2.5 \times (5000)^2\,\text{eV}$

Ans. **[c]**

Q.4 The work function for aluminium is 4.2eV and that for sodium is 2.0 eV. The two metals were illuminated with appropriate radiation so as to cause photo-emission then,
(a) Both aluminium and sodium will have the same threshold frequency.
(b) The threshold frequency of aluminium will be more than that of sodium.
(c) The threshold frequency of aluminium will be less than that of sodium.
(d) The threshold wavelength of aluminium will be more than that of sodium.

Ans. **[b]**

Q.5 If a surface has work function of 3 eV, the longest wavelength of light which will cause the emission of electrons is
(a) 4.8×10^{-7} m (b) 5.99×10^{-7}m
(c) 4.13×10^{-7} m (d) 6.84×10^{-7} m

Ans. **[c]**

Q.6 The photoelectric effect begins in this metal at a frequency of 6×10^{14} Hz. Find the work function for this metal. (Given h = 6.63×10^{-34} Js)
(a) 2.48eV (b) 3eV (c) 2eV (d) 2.3eV

Ans. **[a]**

Q.7 The work function of a metal is ϕ and λ is the wavelength of the incident radiation. There is no emission of photo electrons when?

(a) $\lambda > \frac{hc}{\phi}$ (b) $\lambda = \frac{hc}{\phi}$ (c) $\lambda < \frac{hc}{\phi}$ (d) $\lambda \leq \frac{hc}{\phi}$

Ans. **[a]**

Q.8 A bulb emits light of wavelength 4500Å. The lamp is of power 150W and 8% of the energy appears as emitted light. How many photon are emitted by the lamp per second?
(a) 20.07×10^{-18} (b) 27.07×10^{-18} (c) 27.07×10^{18} (d) 27.17×10^{18}

Ans. **[d]**

Q.9 The frequency of incident light falling on a photosensitive metal plate is doubled, the KE of the emitted photoelectrons is

(a) Double the earlier value (b) Unchanged
(c) More than double (d) Less than double

Ans. **[c]**

Q.10 All electrons ejected from a surface by incident light of wavelength 200 nm can be stopped before travelling 1m in the direction of uniform electric field of 4 N/C. The work function of the surface is?
(a) 4 eV (b) 6.2 eV (c) 2 eV (d) 2.2 eV

Ans. **[d]**

Q.11 Light of two different frequencies whose photons have energies 1eV and 2.5eV respectively, successively illuminate a metal whose work function is 0.5 eV. The ratio of the maximum speeds of the emitted electrons will be.
(a) 1: 5 (b) 1 : 4 (c) 1 : 2 (d) 1 : 1

Ans. **[c]**

Q.12 In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the exciting wavelength is changed to the $\frac{3\lambda}{4}$ speed of the fastest emitted electron will become

(a) $V_1\sqrt{\frac{3}{4}}$ (b) $V_1\sqrt{\frac{3}{4}}$

(c) less than $V_1\sqrt{\frac{3}{4}}$ (d) equal to $V_1\sqrt{\frac{4}{3}}$

Ans. **[d]**

Q.13 The maximum kinetic energy of photoelectrons emitted from a surface when photons of energy 6 eV fall on it is 2 eV. The stopping potential in volts is?
(a) 2 V (b) 4 V (c) 6 V (d) 10 V

Ans. **[a]**

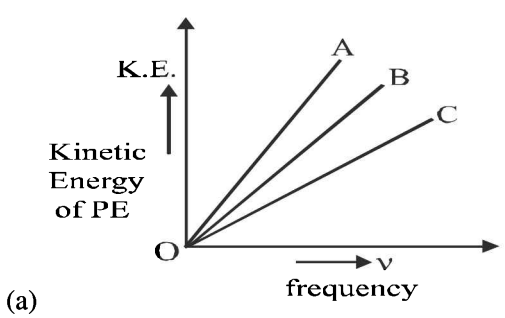
Q.14 When a metallic surface is illuminated with monochromatic light of wavelength λ, the stopping potential is $5V_0$. When the same surface is illuminated with light of wavelength 3λ, the stopping potential is V_0. Then the work function of the metallic surface is?
(a) eV_0 (b) $5eV_0$ (c) V_0 (d) e

Ans. **[a]**

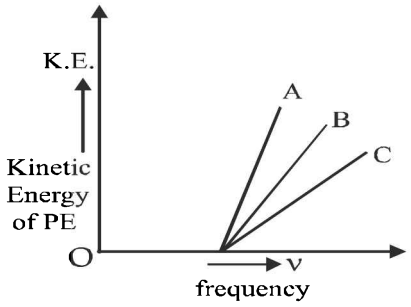
Q.15 A metallic surface is illuminated with monochromatic light of variable wavelength. Above wavelength of 5000Å, electrons are not emitted from the surface. With an unknown wavelength, stopping potential is 3 V. Find the unknown wavelength.
(a) 2252 Å (b) 2262 Å (c) 3452 Å (d) 2342 Å

Ans. **[b]**

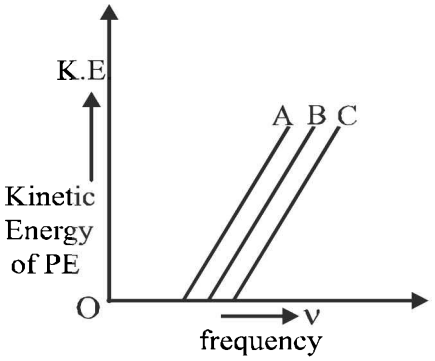
Q 16 Photo-emission is observed for three different metals A, B and C. The kinetic energy E_k, of the fastest photoelectrons versus ν the frequency of incident radiation graph is plotted for each metal. Which one of the following graph shows the phenomenon correctly?



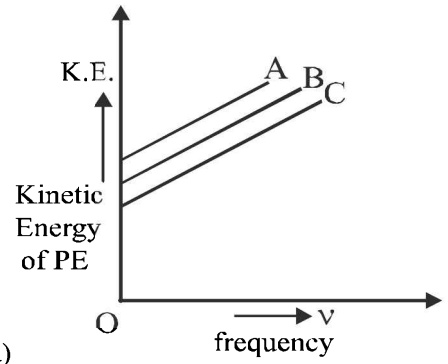
(a)



(b)



(c)



(d)

Ans. **[c]**

Q.17 The kinetic energies of photoelectrons range from zero to 4.0×10^{-19} J when light of wavelength 3000 Å falls on a surface. What is the stopping potential for this light?

(a) 2.5 V (b) 3 V (c) 3.2V (d) 2.1V

Ans. **[a]**

Q.18 Representing the stopping potential V along y-axis and $(1/\lambda)$ along x- axis for a given photocathode, the curve is a straight line, the slope of straight line and the intercept on the y-axis which is equal to

(a) e/hc,ϕ (b) hc/e,-ϕ/e

(c) ec/h,e/ϕ (d) he/c,-e/ϕ

Ans. **[b]**

Q.19 Find the momentum of a 12.0 eV photon?

(a) 6.4×10^{-27} js/m (b) 5.4×10^{-20} js/m

(c) 6×10^{-27} js/m (d) 6.4×10^{-7} js/m

Ans. **[a]**

Q.20 Find the de Broglie wavelength of a 0.01 kg pellet having a velocity of 10 m/s.

(a) 6.63×10^{-23} Å (b) 8.9×10^{-23} Å

(c) 1.97×10^{-23} Å (d) none of these

Ans. **[a]**

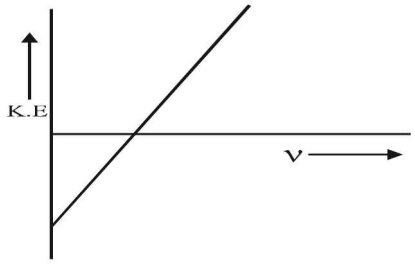
Q.21 An electron of mass m_C and a proton of mass m_P are accelerated through the same potential difference. The ratio of the de Broglie wavelength associated with an electron to that associated with proton is

(a) 1 (b) m_p/m_e (c) m_e/m_p (d) $\sqrt{m_p/m_e}$

Ans. **[d]**

More than One Correct

Q.22 The kinetic energy of the emitted photoelectrons against frequency (ν) of incident radiation is plotted as shown in fig. This graph helps us in determining the physical quantities.



(a) Work function of the cathode – metal
(b) There should frequency
(c) Plank's constant
(d) Charge on an electron.

Ans. **[a], [b], [c]**

Q.23 When photons of energy hc/λ fall on a metal surface, photoelectrons are ejected from it. If the work function of the surface is $h\lambda_{th}$ then.

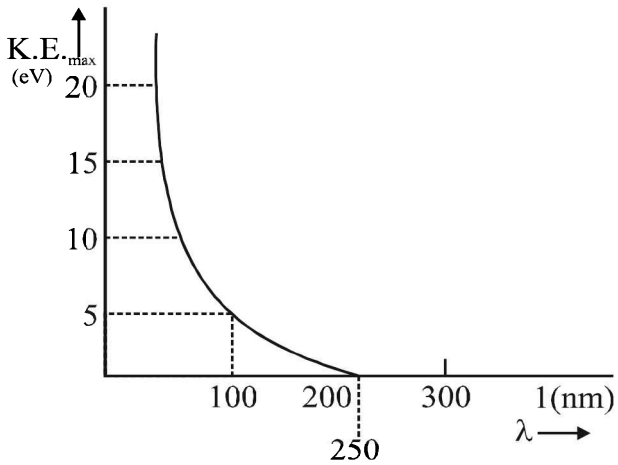
(a) Maximum kinetic energy of the electron is $[(hc/\lambda) - h\upsilon_{th}]$
(b) Maximum kinetic energy of the photoelectron is equal to (hc/λ)
(c) Minimum KE of the photoelectron is zero.
(d) Minimum kinetic energy of the photoelectron is equal to hc/λ.

Ans. **[a], [c]**

Q.24 When the intensity of a light source is increased,
(a) The number of photons emitted by the source in unit time increases.
(b) The total energy of the photons emitted per unit time increases.
(c) More energetic photons are emitted.
(d) Faster photons are emitted.

Ans. **[a], [b], [c]**

Q.25 In a photoelectric effect experiment, the maximum kinetic energy of the ejected photoelectrons is measured for various wavelength of the incident light. Figure shows a graph of this maximum kinetic energy K_{max} as a function of the wavelength of the light falling on the surface of the metal. Which of the following statement/s is /are correct?



(a) Threshold frequency for the metal is $1{\cdot}2 \times 10^{15}$ m
(b) Work function of the metal is 4.968 eV

(c) Maximum kinetic energy of photoelectrons corresponding to light of wavelength 100 nm is nearly 7.4 eV
(d) Photoelectric effect takes place with red light.

Ans. **[a], [b], [c]**

Matrix Type

Q.26 With respect to photoelectric effect experiment, match the entries of Column I with the entries of Column II.

Column I	Column II
(a) If f (frequency) is increased keeping I (intensity) and ϕ (work function) constant	(p) Stopping potential increases (q) Saturation photocurrent increases
(b) If I is increased keepingυ and ϕ constant	
(c) If the distance between anode and cathode increases.	(r) Maximum KE of the photoelectron increases (s) Stopping potential remains the same
(d) If f decreased keeping f and I constant.	

Ans. **[a] → [p],[r];[b] → [q],[s];[c] → [s];[d] → [p],[r]**

COMPREHENSION

Comprehension # Q.26 to Q.28

Threshold wavelength of silver is 3800 Å and wavelength of ultraviolet light is 2600Å which incident on silver surface. (mass of the electron 9.11×10^{-31} kg).

Q.27 Calculate the value of work function in eV?
(a) 1.77 eV (b) 3.27 eV (c) 5.69 eV (d) 2.32 eV

Ans. **[b]**

Q.28 Calculate the maximum kinetic energy (in eV) of the emitted photoelectrons?
(a) 1.5 eV (b) 2.3 eV (c) 3.8 eV (d) 4.2 eV

Ans. **[a]**

Q.29 Calculate the maximum velocity of the photoelectrons in?
(a) 72.89×10^8 V (b) 57.89×10^8 V
(c) 42.93×10^8 V (d) 68.26×10^8 V

Ans. **[a]**

Assertion & Reasoning

STATEMENT

Statement-1 and Statement -II. Each question has 4 choices (A), (B), (C) and (D) out of which ONLY ONE is correct. So select the correct choice. They are:-
(a) Statement - I is True, Statement - II is True; Statement-II is a correct explanation for Statement- I.
(b) Statement - I is True, Statement - II is True; Statement - II is NOT a correct explanation for Statement - I.
(c) Statement - I is True, Statement - II is False.
(d) Statement - I is False, Statement - II is True.

Q.30 **Statement-I:**
In photoelectric effect photoelectrons are emitted when the falling radiation has energy less than work function of metal surface.
Because
Statement-II:
Work function is the minimum energy needed to eject electrons from the surface of metal.

Ans. **[d]**

Q.31 **Statement-I:**
Work function of different metals is different.
Because
Statement-II:
Work function is the minimum energy required by an electron to just escape from the metal surface.

Ans. **[b]**

Q.32 A proton and an electron both have energy 50eV.
Statement-I:
Both have different wavelengths.
Statement-II:
Wavelength depends on energy and not on mass.

Ans. **[c]**

Q.33 **Statement-I:**
An insulated metal plate emits photoelectrons when first illuminated by ultraviolet light but then the number of photoelectrons emitted per unit time decreases until it stops altogether.
Because
Statement-II:
As more and more electrons leave the plate, its potential increases, decreasing the number of free electrons and finally stopping them.

Ans. **[a]**

Q.34 **Statement-I:**
In a photoelectric effect experiment different photoelectrons have different kinetic energy.
Because
Statement-II:
In the incident radiation different photons have different energy.

Ans. **[c]**

Q.35 **Statement-I:**
A photon (having wavelength less than threshold wavelength) is incident on a metallic surface. An electron must eject out of the metallic surface.
Statement-II:

$$KE_{max} = \frac{hc}{\lambda} - \phi$$

When $\frac{hc}{\lambda} > \phi$, then maximum possible kinetic energy of electron is $\frac{hc}{\lambda} - \phi$.

Ans. **[d]**

Q.36 **Statement - I:**
In photoelectric effect, the number photoelectrons emitted is always equal to number of photons incident.
Statement-II:
All the photons falling on the surface will eject photoelectrons, is not necessary.

Ans. **[d]**

Q.37 **Statement-I:**
Though light of a single frequency (monochromatic) is incident on a metal, the energies of emitted photo-electrons are different.
Statement-II:

The energy of electrons emitted from inside the metal surface is redistributed in collision with the other atoms in the metal.

Ans. **[a]**

Q.38 **Statement-I:**
Work function of Aluminium is 4.2 eV. Emission of electron will not be possible if two photons each of energy 2.5 eV strike an electron of Aluminium.
Statement-II:
For photo-electric emission, the energy of each photon should be greater than the work function of Aluminium.

Ans. **[a]**

Q.39 **Statement-I:**
The de-Broglie wavelength of a molecule (in a sample of ideal gas) varies inversely as the square root of absolute temperature.
Statement-II:
The rms velocity of a molecule (in a sample of ideal gas) depends on temperature.

Ans. **[b]**

Subjective Type

Q.40 Illuminating the surface of a certain metal with light of wavelengths λ_1= 0.35 mm and λ_2= 0.54 mm, it was found that the corresponding maximum velocities of photo electrons have a ratio 2:1. Find the work function of that metal.

Ans. ϕ= **5.64 eV**

Q.41 When a surface is illuminated with light of wavelength 4950 Å, a photocurrent appears which vanishes if a retarding potential greater than 0.6 Volt is applied across the photo tube. When a different source of light is used.it is found that the critical retarding potential is changed to 1.1 Volt. Find the work function of the emitting surface and the wavelength of second source.

Ans. $\phi = 3.04\times10^{-19}$, λ_2= **4125 Å**

Q.42 The stopping potential for photoelectrons emitted from a surface illuminated by light wavelength of 5893 Å is 0.36 V. Calculate the maximum kinetic energy of photoelectrons, Find the work function of the surface and threshold frequency?

Ans. $\phi = 2.784\times10^{-19}$ joule , $\nu_{th} = 4.22\times10^{14}$ Hz

Q.43 In an experiment of photo electric emission, following observations were made;
(i)Wavelength of the incident light = 1.98×10^{-7}m,
(ii) Stopping potential = 2.5 volt.
Find:
(A) Kinetic energy of photoelectrons with maximum speed.
(B) Work function and
(C) Threshold frequency

Ans. **(a)** KE_{max} **= 2.5 eV** **(b)** ϕ **= 3.76 eV**
(c) $\nu_{th} \approx 9.1\times10^{14}$ **Hz**

Q.44 The photoelectric threshold wavelength of tungsten is 2730 Å. Ultraviolet light of wavelength 2000Å is incident on a clean metal plate of area $10^{-4}m^2$, with an intensity of photons10 Wm^{-2}. If the efficiency of conversion is 5%, find the photocurrent at saturation. What is the stopping potential?

Ans. **I= 8mA,** V_{SP} **= 1.67 V**

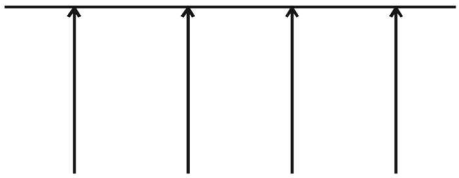
Q.45 A light source, emitting three wave-lengths 5000 Å, 6000 Å and 7000 Å, has a total power of 10^{-3} W and a beam diameter 2×10^{-3} m. The power density is distributed equally amongst the three wavelengths. The beam shines normally on a metallic surface of area $10^{-6}m^2$ and having a work function of 1.9 eV. Assuming that each photon liberates an electron, calculate the charge emitted per unit area in one second.

Ans. **$q = 94.08$ coulomb/m^2 sec.**

Q.46 A beam of light has three wavelengths 4144 Å, 4972 Å and 6216 Å with a total intensity of 3.6×10^{-3} W m^{-2} equally distributed amongst the three wavelengths. The beam falls normally on an area 1.0 cm^2 of a clean metallic surface of work function 2.3 eV. Assuming that there is no loss of light by reflection and that each energetically capable photon ejects one electron. Calculate the number of photo-electrons liberated in 2 seconds.

Ans. **Hence, total electrons emitted in 2 seconds = $2N = 11 \times 10^{11}$**

Q.47 A plate of mass 10 gm. is in equilibrium in air due to the force exerted by light beam on plate. Calculate power of beam. Assume plate is perfectly absorbing.



Ans. **$P = 3 \times 10^7$ W**

Q.48 What voltage must be applied to an electron microscope to produce electron of wavelength0.4 Å?

Ans. **1.42×10^{36} v**

Q.49 The wavelength of light from the spectral emission line of sodium is 589 nm. Find the kinetic energy at which an electron and a neutron would have the same de-Broglie wavelength. Given that mass of neutron = 1.66×10^{-27}kg.

Ans. **$KE_{max} = 3.81 \times 10^{-28}$ J**

Q.50 Determine the De Broglie wavelength of a proton, whose kinetic energy is equal to the rest mass energy of an electron. Given that the mass of an electron is 9.1×10^{-31} kg and the mass of a proton is 1837 times as that of the electron.

Ans. **$\lambda_{de} = 4 \times 10^{-14}$ m**

02 ATOMIC STRUCTURE

2.1 Introduction

In this chapter, we are going to study about atom and its structure. The atomic structure, as we know today, is the result of a gradual development of ideas and experiments conducted throughout the history of science. This development is the result of the hard work of some of the most intelligent and finest minds that humankind has ever produced like J.J Thomson, Rutherford and Bohr etc.

On 5thOct 1906, great scientist, Ludwig Boltzmann, committed suicide due to depression. His theory was that all the things in nature are made up of almost the same type of small particles called **atoms**.

At present, his theory sounds obvious. But in early 20th century, people did not accept this theory. It was very difficult for them to imagine that all the things (like big mountains, small flies, water, animals, human, etc.) that look so different from each other are made up of similar type of small particles. Due to lack of acceptance of his ideas, he got frustrated and committed suicide.



Ludwig Eduard Boltzmann
(1844 - 1906)

Later, Einstein experimentally proved that atoms do exist, so people had to believe him.

Now, once the existence of atoms was established, scientists began to think what is inside an atom? As atoms can't be seen through the naked eyes so the only way to find out more about them was to do - intelligent experiments. So scientists did many experiments and based on those experiments, many

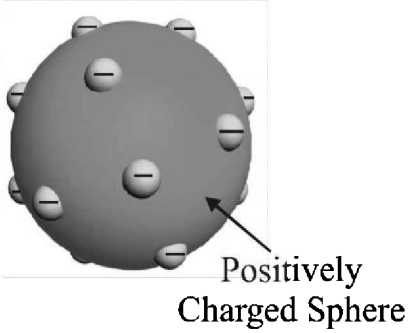
theories were given about atomic structure. Here we will see some of the most significant theories.

2.2 J.J Thomson's Model

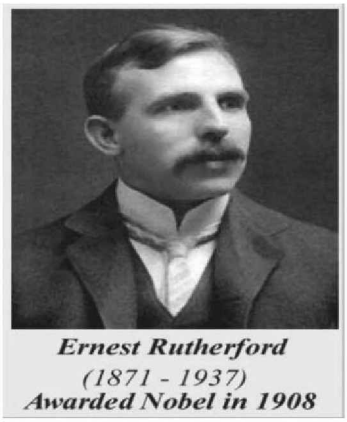


Joseph John Thomson
(1856 - 1940)
Awarded Nobel in 1906

First significant theory about atomic structure was given by J.J. Thomson. In his experiments, he noticed that in some situations, some negatively charged tiny particles were coming out of atoms. He named these particles as electrons. Thus it was established that atoms contain electrons. Further, as it was known that atoms were electrically neutral, so he thought that there must also be an equal amount of positive charge present inside the atom. He said that the positive charge is distributed homogeneously in the atom and negatively charged electrons are distributed throughout this positive charge, just as plums are distributed in a pudding. Pictorial representation of this model is as shown in the figure below. That's why, this model is also known as a ***plum-pudding model of atom.***

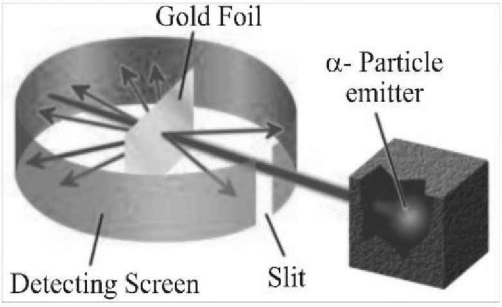


2.3 Rutherford's Model of Atom

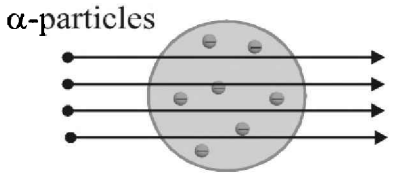


Ernest Rutherford
(1871 - 1937)
Awarded Nobel in 1908

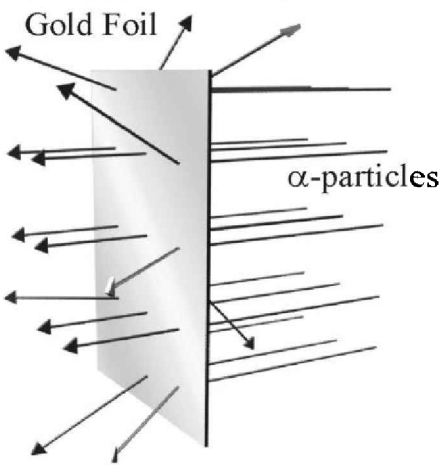
Around 1911, Ernest Rutherford asked his students Geiger and Marsden to perform an experiment know as Gold-foil experiment or Geiger-Marsden scattering experiment. The set up for Gold - foil experiment is as shown in the figure below.



In this experiment, a beam of positively charged 'α-particles' was incident on a thin sheet of gold foil. Thin sheet ensured that almost a single layer of atom was taken. As per Thomson, positive charge is homogeneously distributed inside an atom. So if a heavy alpha particle passes through it, it will not encounter any massive positive concentration of charge. Thus, it should pass through the sheet with minor deflections only or no deflection at all, as shown in the figure below.



As expected, experimentally, it was found that most of the α- particles crossed the gold foil either unaffected or with minor deflections. But a very small percentage of α- particles were deflected through angles much larger than 90°. Very rarely some α- particles were deflected at an angle of 180° i.e. they reversed their direction.

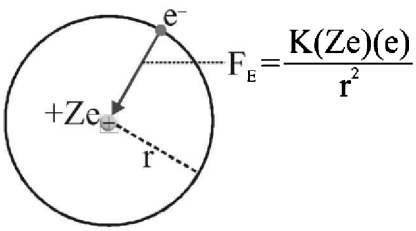


Here, you can see that experimental results contradicted predicted results. It was clear that Thomson's model could not explain the occasional large deflection of the particles. So it was established that a new better model of the atom was needed.

Based on the observations of these experiments, Rutherford predicted what's inside the atom. His predictions were:

- Most of the space in an atom is empty.
- An atom has a very small, heavy and positively charged particle at the centre. This particle is called the nucleus.

- Electrons move around the nucleus in circular orbits in the same way as planets revolve around the sun in the solar system.
- The necessary centripetal force is provided by the electrostatic force between electrons and nucleus. Say, an electron is revolving in n^{th} orbit of an atom as shown in the figure below.

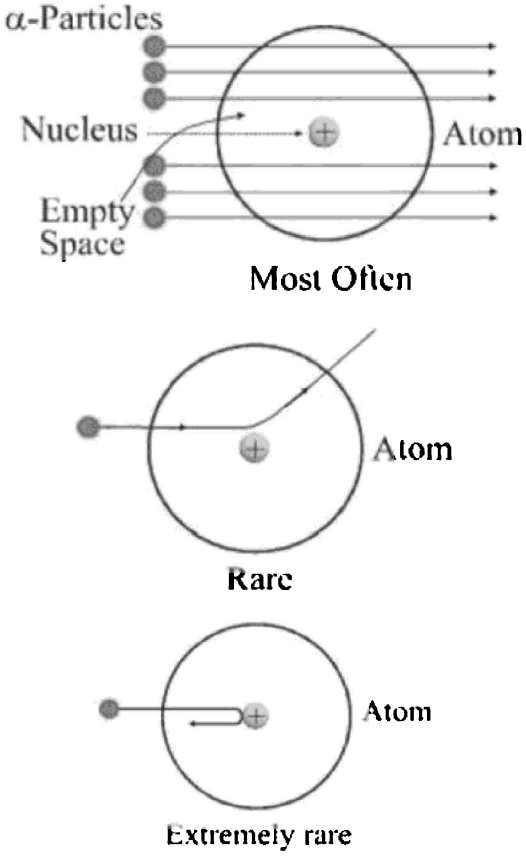


$$\frac{mv_n^2}{r_n} = \frac{K(Ze)(e)}{r_n^2}$$

$$\Rightarrow \quad \mathbf{mv_n^2 = \frac{KZe^2}{r_n}} \qquad \text{... (1)}$$

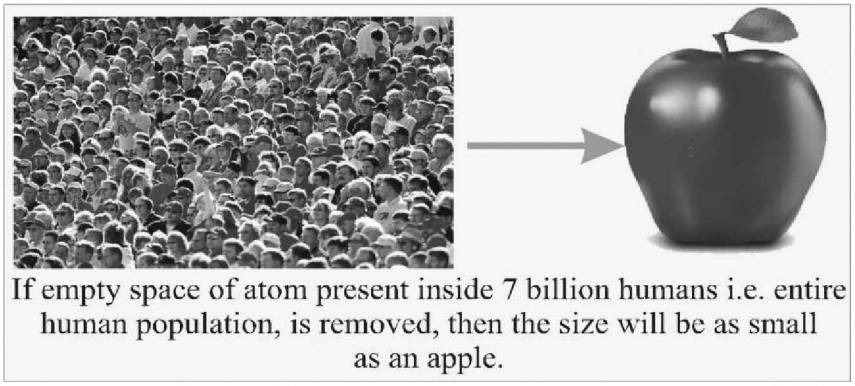
Above equation relates the radius of an orbit and velocity of electrons in that orbit. According to Rutherford, radius of circular orbits (r) can have any value. For each orbit, electron will have a different velocity.

These conclusions easily explained the experimental results as shown in the figure below.



most of the space inside an atom is empty, most of the alpha particles pass through the empty space undeflected. Only when an alpha particle comes close to the nucleus, it faces large repulsion of -

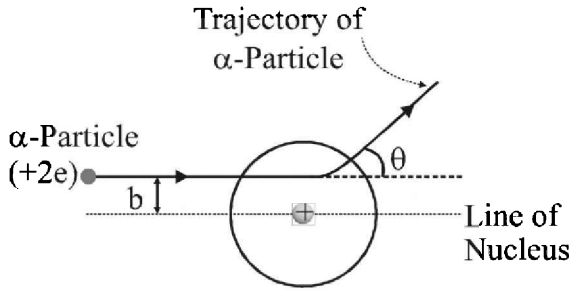
positive nucleus and is thus, deflected by large amount. But this happens rarely as the nucleus is so small in size. Very rarely when an alpha particle collides head-on with heavy nucleus, it bounces back. Later, it was discovered that more than 99.9% mass of the atom is concentrated in the nucleus. It means that most of the space in an atom is empty. For example, if an atom would be about the size of a cricket ground, the nucleus will be even smaller that a cricket ball.



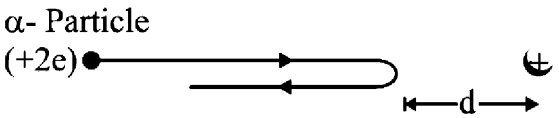
If empty space of atom present inside 7 billion humans i.e. entire human population, is removed, then the size will be as small as an apple.

2.4 Trajectory of α-particles in Gold-Foil Experiment

We have seen that when an α-particle passes close to nucleus then it gets deviated from its path, as shown in the figure below.



Here 'θ' shows the deviation of α-particle from its normal path and 'b' shows the distance between the initial path of α- particle and the line of the nucleus. **'θ'** is called **angle of deviation** and **'b'** is called **impact parameter**. For smaller impact parameter, α-particle suffers larger deflection. This happens because as 'b' decreases, the electrostatic repulsion force on α- particle increases. For 'b' equal to zero, α- particle retraces its path (i.e. θ = 180°) as shown in the figure below.

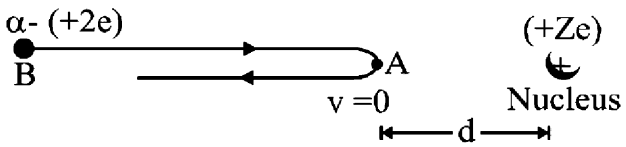


Distance (d) is called **'distance of closest approach'.**

Illustration 1

In the Gold-Foil experiment, find out the distance of closest approach of anα- particle if the kinetic energy of α- particle is 7.7 MeV when it is far from a nucleus of gold. [For gold Z = 79]

Sol.



As alpha particle approaches nucleus, its velocity decreases due to electrostatic repulsion of the nucleus. At A, its velocity becomes zero and it reverses its path.

$$(K.E)_B = 7.7MeV = 1.2 \times 10^{-12}\ J$$

As point B is at infinite distance from the gold foil so

$$(P.E)_B = 0$$

Potential energy at point A –

$$(PE)_A = \frac{1}{4\pi \in_0}\frac{(q_\alpha)(q_{Au})}{d}$$

$$\Rightarrow \quad (PE)_A = \frac{1}{4\pi \in_0}\frac{(2e)(79e)}{d}$$

To find the value of 'd', we will apply concept of energy conservation at points A and B.

$$(K.E)_A + (PE)_A = (K.E)_B + (PE)_B$$

$$\Rightarrow \quad \frac{1}{4\pi \in_0}\frac{(2e)(79e)}{d} = 1.2 \times 10^{-12} + 0\ ,$$

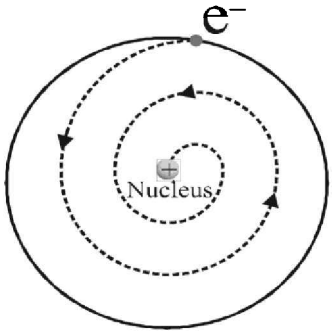
$$\Rightarrow \quad d = 3 \times 10^{-14}\ m$$

2.5 Limitations of Rutherford Model of Atom

Initially, Rutherford's model was widely accepted as many experimental results could be explained using this model. But two major problems were found with this model. Let's discuss them.

- **Not justifying Maxwell's Theory-**

According to Maxwell, accelerating charge radiates energy in the form of EM waves. Now, according to Rutherford's model, the motion of electrons is also accelerated as they move in circular orbits. Thus, electrons should radiate energy like any other accelerating charge. If this happens then energy of an electron will decrease continuously. As a result, the electron will move in a spiral path around the nucleus and finally it will collide with the nucleus as shown in the figure below.



If this happens then the atom will be unstable. But the fact is that the atom is stable. So the Rutherford's model contradicts Maxwell's theory.

- **Spectrum of Gases-**

You must have seen the colourful tube lights outside shops at night as shown below. These tubes contain gases in them and when these gases are heated (in these cases, by use of electricity), they glow with different and unique colours. For example, when neon gas is heated, it gives red light and when hydrogen is heated, it gives blue light.

Scientists were not able to understand why different gases glowed with different colours on heating. But it was clear that this phenomenon must be having some link with atomic structure. So, an appropriate model of atom should be able to explain this phenomenon. But, Rutherford's atomic model fails to explain it. This is the second limitation of Rutherford's model.





These problems indicated that Rutherford's model needed some modifications. These modifications were provided by Niels Bohr.

2.6 Niels Bohr's Model



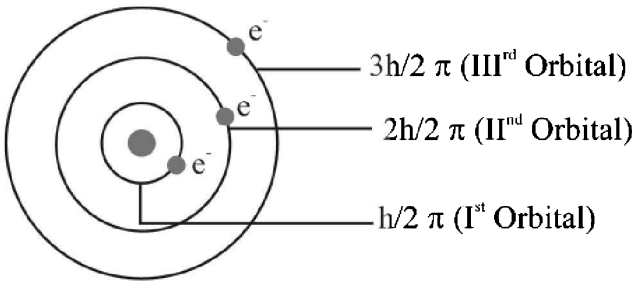
Niels Henrik David Bohr
(1885 - 1962)
Awarded Nobel in 1922

Bohr was an assistant in Rutherford's laboratory at the beginning of his career. In 1913, he hypothesized a new model of atom. Bohr gave three prominent points which are known as three postulates of Bohr. Let's understand them one by one.

1. In his first postulate, Bohr solves the first limitation of Rutherford's model. Bohr was convinced with the idea of Rutherford that electrons move in circular orbits around the nucleus. He said that Maxwell's theory was not valid inside an atom. It means that an electron (in the atom), though accelerated, does not radiate energy. Thus the energy of electrons does not decrease and it continues to revolve in circular orbit.

2. **Quantization of Angular Momentum -**
According to Rutherford, electrons revolve around the nucleus in circular orbits. Radius of circular orbits can have any value. Bohr disagrees here. According to Bohr, only certain value of radius are possible. He postulates that only the orbits, in which the angular momentum (mvr) of electron is an integral multiple of, $\frac{h}{2\pi}$ are possible.

i.e. $$\mathbf{mv_n r_n = n\left(\frac{h}{2\pi}\right);\ n = 1, 2, 3...}$$

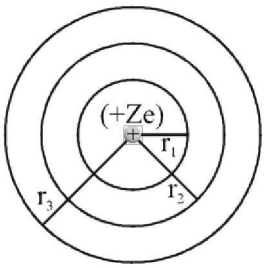


Bohr's Model

Bohr named these possible orbits as stationary orbits or stable orbits. You might be wondering where did this equation of quantization come from? Actually, Bohr had no idea about it. But he suggested that if we assume the quantization rule to be right then the second limitation of Rutherford's model can be solved. Once we see the 3rd postulate later in this chapter, we will see how quantization rule solves the limitation

2.7 Radius of Stationary Orbits for Hydrogen like Atoms -

Atoms (ions) which have only a single electron are called hydrogen like atoms, for example He^{+}, Li^{2+} etc. Below figure shows hydrogen like atom and its stationary orbits.



According to Bohr-

$$mv_n^2 = \frac{KZe^2}{r_n} \qquad \text{... (1)}$$

And $$m\,v_n\,r_n = \frac{n\,h}{2\pi} \qquad \text{... (2)}$$

On solving equation (1) and (2), we get

$$r_n = \left(\frac{h^2}{4\pi^2 m_e K e^2}\right)\frac{n^2}{Z}$$

$$\Rightarrow \quad \mathbf{r_n = (0.529)\frac{n^2}{Z}\ A^o} \qquad \text{... (3)}$$

We can find the radius of stationary orbits for hydrogen like atom with the help of above equation. For example, we can find the radius of the smallest orbit (r_1) for hydrogen (Z =1) by putting n = 1 in the above equation. It comes out to be –

r_1 = 0.529 A°

From equation (3), radius of other orbits for hydrogen will be-

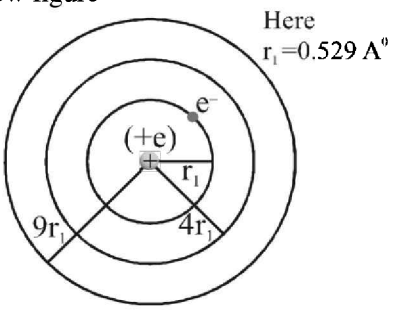
$r_2 = 4\,r_1 = 2.12$ A°

$r_3 = 9\,r_1 = 4.76$ A°

: : :

: : :

These orbits are shown in the below figure –





Quiz-1

Find the smallest radius of stationary orbits for He^+ (Z = 2) and Li^{2+} (Z = 3).

Sol. According to Bohr, an electron can revolve only in those orbits which are permitted by equation–

$$r_n = (0.529)\frac{n^2}{Z}\ \text{Å}$$

For smallest stationary orbit of He^+ n = 1 and Z = 2. On substituting values in above formula we get–

$$r_n = (0.529)\frac{(1)^2}{2}\text{Å} = 0.264\ \text{Å}$$

For smallest stationary orbit of Li^{2+} n = 1 and Z = 3. On substituting values in above formula we get–

$$r_n = (0.529)\frac{(1)^2}{3}\text{Å} = 0.176\ \text{Å}$$

Now, we will talk about the velocity of electron in these stationary orbits. From equation (2), we can see that-

$$v_n = \frac{n\,h}{2\pi m\,r}$$

On putting the value of 'r' from the equation (3), we get

$$v = (2.18\times10^6)\frac{Z}{n}\ \text{m/sec}$$

We can find the velocity of electron in stationary orbits with the help of above equation. For example, the velocity of electron in the first orbit (n = 1) of hydrogen atom will be-

$$v_1 = 2{\cdot}8 \times 10^6\ \text{m/sec}$$

Similarly, velocity of electrons in other orbits of hydrogen atom will be-

$$v_2 = \frac{v_1}{2}$$

$$v_3 = \frac{v_1}{3}$$

Here we can see that velocity of electron decreases as it moves to higher orbits.



Quiz-2

Find velocity of electrons in first, second and third orbit of Li^{2+} (Z = 3).

Sol. As, $v = \left(2.18\times10^{6}\right)\frac{Z}{n}$ m / sec

For first orbit, n = 1

$\Rightarrow \quad v = \left(2.18\times10^{6}\right)\frac{3}{1}$ m / sec = 6.54×10^{6} m/sec

For second orbit, n = 2

$\Rightarrow \quad v = \left(2.18\times10^{6}\right)\frac{3}{2}$ m / sec = 3.27×10^{6} m/sec

For third orbit, n = 3

$\Rightarrow \quad v = \left(2.18\times10^{6}\right)\frac{3}{3}$ m / sec = 2.18×10^{6} m/sec

Energy of Electron In Stationary Orbits -

Kinetic energy of an electron in n^{th} orbit is given as:-

$$K.E. = \frac{1}{2}mv_n^2$$

$$\Rightarrow \quad K.E. = \frac{KZe^2}{2r_n} \quad \text{(using equation 1)} \qquad \ldots.(4)$$

Electrostatic potential energy of electron in n^{th} orbit is given as-

$PE = -eV$

Here 'V' is the potential due to the charge on nucleus (+ Ze) so –

$$P.E = -e\left(\frac{KZe}{r_n}\right) = -\frac{KZe^2}{r_n} \qquad \ldots (5)$$

From equation (4) and (5), we can see that -

PE = – 2 (KE)

The total energy of an electron is given as the sum of its kinetic energy and potential energy. So, total energy is given as:-

$E_T = KE + PE$

$\Rightarrow \quad E_T = KE + (-2\ KE)$

$\Rightarrow \quad E_T = -KE$

$$\Rightarrow \quad E_T = -\frac{KZe^2}{2r_n}$$

Substituting value of r_n

$$\Rightarrow E_T = -\frac{KZe^2}{2\left(\left(\frac{h^2}{4\pi^2 m_e Ke^2}\right)\frac{n^2}{Z}\right)} = -\frac{4\pi^2 m_e K^2 Z^2 e^4}{2h^2 n^2}$$

On putting the values of constants in the above equation, we get -

$$\mathbf{E_T = -13.6 \times \frac{Z^2}{n^2}\ eV} \qquad \text{... (3)}$$

For example, energy of electron in Ist orbit (n = 1) of hydrogen atom (Z = 1) will be:-

$$\mathbf{E_1 = -13.6\ eV}$$

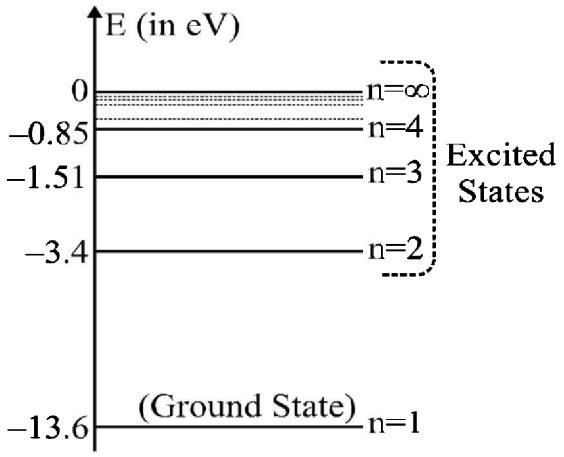
Similarly, the energy of an electron in higher orbits of hydrogen atom will be:-

$$E_2 = \frac{-13.6}{(2)^2} = -3.4\ eV$$

$$E_3 = \frac{-13.6}{(3)^2} = -1.51\ eV$$

: : :
: : :

The diagram below shows all these energy states of the hydrogen atom.



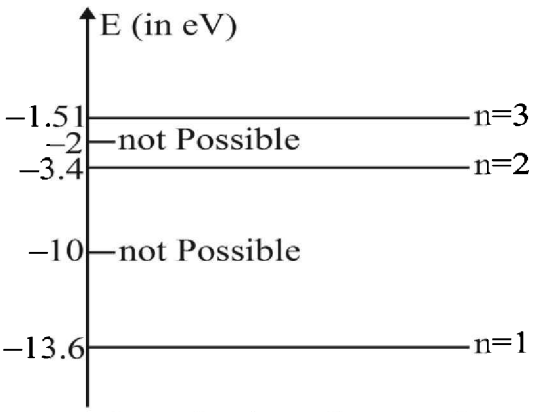
Energy Diagram of Hydrogen atom



Quiz-3

Can an electron possess energy of –10 eV in hydrogen atom?

Sol. As per Bohr's theory, lowest possible energy state of an electron in a hydrogen atom is –13.6eV. After that, next higher allowable energy states for an electron are –3.4 eV, -1.51 eV, –0.85 eV and so on. It means that an electron in a hydrogen atom can be in any of these energy states, but it can never be in between these states. Thus an electron in hydrogen atom can't have energy of – 10 eV An electron (in a hydrogen atom)can have only some particular amounts of energies (i.e. –13.6eV –3.4 eV, –1.51 eV etc.). We can conclude that the energy of an electron in a hydrogen atom is quantized.



Quantization of energy in Hydrogen atom

Let's discuss some important points on the basis of the above energy diagram.

- Here we can see that energy of higher orbits (large r_n) have a smaller numerical value

  i.e. $|E_1| > |E_2| > |E_3|$

But, as all the energies are negative, so energy increases as we move to outer (higher) orbits.

i.e. $E_1 < E_2 < E_3$

Hence the orbit which is closest to the nucleus has lower energy and is called **the ground state**. The higher states E_2, E.....E_n are called excited states.

- From the above energy diagram, we can calculate the value of energy gap between any two successive orbits. Like-

$E_2 - E_1 = 10.2$ eV,

$E_3 - E_2 = 1.80$ eV

$E_4 - E_3 = 0{\cdot}66$ eV and so on

So, we can see that as we move towards outer orbits, energy gap between two successive orbits decreases.

- Notice that energy of electron tends to zero as we move to outermost orbit in Bohr's atom. In other words, we can say that energy of electron is zero when it is infinitely far from the nucleus. At this point, electrostatic force on the electron (due to the nucleus) becomes very weak and electron can be considered free.

Now we will see some important terms related to energy of the electron.

(i) Binding/ Ionisation Energy - We know that the total energy of an electron in Bohr's atom has a negative value. We have seen earlier that an electron becomes free when its energy increases to zero. So, in order to free an electron from an atom, energy must be given to it.

For example, in order to free an electron from the ground state of hydrogen atom, 13.6eV energy must be given to it. Similarly, to remove an electron from the second orbit (n = 2), 3.4 eV energy is needed.

This minimum energy required to free an electron is called **'Binding energy.'** For example, binding energy in the first orbit of the H atom is 13.6 eV and in second orbit is 3.4 eV.

(ii) Excitation Energy –The Excitation energy of an orbit is defined as 'energy required to transit an electron from ground state to that orbit'.

Illustration 2

An electron in H atom needs to transit from ground state (n = 1) to

(i) Second state (n = 2)

(ii) Third state (n=3)

Calculate the excitation energy required in this process.

Sol. (i) In order to transit an electron from ground to second state, the energy of electron must be increased from –13.6 eV to –3.4 eV. So, required energy for this transition or excitation energy of second orbit is–

$$\Delta E = E_2 - E_1 = (-3.4) - (-13.6) = 10.2 \text{ eV}$$

(ii) Similarly, excitation energy of third orbit -

$$\Delta E = E_3 - E_1 = (-1.51) - (-13.6) = 12.09 \text{ eV}$$



Quiz-4

Energy of electron in H atom is negative. What is the significance of energy being negative?

Sol. Negative energy represents that the electron is bound to the nucleus and it cannot escape. If the energy of electron becomes zero or positive, it will become free.

Illustration 3

An electron is revolving in second orbit of a hydrogen atom. Find -

(i) Kinetic energy of electron

Sol. As we know that –

$KE = -E_{Total}$

$\Rightarrow KE = -E_2$ (For IInd orbit)

$\Rightarrow KE = 3.4\text{eV}$

(ii) Potential energy of electron

Sol. We know that –

$PE = -2\,(KE)$

$\Rightarrow PE = -6.8 \text{ eV}$

Illustration 4

Hypothetically, mass of an electron is doubled in a hydrogen atom. How does the following change?

(i) Radius

Sol. We have seen earlier that radius of an orbit is given as -

$$r = \left(\frac{h^2}{4\pi^2 m_e KZe^2}\right)\frac{n^2}{Z}$$

Here we can see that radius is inversely proportional to the mass of the electron.

i.e. $\mathbf{r \propto \frac{1}{m_e}}$

As we double the mass of the electron, radius becomes half.

(ii) Velocity

Sol. We have seen earlier that velocity of an electron is given as -

$$v_n = \frac{nh}{2\pi m_e r_n}$$

From above equation we can see -

$$v_n \alpha \frac{1}{m_e r_n}$$

When mass is doubled, radius becomes half. This keeps velocity unchanged.

(iii) Energy

Sol. We have seen earlier that energy of an electron is given as –

$$E_T = -\frac{KZe^2}{2r_n}$$

From above equation we can see –

$$E_T \alpha \frac{1}{r_n}$$

As mass is doubled, radius becomes half. Thus energy doubles.

2.8 Third Postulate of Bohr

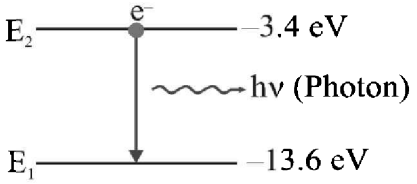
Earlier, we saw that Rutherford's model fails to explain the phenomenon of emission of light by gases. To explain this phenomenon, Bohr incorporated Einstein's and Plank's theory of photon in his model. Now we will understand this in detail –

Suppose, an electron in a hydrogen atom jumps from IInd orbit to Ist orbit. In this transition, energy of electron changes from –3.4 eV to –13.6 eV.

i.e. $\Delta E = E_2 - E_1$

$= 10.2$ eV

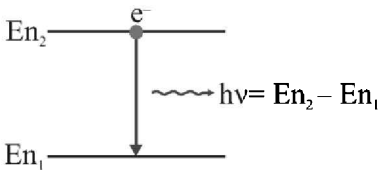
Now, the question is where has this lost energy gone? According to Bohr, this lost energy is converted into a photon of light, as shown in the figure below.



Here $h\nu = E_2 - E_1$

$= 10.2$ eV

We can state the third postulate of Bohr as **"If an electron jumps from a higher energy state (n_2) to a lower energy state (n_1), it emits a photon having energy equal to the energy gap between these states".**



So, $$h\nu = 13.6\left(Z^2\right)\left[\frac{1}{(n_1)^2} - \frac{1}{(n_2)^2}\right]$$

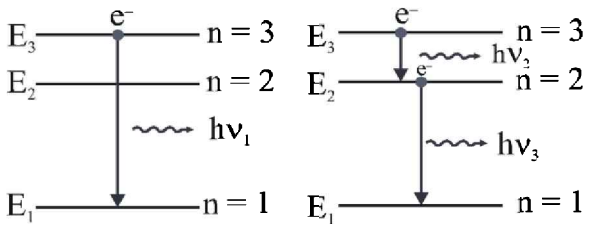
Illustration 5

An electron in hydrogen atom jumps from third orbit to ground state, find-

(i) How many frequencies of photons can be emitted?

Sol. Here, electron can complete the transition from n = 3 to n = 1 in two possible ways as shown in

the figure below.



So, considering these two cases, we can say that three different frequencies of photons can be emitted.

(ii) Find frequencies of emitted photons?

Sol. Here we can see that frequency of emitted photon, for case (a), will be -

$$h\nu_1 = E_3 - E_1$$

$$\Rightarrow \quad h\nu_1 = \frac{13.6}{Z^2}\left[1 - \frac{1}{(3)^2}\right] \quad \text{(here Z = 1)}$$

$$\Rightarrow \quad \nu_1 = \frac{13.6}{h}\left(\frac{8}{9}\right) \qquad \ldots (1)$$

Now, in case (b), two photons are emitted. Here frequencies of emitted photons will be-

$$h\nu_2 = E_3 - E_2$$

$$\Rightarrow \quad \nu_2 = \frac{13.6}{h}\left[\frac{1}{(2)^2} - \frac{1}{(3)^2}\right] \qquad \ldots (2)$$

And $$h\nu_3 = E_2 - E_1$$

$$\Rightarrow \quad \nu_3 = \frac{13.6}{h}\left[1 - \frac{1}{(2)^2}\right] \qquad \ldots (3)$$

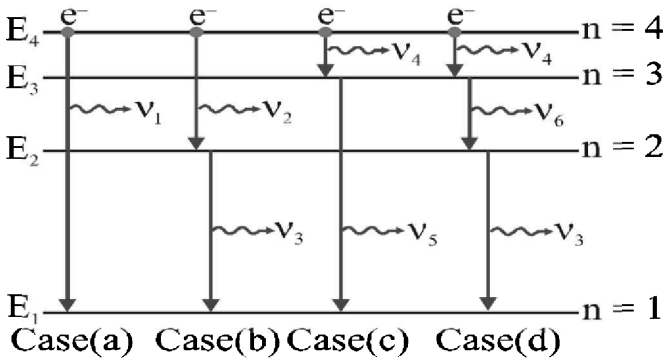
We can calculate frequencies of emitted photons from equations (1), (2) and (3).



Quiz-5

An electron in hydrogen atom jumps from fourth orbit to ground state. Find out, how many frequencies of photons can be emitted?

Sol. Here, electron can complete the transition in four possible ways as shown below.



From the above figure, we can see that six different frequencies of photon can be emitted.

Key points –

Bohr's three postulates can be summarized as–

1. An electron in an atom could revolve in certain stable orbits without the emission of radiant energy.
2. Electron revolves around the nucleus only in those orbits for which the angular momentum is an integral multiple of $h/2\pi$.
3. An electron might make a transition from one of its higher orbits to lower orbits. When it does so, a photon is emitted having energy equal to the energy difference between the initial and final states. The frequency of the emitted photon is then given by $h\nu = \Delta E$.

2.9 Explanation of Radiation of Light by Gases

Now, we will see how Bohr's theory explains the phenomena of radiation of coloured light by gases.
Generally, an electron stays in the lowest energy state (ground state) in an atom. But when the gas is heated, electron gets additional amount of energy and it jumps to higher energy states (excited states). Electron can jump to any excited state depending on the amount of energy absorbed by it. As electron has tendency to remain in ground state so it again jumps from its excited state to ground state by emitting photon(s) of some particular frequency (frequencies). If the frequencies of emitted photons fall under visible range then, we see colour light.
Further, Bohr's theory also explains why different gases, on heating, glows with different colour light. According to Bohr's theory, the frequency of emitted photon by an atom depends on the energy gap between its energy states.

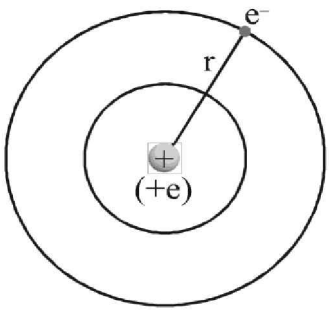
i.e. $$\nu = \frac{\Delta E}{h}$$

As different gases have different energy gaps so frequencies of emitted photons are different for each gas. That's why, different gases glow with different colour light on heating.
This explanation was the biggest success of Bohr's model as other theories around Bohr's time were not able to explain the phenomena of radiation by gases. As a result, Bohr's theory was widely accepted and Bohr was awarded Nobel Prize for this work in 1922.

Illustration 6

In a hypothetical situation, if the potential of H atom is given as $'V = 2\log\frac{r}{5}'$ then find out the radius of stationary orbits in which electron can move?

Sol. First, we will understand the problem. Say, an electron is revolving in an hydrogen atom as shown in the figure below.



H atom

From knowledge of electrostatics we know that, potential due to charge on nucleus is given as

$$V = \frac{Kq}{r} = \frac{Ke}{r}$$

This is the actual potential in hydrogen atom and for this potential, radius of orbits is given as-

$$r_n = 0.529 \times \frac{n^2}{Z}\ A°$$

But here (in above problem), a hypothetical potential is given and we have to find the radius of stationary orbits for this imaginary potential. Here potential is given as-

$$V = 2\ \log \frac{r}{5}$$

So, potential energy of electron will be-

$$U = qV = -2e\ \log \frac{r}{5}$$

Here attraction force between hydrogen nucleus and electron will be –

$$F = -\frac{dU}{dr}$$

$$\Rightarrow \qquad F = \frac{2e}{5r}$$

This force of attraction between nucleus and electron provides the necessary centripetal force to electron so that it can revolve in its orbit.

i.e. $$\frac{mv^2}{r} = \frac{2e}{5r}$$

$$\Rightarrow \qquad v = \sqrt{\frac{2e}{5m}}$$

Now, according to Bohr –

$$mvr = \frac{nh}{2\pi}$$

$$\Rightarrow \qquad r = \frac{nh}{2\pi mv}$$

On substituting value of 'v' in above equation, we get -

$$r = n\left(\frac{\sqrt{5}\,h}{2\sqrt{2}\,\pi\sqrt{m}\,\sqrt{e}}\right)$$

$\Rightarrow$ r = n (constant)

2.10 Emission Spectra of Hydrogen

The Emission spectra of an atom is defined as the set of electromagnetic wavelengths (or frequencies) emitted by an atom.

When an electron in a hydrogen atom jumps from any higher energy state to ground state (n = 1), photons of following frequencies are emitted-

$$\nu = \frac{13.6}{h}\left[\frac{1}{(1)^2} - \frac{1}{(n)^2}\right] \quad \text{.. (1)}$$

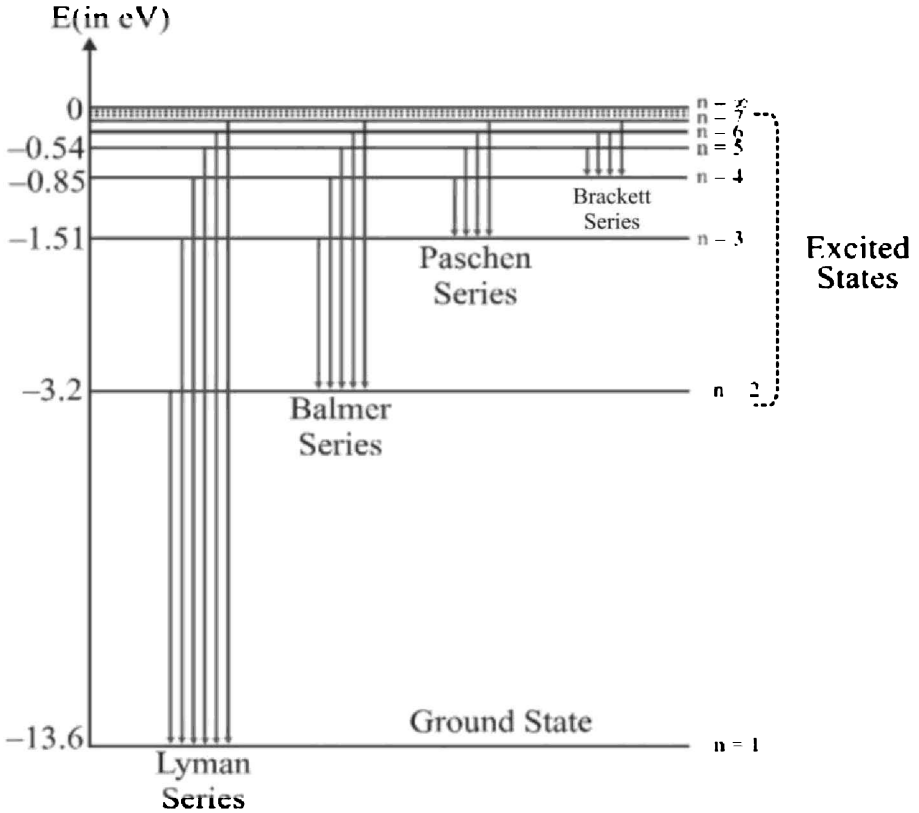
Here n = 1, 2, 3.....

For example, say if electron jumps from the second orbit (n = 2) to ground state then the frequency of emitted photon will be -

$$\nu = \frac{13.6}{h}\left[\frac{1}{(1)^2} - \frac{1}{(2)^2}\right]$$

Similarly, we can find the frequencies of emitted photons for other transitions too (i.e. From n = 3, 4.. to n =1). All these frequencies, given by equation (1), are together known as Lyman series.

In the same way, when electron jumps from any higher energy state to n = 2, n = 3 and n = 4, a unique set of frequencies are emitted by the hydrogen atom which are known as Balmer, Paschen and Brackett series respectively as shown in the figure below.



Here we can formulate frequencies of Balmer series as –

$$\nu = \frac{13.6}{h}\left[\frac{1}{2^2} - \frac{1}{n^2}\right]; \quad n = 3, 4, 5...$$

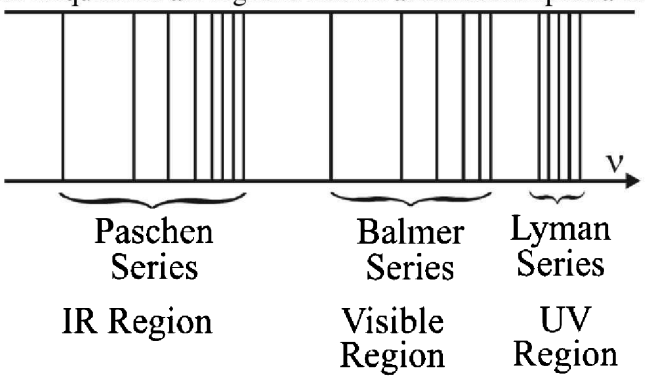
For Paschen series -

$$\nu = \frac{13.6}{h}\left[\frac{1}{3^2} - \frac{1}{n^2}\right]; \quad n = 4, 5, 6...$$

For Brackett series -

$$\nu = \frac{13.6}{h}\left[\frac{1}{4^2} - \frac{1}{n^2}\right]; \quad n = 5, 6, 7...$$

Here, Lyman and Balmer series lies in the ultraviolet and visible region respectively. While Paschen series and onwards lies in the infrared region. All these series (other series are also possible for transitions from higher states to n = 5, 6...) represents the possible frequencies that can be emitted by hydrogen atom and these frequencies are together known as Emission spectra of hydrogen.



Here we can see that only some discrete values of frequencies are emitted. The above graph is also known as **line spectra of hydrogen atom**.

Illustration 7

Find the minimum wavelength that can be emitted by hydrogen atom.

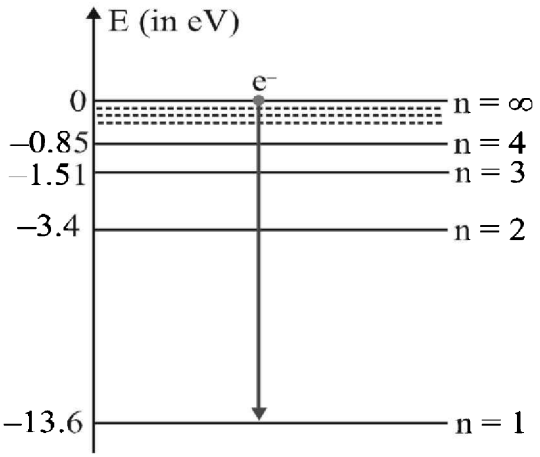
Sol. We know that energy of emitted photons is given as -

$$h\nu = \Delta E$$

$$\Rightarrow \quad \frac{hc}{\lambda} = \Delta E$$

$$\Rightarrow \quad \lambda = \frac{hc}{\Delta E} \qquad \text{.... (1)}$$

From this equation, we can see that value of 'λ' will be minimum for the maximum energy gap (ΔE) between the states. Now, energy gap will be maximum, if electron jumps from very high energy state $(n \to \infty)$ to ground state (n = 1), as shown in the figure below:-



So $(\Delta E)_{max} = 13.6\left[\frac{1}{(1)^2} - \frac{1}{(\infty)^2}\right]$

$\Rightarrow$ $(\Delta E)_{max} = 13.6$ eV

So, from equation (1) -

$$\lambda_{min} = \frac{hc}{(\Delta E)_{max}}$$

$$\Rightarrow \quad \boldsymbol{\lambda_{min} = \frac{hc}{13.6\,(in\,eV)}}$$

Above equation shows the minimum wavelength that can be emitted by a hydrogen atom. It means that wavelength of all the other emitted photons will be greater than 'λ_{min}'

i.e. $\lambda \geq \lambda_{min}$

Key Points -

- Reciprocal of λ_{min} is called Rydberg constant (R).

i.e. $$\boldsymbol{R = \frac{1}{\lambda_{min}}}$$

$\Rightarrow$ $$R = \frac{13.6}{hc}$$

$\Rightarrow$ $R = 1{\cdot}09737 \times 10^7\ m^{-1}$

- We can write energy levels of an atom in terms of Rydberg constant (R) as follows-

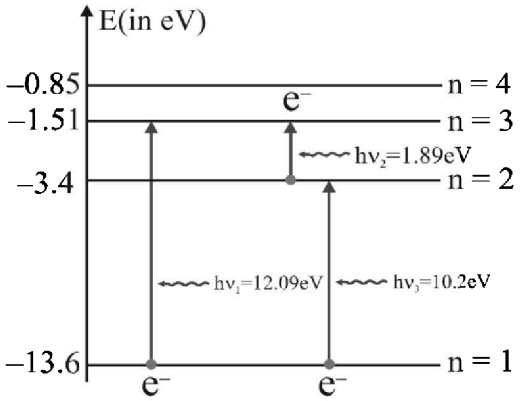
$$E = -13.6 \times \frac{Z^2}{n^2}\ eV$$

$\Rightarrow$ $$\boldsymbol{E = -(Rhc)\frac{Z^2}{n^2}\ eV}$$

2.11 Absorption Spectra of Hydrogen Atom

Earlier, we saw that when an electron jumps from a higher energy state to a lower state, it emits photons of some particular frequency.

Similarly, an electron can absorb photon which can take the electron from its existing state to a higher energy state. For example, an electron can move from ground state to n=3, by absorbing some particular photons (i.e. either 12.09 eV or 10.2eV and 1.89eV) as shown in the figure below.





Quiz-6

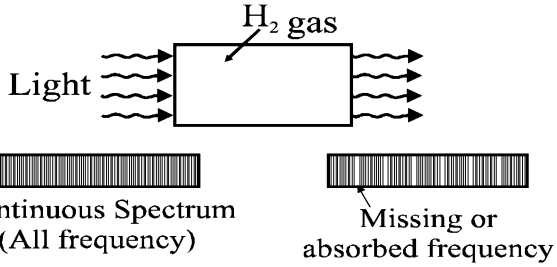
Can an electron in the ground state in hydrogen atom absorb a photon of 11eV energy?

Sol. If the electron absorbs the photon of 11 eV energy, the energy of electron increases to -

$$E = -13.6 + 11 = -2.6 \text{ eV}$$

This energy of electrons does not correspond to any orbit of the hydrogen atom. If the electron absorbs 11eV, it can go in no orbit. So 11eV cannot be absorbed.

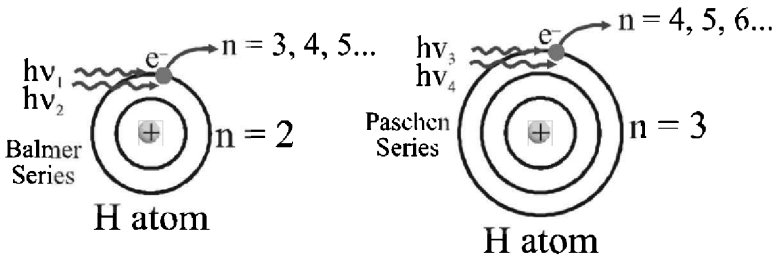
An electron in hydrogen atom can absorb only certain frequencies of photon. That's why, when a continuous spectrum of light (containing all frequencies) passes through the hydrogen gas, only some particular frequencies are absorbed by it as shown in the figure below. Rest of the frequencies come out from the other side.



All these missing or absorbed frequencies are together known as ***absorption spectra of hydrogen.***

It is observed that, at low temperature, only the frequencies of Lyman series are absorbed by hydrogen atom but not the others. At low temperature, electrons exist in the ground state (n = 1) of hydrogen atom. So, electrons absorb only those frequencies which can take them from ground state to higher states (i.e. n = 2, 3 ....) That's why, absorption spectra of hydrogen contains the frequencies of the Lyman series at low temperature.

At high temperature, absorption spectra contain the frequencies of Balmer, Paschen and other higher series. This happens because, at high temperature, electrons exist in different excited states (n=2, 3, 4...) of hydrogen atoms. So, they absorb the photons which can transit it from these excited states to higher states, as shown in the figure below:-



From the above figures, you can clearly see that Balmer and higher series will be absorbed at higher temperatures.

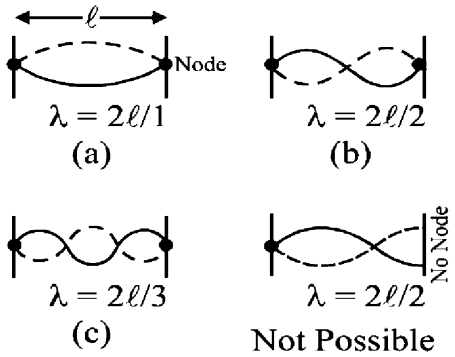
2.12 De- Broglie's Explanation of Bohr's IInd Postulate

Earlier, we have seen that Bohr hypothesized about the quantization of angular momentum of electron in orbits.

i.e. $mv_n r_n = \frac{nh}{2\pi}$; $n = 1, 2, 3...$

This assumption was accepted as many experimental results could be explained on the basis of this. But Bohr could not explain the logic behind the assumption. Later in 1923, De Broglie proposed an explanation. His explanation was based on the theory of wave-particle duality. Now, to understand the De Broglie's explanation, recall the concept of standing waves.

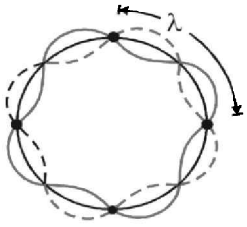
Suppose, a string of length 'ℓ' is tied between two fixed walls. Only some particular standing waves can exist on the string as shown below:



Only those wavelengths of standing waves are possible which have wavelength as -

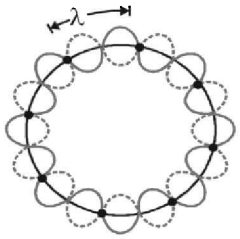
$$\lambda = \frac{2\ell}{n} \; ; \quad n \in I \; ;$$

De Broglie said that the same concept of standing waves can be applied to electrons considering electrons as waves. In case of string, a standing wave is bound in length 'ℓ' where as in case of an electron, a standing wave is bound in a stationary orbit of radius '$\mathbf{r_n}$'. Below figures show some patterns of electron wave in an orbit.



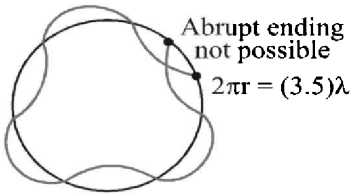
$2\pi r = 4\lambda$

(a)



$2\pi r = 9\lambda$

(b)



(c)

Here, only certain wavelengths for which the wave closes smoothly in the orbit can exist. Wavelength in case (a) and (b) is possible, but wavelength in case (c) is not possible as it doesn't end smoothly. From the above figure, we can understand that a wave will close by itself if it contains an integral number of wavelengths.

Like, in case (a) $2\pi r = (4)\,\lambda$

In case (b) $2\pi r = (9)\,\lambda$

So, in generalized form, we can write wavelength of standing waves (for an electron) as-

$$\mathbf{2\pi r_n = n\lambda}\ ; \quad n = 1, 2, 3...$$

$$\Rightarrow \qquad \lambda = \frac{2\pi r_n}{n}$$

Now, according to De Broglie, wavelength of electron can be written as –

$$\lambda = \frac{h}{P}$$

$$\Rightarrow \qquad \lambda = \frac{2\pi r_n}{n} = \frac{h}{mv_n}$$

$$\Rightarrow \qquad mv_n r_n = \frac{n\,h}{2\pi}$$

Thus, De Broglie proved the Bohr's hypothesis by considering electrons as a wave.

2.13 Limitations of Bohr Model

Although Bohr's theory solved the major limitations of Rutherford's model, but it did not explain a lot of things accurately. Let's see them–

(i) Bohr's theory is only applicable for hydrogen like atoms (single electron atoms). It cannot explain the structure of atoms with more than one electron.

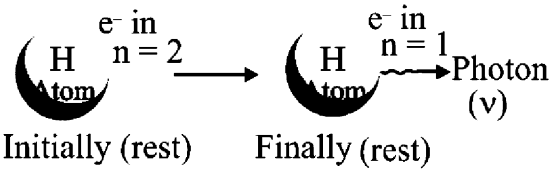
(ii) It was observed in the emission spectra of hydrogen that certain frequencies of photons are emitted more frequently than others. Although Bohr's theory predicts about the emission of photons of different frequencies, but it does not explain why certain frequencies of photons are favoured by the hydrogen atom.

(iii) Bohr's theory gives no reason why Maxwell's theory is not valid inside an atom.

So these were some limitations of Bohr's model. Bohr himself admitted that his theory was not perfect. Later in 1920, Bohr, along with Schrodinger, proposed a new theory which gives a different view of the atom. This theory is studied under different branch of physics called **'Quantum mechanics'** which is beyond the scope of this book.

2.14 Momentum Conservation in Photo Emission

Say, an electron in a hydrogen atom jumps from second state (n = 2) to ground state, as shown in the figure below.

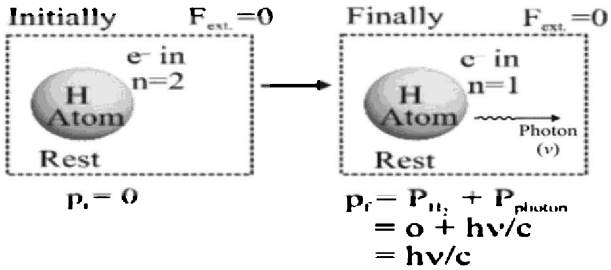


Here frequency of emitted photon will be -

$$h\nu = E_2 - E_1$$

$$\Rightarrow \quad \nu = \frac{10.2(\text{in eV})}{h} \qquad \ldots (1)$$

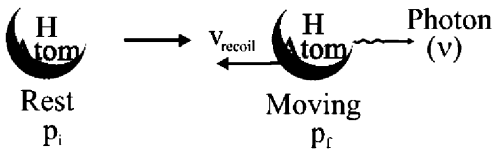
Till now, we have been considering the above solution as correct, but actually the above equation gives only the approximate value of frequency of the emitted photon. This is because the above solution violates ***the law of conservation of momentum.***



If no external force is acting on a system, momentum of the system remains conserved. Here, $P_i \neq P_F$ (violation of momentum conservation law)

So, $$\nu \neq \frac{10.2(\text{in eV})}{h}$$

To find the accurate value of frequency of emitted photons, we must consider the law of conservation of momentum. Actually, when an atom emits a photon, it recoils in backward direction, same as the gun recoils on firing.



Now, by momentum conservation law -

$$P_i = P_f$$

$$\Rightarrow \quad 0 = m(v_{recoil}) - \frac{h\nu}{c}$$

$$\Rightarrow \quad v_{recoil} = \frac{h\nu}{mc} \qquad \ldots (2)$$

By energy conservation law -

$$\Delta E = \frac{1}{2} m \left(v_{recoil}\right)^2 + h\nu \qquad \ldots (3)$$

Here ΔE is the energy released in electron transition. Equation (3) shows that some amount of this released energy goes to the atom which appears as its kinetic energy while remaining energy goes to photon.

From equation (2) and (3), we can calculate the actual value of frequency of the emitted photon.

$$\Delta E = h\nu + \frac{1}{2} m_H \frac{h^2\nu^2}{c^2 m_H{}^2}$$

$$\Rightarrow \quad h\nu_0 = h\nu + \frac{1}{2}\frac{h^2\nu^2}{c^2 m_H}$$

$$\Rightarrow \quad \left(\nu_0 - \nu\right) = \frac{1}{2}\frac{h^2\nu^2}{c^2 m_H}$$

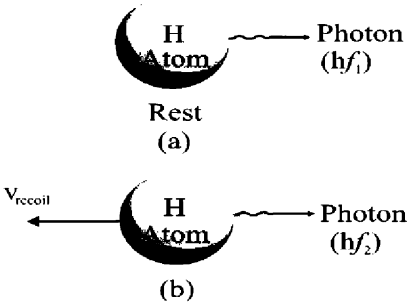
As $\nu_0 \cong \nu$

$$\therefore \quad \frac{\nu_0 - \nu}{\nu_0} = \frac{1}{2}\frac{h\nu_0}{c^2 m_H}$$

$$\Rightarrow \quad \frac{\Delta\nu}{\nu_0} = \frac{1}{2}\frac{h\nu_0}{c^2 m_H}$$

Illustration 8

Below figure shows two frequencies (i.e. f_1 and f_2) of emitted photons for the transition of electron from n = 2 to n = 1 in hydrogen atom. 'f_1' and 'f_2' are the calculated frequencies for the cases when momentum conservation is not considered and when it is considered respectively. Check the right option.



(i) $f_1 > f_2$ (ii) $f_1 < f_2$ (iii) $f_1 = f_2$

Sol. Here we can see that energy of photon in case (a) will be given as -

$$hf_1 = E_2 - E_1$$

$$\Rightarrow \quad hf_1 = \Delta E \qquad \ldots (1)$$

While for case (b), by energy conservation theorem-

$$\Delta E = \frac{1}{2} m \left(V_{recoil} \right)^2 + h f_2$$

So, from above equation, we can write that –

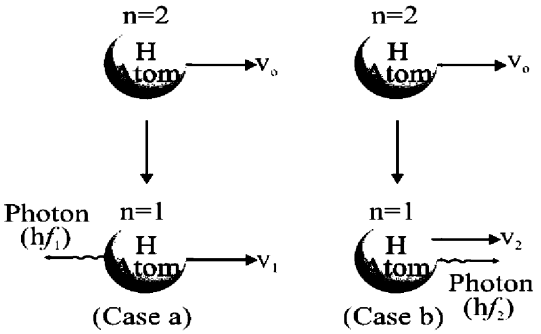
$$h f_2 < \Delta E \qquad \text{... (2)}$$

From equation (1) and (2), we can clearly see that

$$f_2 < f_1$$

Illustration 9

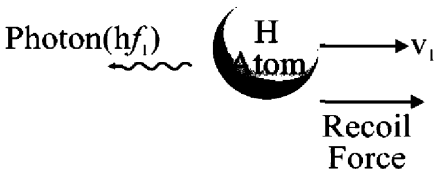
Two identical hydrogen atoms are moving in the same direction with velocity 'v_0'. When an electron transit from n = 2 to n = 1, their velocity changes to 'v_1' and 'v_2' as shown in the figure below.



(i) Compare the velocities of hydrogen atoms in both the cases?

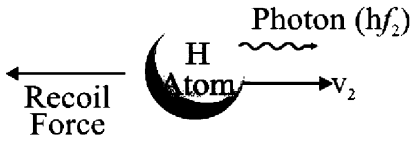
Sol. Here we can see that both the atoms are moving towards the right direction. In case (a), recoil force on atom will act towards the right direction as a photon is emitted towards the left direction. Here recoil force supports the motion of atom so its velocity will increase

i.e. $v_1 > v_0$ ... (1)



While, in case (b), recoil force on atom will act in left direction as a photon is emitted towards the right direction. Here recoil force opposes the motion of atom so its velocity will decrease.

i.e., $v_2 < v_0$ ... (2)



So from equation (1) and (2), we get -

$$v_1 > v_2$$

(ii) Compare the frequencies of emitted photons in both the cases.

Sol. In both the cases, when an electron jumps from n=2 to n=1, it releases some amount of energy- i.e. $\Delta E = E_2 - E_1$

In case (a), some of this released energy is used to increase the K.E. of the atom.

$$h f_1 = \Delta E - |\Delta K.E.| \qquad \text{... (1)}$$

In case (b), K.E. of atom decreases. This decreased energy also goes to photon.

$$h f_2 = \Delta E + |\Delta K.E.| \qquad \text{... (2)}$$

From equation (1) and (2), we can clearly see that -

$hf_1 < hf_2$

$\Rightarrow f_1 < f_2$

So, frequency of emitted photon will be more in case (b).

2.15 Motion of Nucleus-

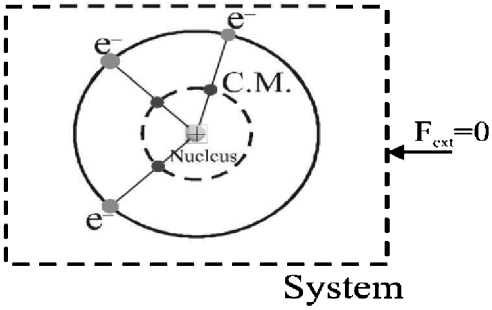
Till now we have been discussing about the atomic structure considering nucleus at rest and electrons orbiting around the nucleus. But in reality, the situation is a bit different. Nucleus inside an atom is not stationary. Let's understand, why?

We know that, if no external force acts on a system, then the acceleration of the centre of mass of the system is zero.

I.e. If $\mathbf{F_{ext} = 0}$

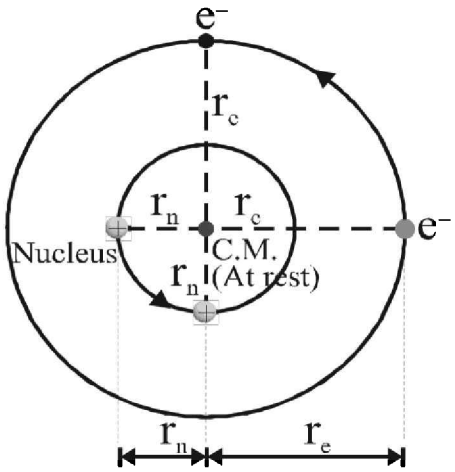
Then $\mathbf{a_{CM} = 0}$

But if we consider nucleus at rest, then the above law is violated, as shown in the figure below:



From the above figure, we can see that as the electron moves around the nucleus, centre of mass of nucleus-electron system moves in a circle. So $a_{CM} \neq 0$ although $F_{ext} = 0$.

That's why the situation that nucleus remains at rest is not possible. The actual situation inside an atom is that centre of mass remains at rest. Both nucleus and electron revolve around the centre of mass as shown in the figure below:

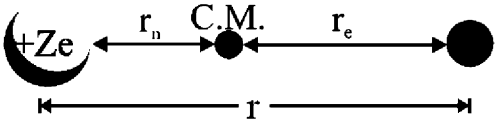


As a nucleus is much heavier than electrons, so centre of mass lies very close to the nucleus (almost situated in the nucleus). In other words, we can say that nucleus moves in a circle of very small radius compared to electron. That's why, we generally neglect the motion of nucleus and consider it at rest. But to find more accurate results, we need to consider the motion of the nucleus as well.

Illustration 10

Find out value of 'r_n' and 'r_e' if radius of atom is 'r'.

Sol. Radius of atom(r) is defined as the distance between nucleus and electron.



So, $r = r_n + r_e$ ... (1)

Now, by the concept of centre of mass,

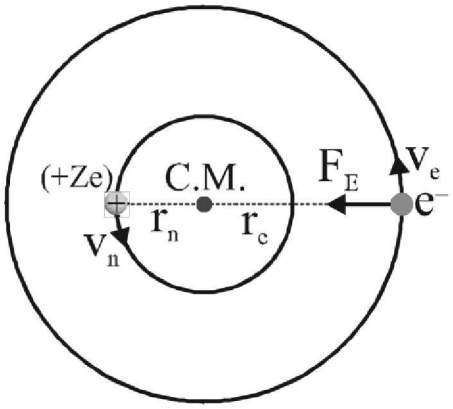
$m_n r_n = m_e r_e$ ...(2)

From equation (1) and (2), we get -

$$r_e = \frac{m_n r}{m_e + m_n} \text{ and } r_n = \frac{m_e r}{m_e + m_n}$$

$$E_n = -13.6 \times \frac{z^2}{n^2} eV$$

Here we have to find the same while taking the motion of nucleus into account. To find this, we will go through the same process which we followed earlier.



Now here –

$$\begin{pmatrix}\text{Attraction force}\\ \text{between nucleus}\\ \text{and electron } (F_E)\end{pmatrix} = m_e \begin{pmatrix}\text{Centripetel}\\ \text{acceleration}\\ \text{of electron}\end{pmatrix}$$

$$\Rightarrow \quad \frac{K(Z_e)(e)}{(r_n + r_e)^2} = m_e\left(\frac{V_e^2}{r_e}\right) \quad \text{... (3)}$$

Here nucleus is also revolving around the centre of mass. For nucleus:

$$\frac{K(Z_e)(e)}{(r_n + r_e)^2} = \frac{m_n V_n^2}{r_n} \quad \text{... (4)}$$

According to Bohr, total angular momentum in any orbit of an atom is given as:

$$L_T = \frac{nh}{2\pi}$$

$$\Rightarrow \quad L_e + L_n = \frac{n\,h}{2\pi}$$

$$\Rightarrow \quad m_e v_e r_e + m_n v_n r_n = \frac{nh}{2\pi} \qquad ...(5)$$

On solving equation (3), (4) and (5), we get –

$$\mathbf{E_n = -13{\cdot}6\frac{z^2}{n^2}\left(\frac{m_n}{m_n + m_e}\right)} \qquad ...(6)$$

Above equation shows the energy of an electron in n^{th} orbit of Bohr atom when we take motion of nucleus into account. We can write equation (6) as:-

$$\mathbf{E_n = -13{\cdot}6\times\frac{Z^2}{n^2}\left(\frac{1}{1+\frac{m_e}{m_n}}\right)}$$

If we take $m_e << m_n$; $\frac{m_e}{m_n} \cong 0$ and $E_n = -13.6\times\frac{z^2}{n^2}$

So we can see that with the approximation of $m_e << m_n$, we get our old equation for energy.

Illustration 11

Electron in hydrogen $\left({}_1^1H\right)$ and Deuterium $\left({}_1^2H\right)$ atom jumps from the second state to ground state by emitting photons of frequencies 'f_H' and 'f_D' respectively. Check the correct option considering motion of nucleus.

(A) $f_H = f_D$ (B) $f_H > f_D$ (C) $f_H < f_D$

Sol. Here energy of photon will be given as -

$hf = \Delta E$

$$\Rightarrow \quad hf = -13{\cdot}6\,Z^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right]\left[\frac{1}{\frac{m_e}{m_n}+1}\right]$$

$$\Rightarrow \quad hf = -10{\cdot}2\,Z^2\left(\frac{1}{1+\frac{m_e}{m_n}}\right)$$

Now, atomic number (Z) and 'm_e' is same for hydrogen and deuterium. So, from above equation, we can see that frequency of emitted photon will depend on 'm_n'.

As, $(m_n)_{hydrogen} < (m_n)_{Deuterium}$

So, $hf_H < hf_D$

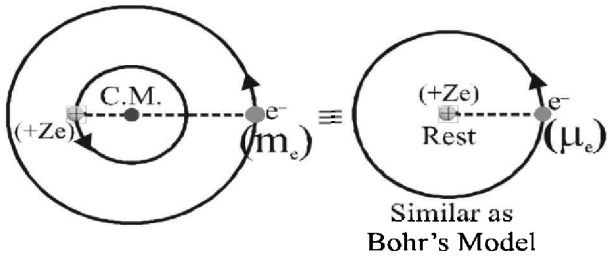
$\Rightarrow \quad f_H < f_D$

Option C is right.

2.16 Equivalent Model of Atom

It is very complicated to analyse mechanics of atom when we consider the motion of nucleus. So, sometimes we use an equivalent model of atom. In this equivalent model, we assume nucleus to be at

rest. But here, we replace the mass of the electron by its reduced mass 'μ_e' as shown in the figure below:



$$\text{Here } \mu_e = \frac{m_n m_e}{m_n m_e}$$

Now, for this equivalent model, we can use all those results that we derived considering nucleus at rest. We just need to replace the mass of the electron (m_e) by its reduced mass (μ_e). For example, radius of nth stationary orbit of Bohr's model is given as -

$$r_n = \frac{n^2h^2}{4\pi^2 m_e KZe^2}$$

But here, if we consider motion of nucleus also, then radius of nth orbit will be given as -

$$r_n = \frac{n^2h^2}{4\pi^2 \mu_e KZe^2}$$

Similarly, we can also find the velocity and energy of electrons in n^{th} orbit of Bohr model.

Illustration 12

Compare the radius of first orbit for hydrogen and deuterium considering the motion of nucleus.

Sol. We have seen that the radius of nth orbit considering the motion of the nucleus is given as:-

$$r_n = \frac{n^2h^2}{4\pi^2 \mu_e KZe^2}$$

$$r_n \alpha \frac{1}{\mu_e}$$

$$\Rightarrow \quad r_n \alpha \frac{m_n + m_e}{m_n m_e}$$

$$\Rightarrow \quad r_n \alpha \frac{1 + \frac{m_e}{m_n}}{m_e} \qquad \ldots (1)$$

As, $(m_n)_{\text{hydrogen}} < (m_n)_{\text{Deuterium}}$

$(r_n)_{\text{hydrogen}} > (r_n)_{\text{Deuterium}}$

2.17 Atomic Collision

In this section, we will study about the collision of atom with another atom or particles like electron, neutron etc. But to understand about atomic collision, we need to recall some basic concepts of collision that we studied in mechanics. Say, two balls collide with each other, as shown in the figure below ($v_1 > v_2$).

- In a collision between these two balls, momentum of the system remains conserved.



- The loss of kinetic energy in collision depends on the coefficient of restitution (e). Maximum kinetic energy than can be lost in a collision is given as -

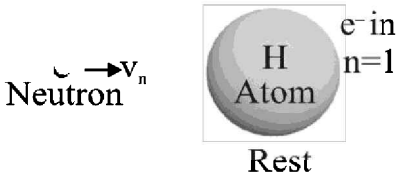
$$\Delta KE_{max} = \frac{1}{2}\mu (V_{rel})^2$$

Here μ (reduced mass) $= \dfrac{m_1 m_2}{m_1 + m_2}$

In normal collisions, this lost kinetic energy is converted into some other form of energy like heat energy, sound energy etc. But, in case of atomic collision, this lost kinetic energy is used by electrons to jump to a higher energy state. The following problems will further clarify the concept of atomic collisions.

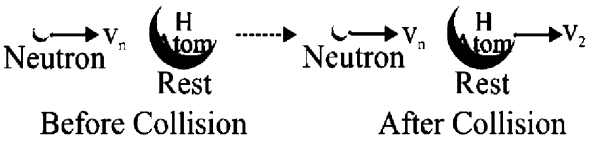
Illustration 13

A neutron collides with a hydrogen atom which is at rest. An electron is revolving in the ground state of the hydrogen atom-



(i) Find out the velocity of the neutron and hydrogen atom after collision if the initial kinetic energy of the neutron is 10 eV.

Sol. Say, velocity of the neutron and hydrogen atom becomes v_1 and v_2 after collision. So, by law of conservation of momentum-



$$P_i = P_f$$

$\Rightarrow m_n v_n + m_H (0) = m_n v_1 + m_H v_2$

As, the hydrogen atom contains only one proton and one electron, so $\mathbf{m_n \cong m_p \cong m_H}$

$\Rightarrow v_n = v_1 + v_2$ ...(1)

Maximum KE that can be lost in the collision:

$$\Delta KE_{max} = \frac{1}{2}\mu (V_{rel})^2$$

$$= \frac{1}{2}\left(\frac{m_n . m_n}{m_n + m_n}\right)(v_n - 0)^2$$

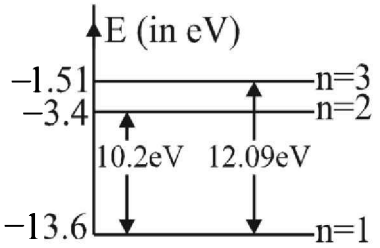
$$= \frac{1}{2}\left[\frac{1}{2} m_n (v_n)^2\right] \qquad (\text{As } m_n \cong m_H)$$

$$= \frac{1}{2}\left[(KE)_n\right] \qquad \text{..(2)}$$

$\Rightarrow \quad \Delta KE_{max} = 5$ eV

Above equation shows that loss in kinetic energy will not be greater than 5 eV.

We have discussed earlier that this lost energy has to be absorbed by electrons in the H atom to jump to a higher energy state. But we know that minimum energy that can excite an electron in the H atom to the nearest energy state (i.e. n = 2) is 10.2 eV, as shown below.



Here maximum lost energy is 5eV only. By absorbing this energy, electron can't transit to a higher energy state, so it will not absorb this energy. Thus, there will be no loss of kinetic energy in the collision. It means that kinetic energy will remain conserved.

i.e. $\mathbf{(KE)_i = (KE)_f}$

$$\Rightarrow \quad 10 \text{ eV} = \frac{1}{2} m_n v_1^2 + \frac{1}{2} m_H v_2^2 \qquad \text{... (3)}$$

From equation (1) and (3), we can find the value of V_1 and V_2.

(ii) Find out the velocity of the neutron and hydrogen after collision. The initial kinetic energy of the neutron is 22 eV.

Sol. In this case, equation of conservation of momentum will be same as in the first part of this problem.

i.e. $V_n = V_1 + V_2 \qquad \text{...(1)}$

Maximum KE that can be lost in a collision, will be -

$$\Delta KE_{max} = \frac{1}{2}\left[(KE)_n\right]$$

$\Rightarrow \Delta KE_{max} = 11$ eV

$\Rightarrow \Delta KE \leq 11$ eV

There are two possibilities here. Either electron will absorb 10.2 eV energy and jump to first excitation state (n=2) or it won't absorb any energy and remain in the ground state.

Let's discuss both the possibilities–

(a) If the electron absorbs 10.2 eV energy during collisions and jumps to first excitation state (n = 2) then remaining energy (22 – 10.2 = 11.8 eV) will appear as kinetic energy of neutron and hydrogen atom. So, by law of conservation of energy–

$$\frac{1}{2} m_n v_n^2 - 10.2\,\text{eV} = \frac{1}{2} m_n v_1^2 + \frac{1}{2} m_H v_2^2$$

$$\Rightarrow \qquad (22 - 10.2) \text{ eV} = \frac{1}{2} m_n v_1^2 + \frac{1}{2} m_H v_2^2$$

$$\Rightarrow \qquad 11.8 \text{ eV} = \frac{1}{2} m_n v_1^2 + \frac{1}{2} m_H v_2^2 \qquad \text{... (2)}$$

(b) If electron does not absorb any energy

$(KE)_i = (KE)_f$

$$\Rightarrow \quad \frac{1}{2}m_n v_n^2 - 0 = \frac{1}{2}m_n v_1^2 + \frac{1}{2}m_H v_2^2$$

$$\Rightarrow \quad 22\text{ eV} = \frac{1}{2}m_n v_1^2 + \frac{1}{2}m_H v_2^2 \qquad \text{... (3)}$$

We can see that on solving equation (1) with equation (2) and (3) separately, we will get the values of v_1 and v_2 for case (a) and case (b) respectively.



Quiz-7

An electron with initial kinetic energy 12.5eV collides with a hydrogen atom at rest. Discuss about the possible states where electrons can exist after collision.



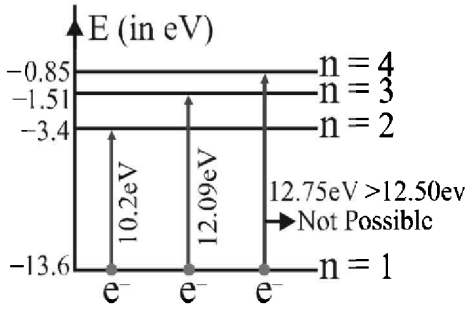
Sol. Here maximum kinetic energy that can be lost in a collision is given as -

$$\Delta KE_{max} = \frac{1}{2}\left(\frac{m_e.m_H}{m_e + m_H}\right)(v_e - 0)^2$$

$$= \frac{1}{2}\left(\frac{m_e}{\frac{m_e}{m_H} + 1}\right)(v_e)^2$$

$$\Rightarrow \quad \Delta KE_{max} = \frac{1}{2}m_e v_e^2 \text{ (as } m_e << m_H)$$

$$\Rightarrow \quad \Delta KE_{max} = (KE)_e = 12{\cdot}5\text{ eV}$$



Here, there are three possible situations–

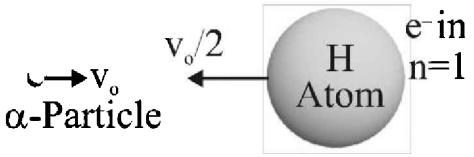
(a) The electron absorbs 10.2 eV energy and jumps to the second energy state.

(b) The electron absorbs 12.09 eV energy and jumps to the third energy state.

(c) The electron absorbs no energy and remains in the ground state.

Illustration14

In the figure shown below, find out the maximum energy of α- particle for excitation of electron from n = 1 to n = 2 after collision.



Sol. The required energy for excitation of electron from n = 1 to n = 2 will be-

$$\Delta E = E_2 - E_1$$
$$= 10.2 \text{ eV}$$

Thus, electron may jump from n = 1 to n = 2 if-

$$\Delta KE_{max} \geq 10.2 \text{ eV}$$

$$\Rightarrow \quad \frac{1}{2}\mu\left(v_{rel}\right)^2 \geq 10.2\text{eV}$$

$$\Rightarrow \quad \frac{1}{2}\left(\frac{m_\alpha . m_H}{m_\alpha + m_H}\right)\left[v_0 - \left(\frac{v_0}{2}\right)\right]^2 \geq 10.2\text{eV} \qquad \text{... (1)}$$

We know that α-particle contains two neutrons and two protons. So -

$$m_H = \frac{m_\alpha}{4}$$

So from equation (1) -

$$\frac{1}{2}\left(\frac{m_\alpha \cdot \frac{m_\alpha}{4}}{m_\alpha + \frac{m_\alpha}{4}}\right)\left(\frac{3v_0}{2}\right)^2 \geq 10.2\text{eV}$$

$$\Rightarrow \quad \frac{1}{2}m_\alpha\left(v_0\right)^2 \geq 22.67\text{eV}$$

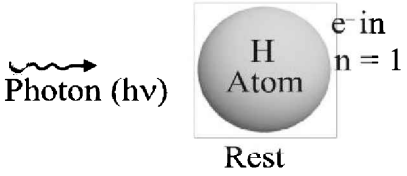
$$\Rightarrow \quad (\text{K.E})_\alpha = 22.67 \text{ eV}$$

$$\Rightarrow \quad (\text{K.E})_{min} = 22.67 \text{ eV}$$

Minimum energy of α- particle must be 22.67 eV in order to excite electron to n = 2 after collision.

Illustration 15

A photon collides with a hydrogen atom as shown in the figure below. Assume atom to be at rest throughout. Discuss the possible outcomes of collision if –



(i) Energy of photon is 10.2 eV.

Sol. We have seen earlier in the photoelectric effect that when a photon collides with an atom, either it is fully absorbed (by an electron) or it is not absorbed at all.If the energy of the photon correspond to the energy gaps between different energy states of H atom,then only it is absorbed by an electron. Here, energy of a photon is 10.2 eV. By absorbing that much energy, electrons can jump to n = 2. So electron can absorb the photon

(ii) Energy of photon is 11 eV.

Sol. By absorbing 11eV energy, electron can't transit to any stationary orbit. So electron will not

absorb the photon.

(iii) Energy of photon is 12.09 eV.

Sol. By absorbing 12.09 eV, electron can transit to n = 3. So, electron will absorb the photon.

(iv) Energy of photon is 13.1 eV.

Sol. Electron will not absorb the photon.

(v) Energy of photon is 13.6 eV.

Sol. In this situation, if the electron absorbs the photon then its energy will increase to zero. It means that the electron will become free i.e. it will come out from hydrogen atoms. We can say that hydrogen atom will get ionized.

(vi) Energy of photon is 15 eV

Sol. Electron will absorb the photon and use 13.6eV of energy to come out of hydrogen atom. The remaining 1.4eV will become KE of this electron.

Exercise of Atomic Structure

Multiple Choice Types

Only one option correct

Q.1 The angular momentum of an electron in the ground state of the Li^{++} ion is –

(a) $\frac{3h}{2\pi}$ (b) $\frac{9h}{2\pi}$ (c) $\frac{h}{2\pi}$ (d) $\frac{h}{6\pi}$

Ans. **[c]**

Q.2 An atom emits a spectral line of wavelength λ when an electron makes a transition between levels of energy E_1 and E_2. Which expression correctly relates λ, E_1 and E_2?

(a) $\lambda = \frac{hc}{E_1 + E_2}$ (b) $\lambda = \frac{2hc}{E_1 + E_2}$

(c) $\lambda = \frac{2hc}{E_1 - E_2}$ (d) $\lambda = \frac{hc}{E_2 - E_1}$

Ans. **[d]**

Q.3 If the radius of an orbit is r and the velocity of the electron in it is v, then the frequency of electron in the orbit will be –

(a) $2\pi rv$ (b) $\frac{2\pi}{vr}$ (c) $\frac{vr}{2\pi}$ (d) $\frac{v}{2\pi r}$

Ans. **[d]**

Q.4 When an electron jumps from n_3^{th} orbit to n_2^{th} orbit, the energy radiated is given as –

(a) $h\nu = E_2/E_3$ (b) $h\nu = E_3/E_2$

(c) $h\nu = E_2 - E_3$ (d) $h\nu = E_3 - E_2$

Ans. **[d]**

Q.5 As the electron in Bohr orbit of the hydrogen atom jumps from state n = 2 to n = 1, the K.E. and P.E. change as

(a) K.E. two times, P.E. also two times

(b) K.E. four times, P.E. also four times

(c) K.E. four times, P.E. two times

(d) K.E. two times, P.E. four times

Ans. **[b]**

Q.6 In a hydrogen atom following the Bohr's postulates the product of linear momentum and angular momentum is proportional to $(n)^X$ where 'n' is the orbit number. Then 'x' is.

(a) 0 (b) 2 (c) –2 (d) 1

Ans. **[a]**

Q.7 According to Bohr model, the minimum energy (in eV) required to remove an electron from the ground state of a doubly ionised Li atom (Z = 3) is –

(a) 1.51eV (b) 13.6 eV (c) 40.8 eV (d) 122.4 eV

Ans. **[d]**

Q.8 The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume the Bohr model is valid in this case.

The frequency of the electron in the initial state is 1/27 of that in the final state. The possible values of n_1 and n_2 are

(a) $n_1 = 6, n_2 = 3$ (b) $n_1 = 4, n_2 = 2$

(c) $n_1 = 8, n_2 = 1$ (d) $n_1 = 3, n_2 = 1$

Ans. **[d]**

Q.9 When an electron in a hydrogen atom jumps from the ground state to a higher state:

(a) Both K.E. and P.E. increase

(b) Both K.E. and P.E. decrease

(c) P.E. increases and K.E. decrease

(d) P.E. decreases and K.E. increases

Ans. **[d]**

Q.10 An electron in a hydrogen atom jumps from n = 4 to the ground state (n=1). Then, the number of spectral lines observed can be-

(a) 3 (b) 6 (c) 5 (d) 2

Ans. **[b]**

Q.11 Which of the following parameters is the same for all hydrogen like atoms and ions in their ground states?

(a) Radius of the orbit

(b) Speed of the electron

(c) Energy of the atom

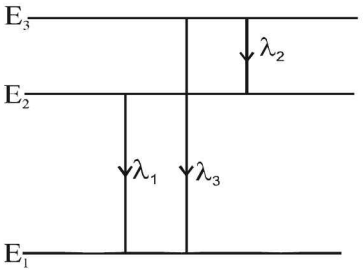
(d) Orbital angular momentum of the electron

Ans. **[d]**

Q.12 Transitions between three energy levels in a particular atom give rise to three spectral lines of wavelengths λ_1, λ_2 and λ_3. Which one of the following equations correctly relates λ_1, λ_2 and λ_3?

(a) $\lambda_1 = \lambda_2 - \lambda_3$ (b) $\lambda_1 = \lambda_3 - \lambda_2$

(c) $\frac{1}{\lambda_1} = \frac{1}{\lambda_2} + \frac{1}{\lambda_3}$ (d) $\frac{1}{\lambda_1} = \frac{1}{\lambda_3} - \frac{1}{\lambda_2}$



Ans. **[c]**

Q.13 An electron jumps from the 4th orbit to the 2nd orbit of the hydrogen atom. Given: the Rydberg's constant R = 105 cm^{-1}. The frequency (in Hz) of the emitted radiation will be

(a) $\frac{3}{16}\times 10^5$ (b) $\frac{3}{6}\times 10^{15}$ (c) $\frac{9}{16}\times 10^5$ (d) $\frac{9}{16}\times 10^{15}$

Ans. **[d]**

Q.14 Find the ratio of the ionization energy for hydrogen atom and doubly ionized lithium ion (Li^{2+}).

(a) $\frac{1}{9}$ (b) $\frac{4}{9}$ (c) $\frac{5}{9}$ (d) 9

Ans. **[a]**

Q.15 Whenever a hydrogen atom emits a photon in the Balmer series.

(a) It need not emit any more photon.
(b) It may emit another photon in the Paschen series.
(c) It must emit another photon in the Lyman series.
(d) It may emit another photon in the Balmer series.

Ans. **[c]**

Q.16 In terms of Rydberg constant R, the shortest wavelength in the Balmer series of hydrogen atom spectrum will have a wavelength -

(a) 1/R (b) 4/R (c) 3/2 R (d) 9/R

Ans. **[b]**

Q.17 If λ_1 and λ_2 are the wavelength of the first members of the Lyman and Paschen series, respectively, then $\lambda_1 : \lambda_2$ is

(a) 1: 3 (b) 1: 30 (c) 7: 50 (d) 7: 108

Ans. **[d]**

Q.18 If R is the Rydberg constant for hydrogen, then the wave number of the first line in the Lyman series is –

(a) $\frac{R}{2}$ (b) 2 R (c) $\frac{R}{4}$ (d) $\frac{3R}{4}$

Ans. **[d]**

Q.19 Let ν_1 be the frequency of series limit of Lyman series, ν_2 the frequency of the first line of Lyman series and ν_3 the frequency of series limit of Balmer series. Then which of the following is correct?

(a) $\nu_1 - \nu_2 = \nu_3$ (b) $\nu_2 - \nu_1 = \nu_3$

(c) $\nu_3 = \frac{1}{2}(\nu_2 + \nu_2)$ (d) $\nu_2 + \nu_1 = \nu_3$

Ans. **[a]**

Q.20 The wavelength of the third line of the Balmer series in the hydrogen spectrum is 5940Å. The wavelength of the first line is –

(a) $\frac{189}{125} \times 5940\text{Å}$ (b) $\frac{125}{100} \times 5940\text{Å}$ (c) 125 × 5940 Å (d) 5940 Å

Ans. **[a]**

Q.21 The ratio of minimum to maximum wave length in the Balmer series is –

(a) 5: 9 (b) 5: 36 (c) 1: 4 (d) 3: 4

Ans. **[a]**

Q.22 The angular momentum of an electron in a hydrogen atom is proportional to:

(a) $1/\sqrt{r}$ (b) $1/r$ (c) $\sqrt{r}$ (d) r^2

Ans. **[c]**

Q.23 A hydrogen atom in the ground state absorbs 10.2 eV of energy. The orbital momentum of the electron is increased by

(a) 1.05×10^{-34} Js (b) 2.11×10^{-34} Js

(c) 3.16×10^{-34} Js (d) 4.22×10^{-34} Js

Ans. **[a]**

Q.24 What is the angular momentum of an electron in Bohr's hydrogen atom whose energy is – 3.4 eV?

(a) $\frac{h}{2\pi}$ (b) $\frac{h}{3\pi}$ (c) $\frac{h}{\pi}$ (d) $\frac{2h}{\pi}$

Ans. **[c]**

Q.25 The hydrogen atom emits a photon when an electron jumps from n = 5 to n = 1 state, recoil speed of hydrogen atom is nearly.

(a) 10^{-4} ms^{-1} (b) 2×10^{-2} ms^{-1}

(c) 4 ms^{-1} (d) 8×10^2 ms^{-1}

Ans. **[c]**

Multi Option Correct

Q.26 In Bohr's model of hydrogen atom-

(a) The radius of nth orbit is proportional to n^2

(b) The total energy of electron in n^{th} orbit is proportional to n.

(c) The angular momentum of the electron in an orbit is an integral multiple of $h/2\pi$

(d) The magnitude of the potential energy of an electron in any orbit is greater than its kinetic energy.

Ans. **[a], [c], [d]**

Q.27 According to the Bohr's theory of a hydrogen atom, for the electron in the nth permissible orbit.

(a) Linear momentum $\propto \frac{1}{n}$ (b) radius of orbit $\propto n$

(c) Kinetic energy $\propto \frac{1}{n^2}$ (d) angular momentum $\propto n$

Ans. **[a], [c] [d]**

Q.28 Which of the following products in a hydrogen atom are independent of the principal quantum number n? The symbols have their usual meanings.

(a) E^2r (b) $\frac{E}{v^2}$ (c) $v^2 r$ (d) $\frac{E}{r}$

Ans. **[b],[c]**

Q.29 Suppose (hypothetical situation) the potential energy between an electron and a proton at a distance r is given by $\frac{Ke^2}{3r^3}$. Application of Bohr's theory to hydrogen atom in this case shows that -

(a) Energy in the nth orbit is proportional to n^6

(b) Energy is proportional to m^{-3} (m = mass of electron)

(c) Energy in the nth orbit is proportional to n^{-2}

(d) Energy is proportional to m^3 (m = mass of electron)

Ans. **[a], [b]**

Matrix Type

Q.30 Match the following table

Column – I	Column – II
(a) Frequency of orbiting electron	(p) Is directly proportional to n
(b) Angular momentum of orbiting electron	(q) Is directly proportional z^2
(c) Radius of orbital depends on principal quantum number as	(r) Increases
(d) If electron is going from lower energy level to higher energy level, then total energy of e^- will	(s) Proportional to n^2

Ans. **[a] → [q] , [b] →[p], [c] →[s], [d] → [r]**

Q.31 Regarding transition of electrons matches the following table:

Column – I	Column – II
(a) n = 5 to n = 2	(p) Lyman Series
(b) n = 8 to n = 4	(q) Brackett Series
(c) n = 3 to n = 1	(r) Paschen Series
(d) n = 4 to n = 3	(s) Balmer Series

Ans. **[a] →[s],** **[b] →[q],** **[c] →[p],** **[d] →[r]**

Compression Type

For Problem 32& 33

The energy levels of a hypothetical one electron atom are given by

$E_n = -\frac{18.0}{n^2}$ eV , where n = 1, 2, 3.....

Q.32 Compute the four lowest energy levels and construct the energy level diagram.

(a) –18.0 eV, – 4.5 eV, – 2.0 eV, –1.125 eV

(b) – 4.5 eV, –19 eV, –1.125 eV, 2 eV

(c) 2 eV, – 4.6 eV, –1.125 eV, 18 eV

(d) –1.125 eV, 18 eV, 2 eV, –4.5 eV

Ans. **[a]**

Q.33 What is the photoelectric threshold wavelength of this atom?

(a) 700Å (b) 680 Å (c) 688.9 Å (d) 780 Å

Ans. **[c]**

For Problem 34to 38

A single electron orbits a stationary nucleus of charge + Ze, where Z is atomic number and e is the charge on the electron. It requires 47.2 eV energy to excite the electron from the second orbit to third Bohr orbit. Find:

Q.34 The value of Z.

(a) 2 (b) 3 (c) 5 (d) 1

Ans. **[c]**

Q.35 The energy required to excite the electron from the third to the fourth Bohr orbit.

(a) 15 eV (b) 16.53 eV (c) 16 eV (d) 15.53 eV

Ans. **[b]**

Q.36 The wavelength of electromagnetic radiation required to remove the electron from atom.
(a) 36.5 Å (b) 36 Å (c) 35.5 Å (d) 35 Å

Ans. **[a]**

Q.37 Find the K.E., P.E. of electron in the 1st Bohr orbit?
(a) 340 eV, 680 eV (b) -340eV,-680 eV
(c) -340 eV, 680 eV (d) 340eV, – 680 eV

Ans. **[d]**

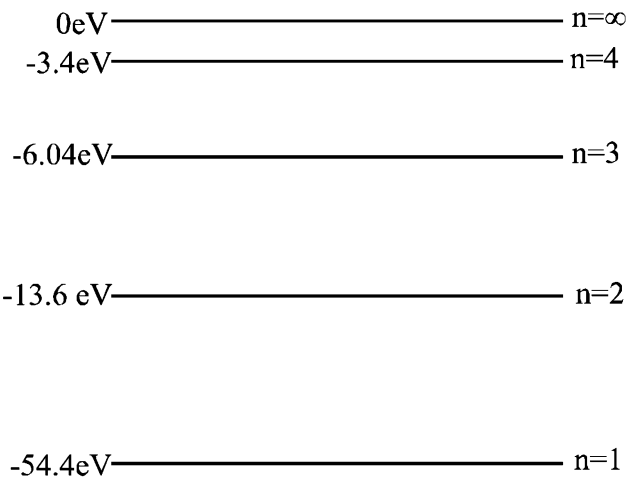
Q.38 The radius of the first Bohr orbit.

[The ionization energy of hydrogen atom = 13.6 eV, Bohr radius = 5.3×10^{-11} m]

Ans. **$0{\cdot}106 \times 10^{-10}$ m**

For Problem 39 to 42

The energy level diagram for hydrogen- like atom is shown in Fig.



Q.38 What is the atomic number of atom?
(a) 4 (b) 3 (c) 1 (d) 2

Ans. **[d]**

Q.39 Can an electron, in ground state, absorb a photon of 42 eV?
(a) Electron can't absorbs photon (b) Electron absorbs photon
(c) Energy level exists (d) None of these

Ans. **[a]**

Q.40 Can it absorb a photon of 56 eV?
(a) Electron can't absorbs photon (b) Electron absorbs photon
(c) Energy level doesn't exists (d) None of these

Ans. **[b]**

Q.41 Calculate the radius of its first Bohr orbit.
(a) 0.200Å (b) 0.150 Å (c) 0.260 Å (d) 0.265 Å

Ans. **[d]**

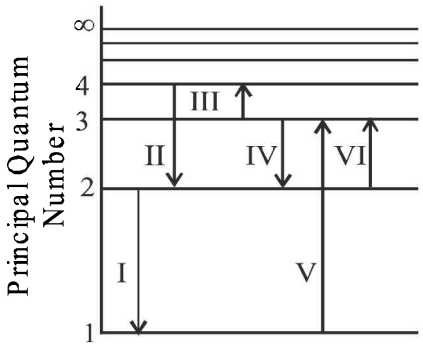
Q.42 Calculate the kinetic energy and potential energy of an electron in the first orbit.

(a) 54.4 eV, –108.8 eV (b) –54.4eV,–108.8 eV
(c) –54.4 eV, 108.8 eV (d) 54.4 eV, 108.8 eV

Ans. **[a]**

For Problems 43 to 44

Below figure shows an energy level diagram of the hydrogen atom. Several transitions are marked as I, II, III. The diagram is only indicative and not to scale.



Q.43 The wavelength of the radiation involved in transition II is -
(a) 291 nm (b) 364 nm (c) 487 nm (d) 652 nm

Ans. **[c]**

Q.44 Which transition will occur when a hydrogen atom is irradiated with radiation of wavelength 103 nm?
(a) I (b) II (c) IV (d) V

Ans. **[d]**

Statement and Assertion Type

(a) Statement-I is True, Statement-II is True; statement - II is a correct explanation for statement-I
(b) Statement -I is True, Statement -II is True; Statement - II is NOT a correct explanation for statement -I.
(c) Statement-I is True, Statement-II is False.
(d) Statement - 1 is False, Statement-II is true.

Q.45 **Statement I** In a hydrogen atom, the energy of emitted photon corresponding to transition from n = 2 to n = 1 is much greater as compared to the transition from n = ∞ to n = 2.
Statement II
The Wavelength of photon is directly proportional to the energy of emitted photon.

Ans. **[c]**

Q.46 **Statement-I** Balmer series lies in the visible region of the electromagnetic spectrum.
Statement-II for Balmer series –

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{K^2}\right)$$, where K=3, 4, 5....

Ans. **[b]**

Subjective Type

Q.47 The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 the principal quantum numbers of the two energy states. Assume the Bohr's model to be valid. The time

period of the electron in the initial state is eight times that in the final state. What are the possible values of n_1 and n_2?

Ans. **[$n_1 = 2n_2$]**

Q.48 Find the kinetic energy, potential energy and total energy in first and second orbit of hydrogen atom?

Ans. **[$K.E._1 = -E_1 = 13.60$ eV, $P.E_1 = -27.20$ eV, $K.E._2 = 3.40$ eV, $P.E._2 = -6.80$ eV]**

Q.49 The electron in a hydrogen atom at rest makes a transition from n = 2 energy state to the n = 1 ground state. Find the energy (in eV) of the emitted photon.

Ans. **[E = 10.2 eV]**

Q.50 (a) Find the wavelength of the radiation required to excite the electron in Li^{++} from the first to third Bohr orbit.
(b)How many spectral lines are observed in the emission spectrum of the above excited system?

Ans. **(a) $\lambda = 11.4$ nm**
(b) Three lines will be observed.

Q.51 Determine the minimum wavelength of a photon that can cause ionization of He^+ ion.

Ans. **[$\lambda_{min} = 22.8$ nm]**

Q.52 Find the wavelength of the radiation required to excite the electron in Li^{++} from the first to third Bohr orbit. The ionization energy of the hydrogen atom is 13.6 eV.

Ans. **[$\lambda = 113.97$Å]**

Q.53 Find the largest and shortest wavelengths in the Lyman series for hydrogen.

Ans. **[911 Å]**

Q.54 (a) Find the longest wavelength photon emitted in Balmer series and determine its energy.
(b) Find the shortest wavelength photon emitted in the Balmer series.

Ans. **(a) 656.3 nm, 1·89 eV, (b) 364.6 nm**

Q.55 Calculate the wavelength and the frequency of the H_b line of the Balmer series for hydrogen.

Ans. **[6.12×10^{14} Hz]**

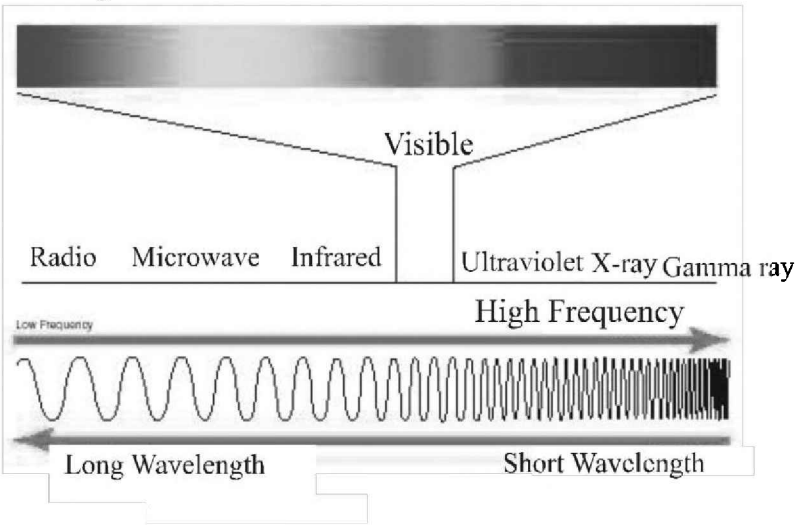
Q.56 The radiation, emitted when an electron jumps from n = 3 to n = 2 orbit in a hydrogen atom, falls on a metal to produce photo-electrons. The electrons from the metal surface with maximum kinetic energy are made to move perpendicular to a magnetic field of T in a radius of 10^{-3}m. Find
(i) The kinetic energy of electrons
(ii) The work function of the metal and
(iii) The wavelength of the radiation.

Ans. **(i) $K = \frac{100}{116.48} eV = 0.86 eV$**
(ii) W = 1.03 eV
(iii) $l = 6.563 \times 10^{-7} m = 6563$Å

03 X – RAY

3.1 Introduction

We all have studied electromagnetic spectrum in previous chapters. There are various types of electromagnetic waves, namely radio waves, microwaves, infrared rays, visible light, UV rays, X-rays and gamma rays. They all have different frequencies as well as different wavelengths. Radio waves have the lowest frequency while gamma rays have the highest frequency among all electromagnetic waves as shown in the figure below:



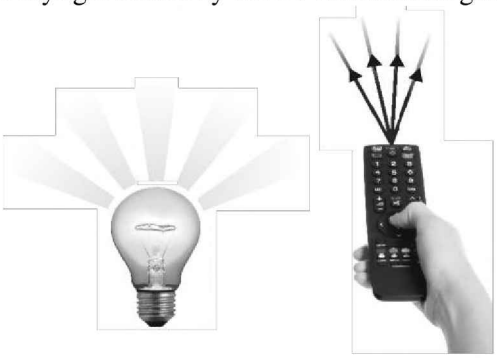
We know that,

$$c = \nu \lambda$$

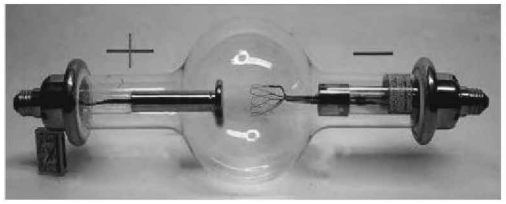
Here, ν = frequency of electromagnetic wave
λ= wavelength of electromagnetic wave
c = speed of light

The frequency of an electromagnetic wave is inversely proportional to its wavelength i.e. if frequency of any electromagnetic wave is higher, its wavelength will be smaller and vice versa. As frequency or wavelength of electromagnetic waves changes, they show different properties. In this chapter, we will study about X- rays.

As we all know that a household electric bulb emits visible light, thus we can say that this bulb is an apparatus or device which generates visible light. Similarly, there is a small apparatus in a TV remote which generates infrared rays. When we press button on the remote to change channel, infrared rays come out of the remote. These rays get sensed by the TV and channel gets changed.



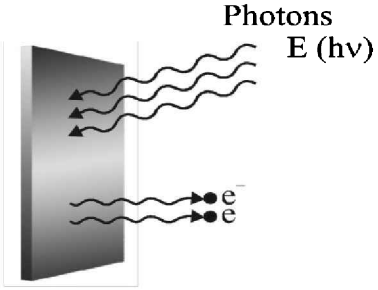
Similarly, X-rays are produced by a device known as **Coolidge tube**. This instrument was named after the scientist who invented it. Coolidge tube is shown below.



Before studying the working principle of the Coolidge tube in detail, let us understand the term "**Thermionic Emission".**

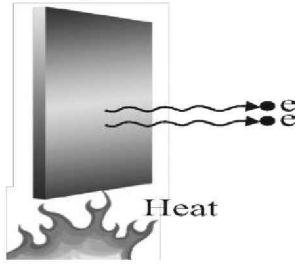
3.2 Thermionic Emission

We have seen in the photo electric effect that when photons fall on a metal surface, electrons are emitted from it. This phenomenon is known as **photo emission.**



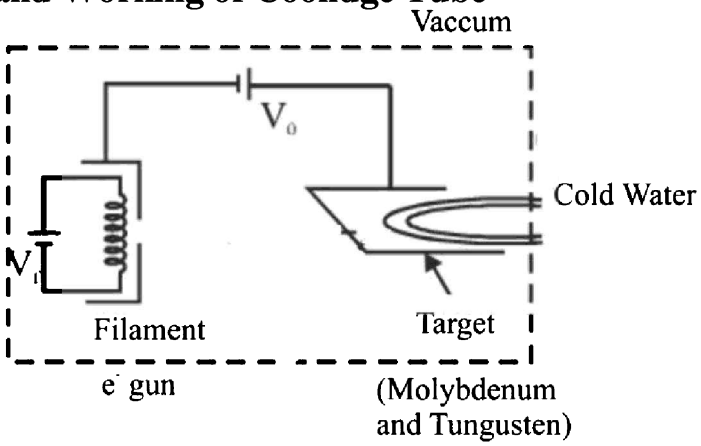
(Photo emission)

Similar phenomenon occurs when we heat a metal plate. Electrons inside the metal plate absorb the supplied heat energy. When their energy increases by a certain amount, electrons are emitted. This phenomenon is called **thermionic emission** (Emission of electrons on supplying thermal energy).



(Thermionic emission)

3.3 Structure and Working of Coolidge Tube

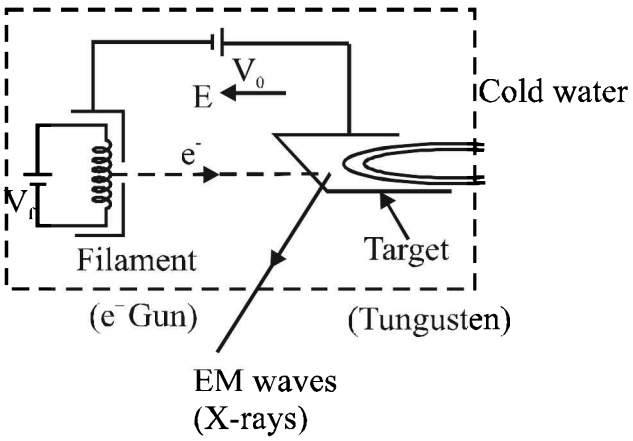


The above figure shows the structure of a Coolidge tube. On the left side of the Coolidge tube, a battery is connected across a resistor. This resistor is called **filament**. When current flows in this resistor, the filament gets heated and emits electrons due to thermionic emission. This filament is also called **electron gun**.

The metallic substance on the right side is called the **target.** Generally, it is made up of *tungsten or molybdenum*.

When this device works, the target becomes very hot. To cool down the target, a cooling system is set up. Cold water continuously passes through the target via a pipe. This cold water takes away heat from the target.

A constant potential difference is maintained across the filament and the target through a battery V_0. Due to this, an electric field is setup between the filament and target. Electrons emitted by filament get accelerated and strike the target. Due to very high speed, electrons penetrate inside the target. As electrons go inside the target, electromagnetic waves, mainly X-rays, are emitted. Now why X-rays are emitted is the main topic of study in this chapter.

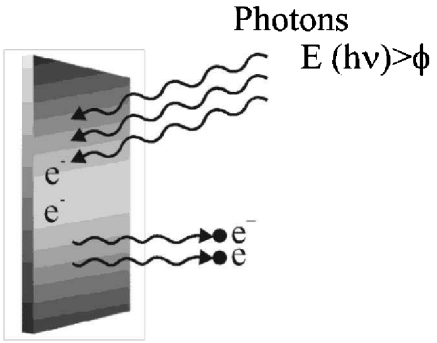


We know that in photo-emission, each photon carries a certain amount of energy i.e. **hν**. Work function (ϕ) amount of energy is needed to eject an electron from the metal surface.

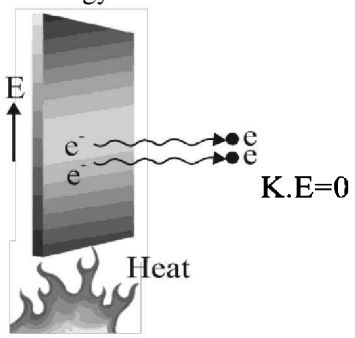
Emission of electrons takes place only when

$$h\nu \geq \phi$$

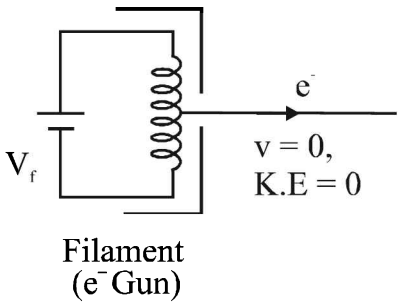
When an electron absorbs energy from the photon, its energy suddenly increases by **hν.** Then it loses work function amount of energy during emission from metal surface. The rest of the energy **(hν – ϕ)** gets converted into the kinetic energy of the electron.



In thermionic emission, energy is supplied to a metal surface by heating it, so the energy of electrons does not increase suddenly, it increase slowly. When electrons absorb the work function amount of energy, they get emitted out from the metal surface. Electrons use this entire energy absorbed to escape from the metal so, their kinetic energy remains almost zero while coming out from metal.



Thus, when the filament of the Coolidge tube emits electrons, they have zero initial velocity.
So, how do electrons move towards the target?



This is due to the electric field setup between the filament and target. When an electron moves from filament to target, potential changes by $\mathbf{V_0}$, so a change in potential energy will be

$$\Delta V = eV_0$$

By conservation of energy

$$\Delta KE = eV_0$$

So electrons hit the target with KE of $\mathbf{eV_0}$.

When electrons enter the target, two types of processes take place.

(a) Process 1-

The emitted electrons collide with other electrons and nucleus inside the target and lose their kinetic energy. Some of the lost kinetic energy is converted into heat energy. As a result, target becomes hot. Rest of the lost kinetic energy is converted into photon. However, it is not possible to calculate how much of the lost kinetic energy is converted to heat and how much to photon. Also, at times, an electron may lose all its KE in a single collision. At other times, an electron may take several collision

to lose all its KE.

In any collision, loss of energy (E_e) by electron

$$\mathbf{0 < E_{e^-} \leq eV_0} \quad \text{... (1)}$$

Thus energy of photon (E_{photon}) produced in any collision is

$$\mathbf{0 < E_{photon} \leq eV_0} \quad \text{... (2)}$$

Photons can have energy equal to $\mathbf{eV_0}$ only when an electron loses entire energy in a single collision and this entire energy is converted into a photon.

The above equation can be written as:

$$0 < \frac{hc}{\lambda} \leq eV_0$$

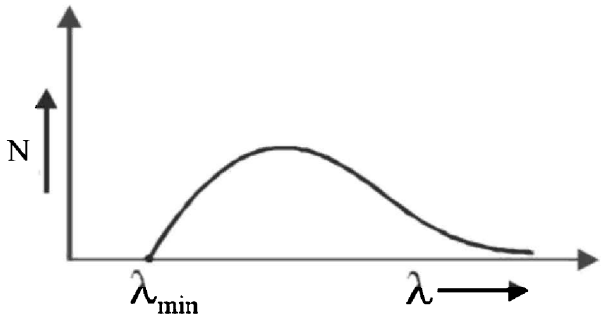
$$\Rightarrow \quad 0 < \frac{1}{\lambda} \leq \frac{eV_0}{hc}$$

$$\Rightarrow \quad \infty > \lambda \geq \frac{hc}{eV_0}$$

This equation tells us that target emits photons of different wavelengths but their wavelengths are always greater than or equal to $\frac{hc}{eV_0}$,

$$\Rightarrow \quad \lambda_{min} = \frac{hc}{eV_0}$$

The graph given below shows the variation in number of photon emitted per unit wavelength (N) with wavelength. Area under this graph gives total number photons emitted.





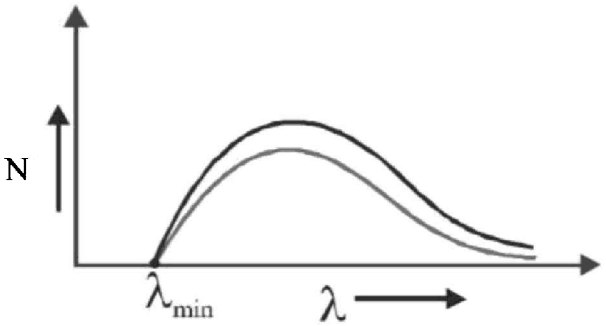
Quiz-1

(I) If we increase the potential (V_f) of the battery attached with the filament then how does the graph change?

***Sol*.** An increase in the potential of the battery increases the flow of current in the filament. Thus filament gets more heated emitting more number of electrons. Now more electrons hit the target and do more collisions. Therefore a larger number of photons are emitted from the target. Thus, area under curve should increase. But what happens with λ_{min}?

Since, $\lambda_{min} = \frac{hc}{eV_0}$

Where, h, c and e are all constants. V_0 is not being changed here. Hence λ_{min} remains same with increase in V_f.

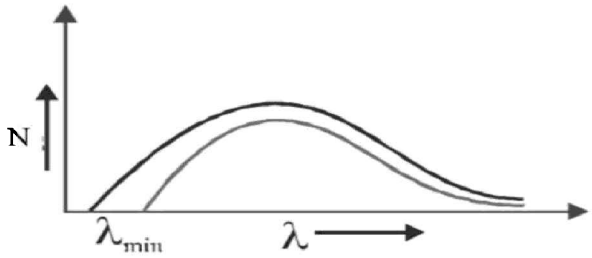


(II) If we increase the potential of a battery fixed between filament and target material, how does the graph change?

Sol. As, $\lambda_{min} = \frac{hc}{eV_0}$

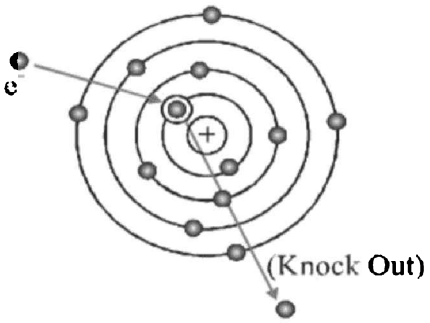
So by increasing the accelerating voltage (V_0), λ_{min} decreases.

On the other hand, an increase in the potential of the battery increases the electric field and therefore increases the kinetic energy of electrons ($\because$ K.E = eV_0). Now when electrons hit the target with more kinetic energy, number of collisions are more and hence more number of photons are emitted from the target. Thus the new graph is –



(b) Process 2 -

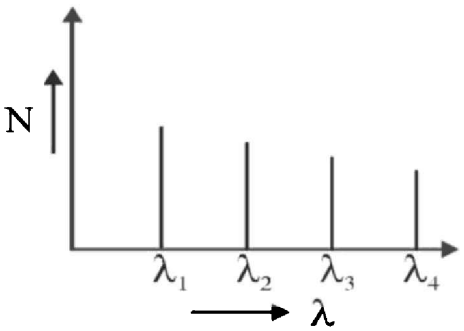
As we saw previously that the target of a Coolidge tube is made up of tungsten or molybdenum (heavy material), so every atom of target material contains lots of electrons. Suppose, an accelerated electron with high kinetic energy collides with an electron of this atom (say in n = 1) and knocks it out. Now a vacancy is created. An electron from a higher orbit makes a transition to fill this vacancy. During this transition, a photon is radiated.



The two processes (process 1 and 2) are different from each other. In process 1, accelerated electrons collide inside the target but they do not knock out any orbital electron. During these collisions, photons are emitted. While in process 2, electrons knock out orbital electrons. When an electron from a higher orbit makes a transition to fill the created vacancy, photons are emitted out. The energy of these photons, in process 2, must be equal to energy gap between the orbits of the atom (ΔE). i.e.:

$$\frac{hc}{\lambda} = \Delta E$$

Note that photons of all wavelengths cannot be emitted in process 2. Only those wavelengths can be emitted which correspond to energy gaps in the atom. So for process 2 we get the graph as shown. We can see spikes at some wavelengths. In process 2, photons of only these wavelengths are emitted. For other wavelengths, there is no emission.





Quiz-2

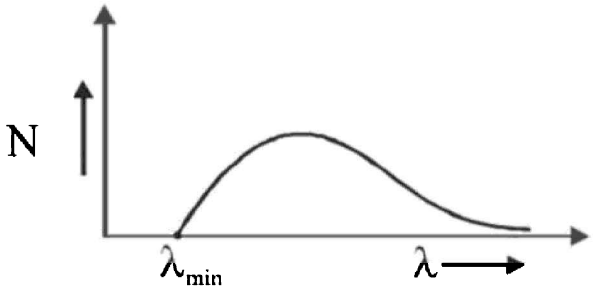
What happens with the graphs (of process 1 and process 2) on changing the target material?

Sol. **Process 2 -** If we use different material for target, energy gaps will also be different.

$$\frac{hc}{\lambda} = \Delta E$$

As ΔE will change, graph will also have different values for λ_1, λ_2, λ_3,λ_4. Thus changing material of target will change the graph for process 2.

Process 1- The graph of process 1 is –



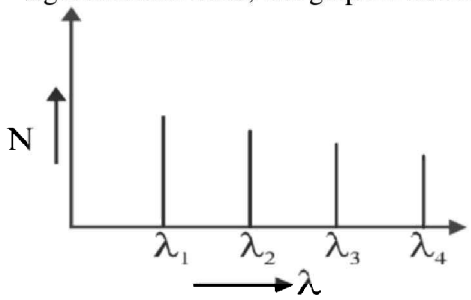
$$\because \quad \lambda_{min} = \frac{hc}{eV_0}$$

Here, h, c and e all are constants. V_0 is the potential difference across the filament and the target.

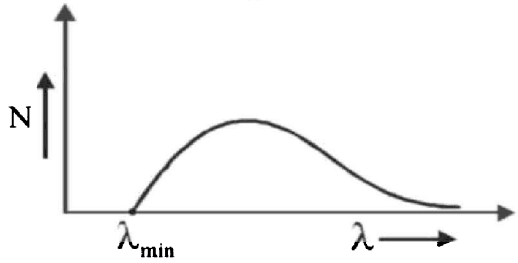
As λ_{min} does not depend on target material, so the graph of process 1 will remain same.

Key Points –

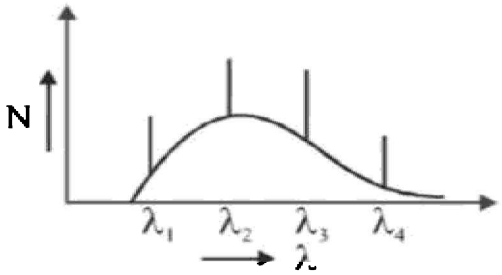
- For different targets, the graph of process 2 will be different. So we can say that this graph is a characteristic of the target material. Thus, this graph is called **characteristic spectra.**



- Graph of process 1 is called **continuous spectra**.



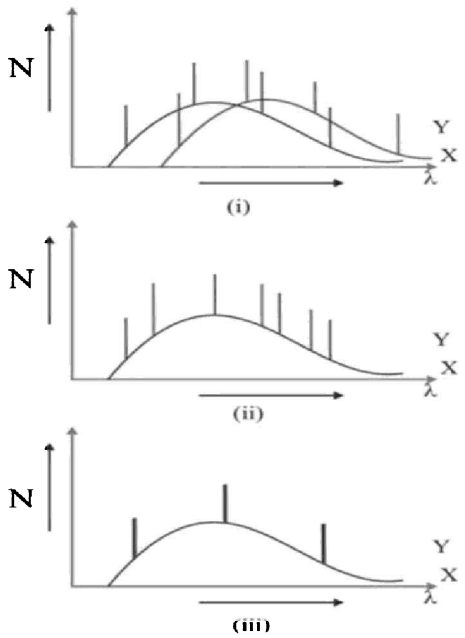
We have studied about two types of processes and their respective graphs. But these processes are going on simultaneously in Coolidge tube. The **overall graph** of these two processes combined will be –



Radiation of photons of wavelength λ_1, λ_2, λ_3 and λ_4 takes place due to both these processes (process 1 and process 2). In any range from λ_1 to λ_2 or λ_2 to λ_3 etc., photons are emitted due to process 1 only. Thus we can say that maximum photons are emitted at λ_1, λ_2, λ_3. But these wavelengths change when target metal is changed. When tungsten or molybdenum is chosen as the target metal, electrons radiate the photons of wavelengths which lie in the range of X- ray.

Illustration 1

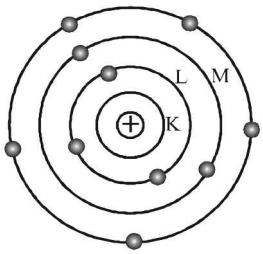
If x and y are taken as different target material in a Coolidge tube without changing anything else then which of the following graph is correct in the two cases?



Sol. As continuous spectrum is independent of target material so it remains unchanged in both the cases. But characteristic spectra depend on target material so it will be different in both the cases. Hence, option (ii) is correct.

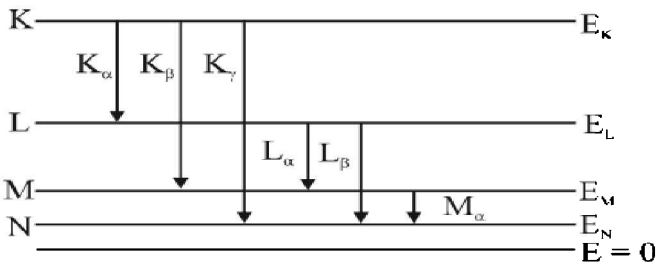
3.4 The Energy Levels

Energy levels of an atom are named as **K shell, L Shell, M shell and so on.**



As we have seen (for process 2), when electron jumps from higher orbit to lower orbit, it radiates photons.

- If electron jump from n = 2 to n = 1, then photon emitted is called $\mathbf{K_\alpha}$.
- If electron jump from n = 3 to n = 1, photon emitted is called $\mathbf{K_\beta}$.
- Similarly, if an electron makes a jump from n = 4 to n = 1, photon emitted is called $\mathbf{K_\gamma}$.
- Similarly, when an electron jumps from n = 3 to n = 2, photon emitted is called $\mathbf{L_\alpha}$.
- If electron jumps from n = 4 to n = 2, photon emitted is called $\mathbf{L_\beta}$ and so on.



- For K-series, energy of emitted photon is given as–

$$\frac{hc}{\lambda} = 13.6(Z-1)^2\left[\frac{1}{1^2} - \frac{1}{n^2}\right]$$

- For L- series,

$$\frac{hc}{\lambda} = 13.6(Z-7.4)^2\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$$

Here, n is the quantum number of shell from which transition is made.



Quiz-3

Find out the wavelength of K_α.

Sol. K_α is the photon emitted due to transition of electron from n = 2 to n = 1.

Energy of photon emitted is

$$\frac{hc}{\lambda} = \Delta E = 13.6(Z-1)^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

Here, Z is the atomic number of element.

Illustration 2

If K_α radiation of Z = 11 has wavelength of λ_0 then calculate the value of Z for which K_α has wavelength of $4\lambda_0$.

Sol. For K_α radiation

$$\frac{hc}{\lambda} = 13.6(Z-1)^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

When wavelength is λ_0,

$$\frac{hc}{\lambda_0} = 13.6(10)^2\left[\frac{3}{4}\right] \quad \text{... (1)}$$

When wavelength is $4\lambda_0$,

$$\frac{hc}{4\lambda_0} = 13.6(Z-1)^2\left[\frac{3}{4}\right] \quad \text{... (2)}$$

Dividing equation (1) by (2)

$$4 = \frac{10^2}{(Z-1)^2}$$

$\Rightarrow$ 2 Z – 2 = 10

$\Rightarrow$ Z = 6

3.5 Moseley's Law –

As equation of K_α radiation is:

$$\frac{hc}{\lambda} = 13.6(Z-1)^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

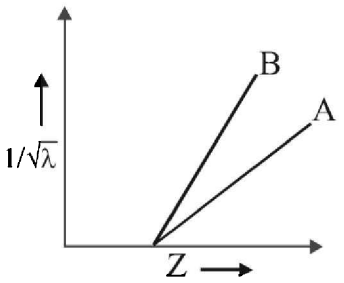
$$\Rightarrow \quad \frac{1}{\lambda} \propto (Z-1)^2$$

$$\Rightarrow \quad \frac{1}{\sqrt{\lambda}} \propto (Z-1)$$

This equation is called **Moseley's Law**.

Illustration 3

Figure shows a Moseley's graph between $\frac{1}{\sqrt{\lambda}}$ and Z. Which of the following slope in the graph represents K_α and K_β?



Sol .For K_α;

$$\frac{hc}{\lambda} = 13.6(Z-1)^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

$$\Rightarrow \quad \frac{hc}{\lambda} = 13.6(Z-1)^2\frac{3}{4}$$

$$\Rightarrow \quad \frac{1}{\sqrt{\lambda}} = \sqrt{\frac{13.6}{hc}\frac{3}{4}}(Z-1) \qquad \text{... (1)}$$

For K_β;

$$\frac{hc}{\lambda} = 13.6(Z-1)^2\left(\frac{1}{1^2} - \frac{1}{3^2}\right)$$

$$\Rightarrow \quad \frac{h}{\lambda} = 13.6(Z-1)^2\left(\frac{8}{9}\right)$$

$$\Rightarrow \quad \frac{1}{\sqrt{\lambda}} = \sqrt{\frac{13.6}{hc}\frac{8}{9}}(Z-1) \qquad \text{... (2)}$$

Equations (1) and (2) are clearly equations of straight line.

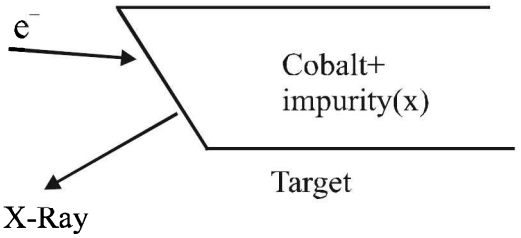
$\because \quad (\text{Slope})_{K_\alpha} < (\text{Slope})_{K_\beta}$

Hence graph B will represent K_β and graph A will represent K_α.

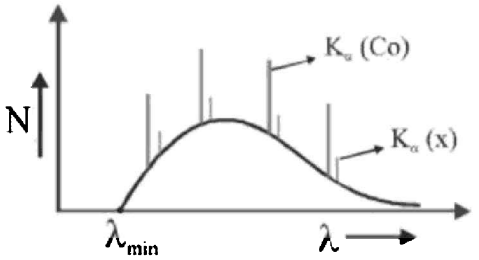
Illustration 4

An impure cobalt target is bombarded with e^- and characteristic spectrum is observed. Wavelength of K_α for cobalt and impurity are 5.9 pm and 142.5 (impurity). Identify the impurity.

Sol. When impure cobalt target is bombarded with high energy electrons, it will radiate X-rays.



Now the graph between N and λ will be seen as



Small peaks represent characteristic spectra of impurity x and large peaks represent characteristic spectra of cobalt. Continuous spectra will be same for cobalt and impurity.

Given,

$K_\alpha(Co) = 178.9$ pm

$K_\alpha(x) = 143.5$ pm

According to Moseley's Law

$$\frac{1}{\sqrt{\lambda}} \propto (Z-1)$$

For cobalt,

$$\frac{1}{\sqrt{178.9}} = k(Z_{co} - 1) \quad \text{... (1)}$$

For impurity x,

$$\frac{1}{\sqrt{143.5}} = k(Z_x - 1) \quad \text{... (2)}$$

Dividing equation (1) by (2)

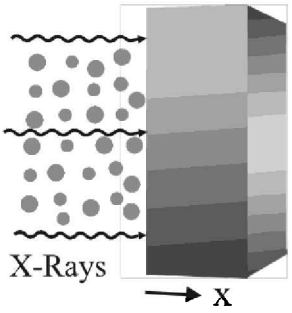
$$\sqrt{\frac{143.5}{178.9}} = \frac{Z_{Co} - 1}{Z_x - 1}$$

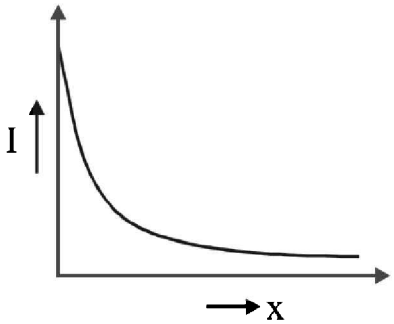
$$\Rightarrow \sqrt{\frac{143.5}{178.9}} = \frac{27-1}{Z_x - 1} \Rightarrow Z_X = 30$$

It means the impurity mixed with cobalt is zinc.

3.6 Absorption of X-rays in Heavy Metal

1. When photons of X-ray hit a metal surface, they get absorbed by it.





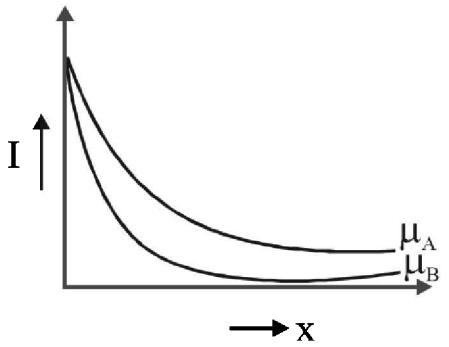
The intensity versus distance graph shows that as we go deeper inside the metal, intensity of photons decreases exponentially. This is because every layer of metal absorbs some photons and thus lesser number of photons is able to penetrate further. Therefore, intensity keeps on decreasing with distance.

$$I = I_0 e^{-\mu x}$$

Here, μ = coefficient of absorption.
If a metal has higher value of μ then it will absorb more X-rays.

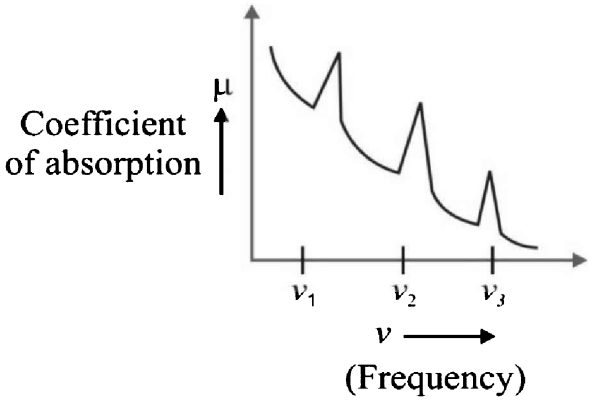
Illustration 5

A graph of intensity versus distance for two materials A and B is shown. Is $\mu_A > \mu_B$?

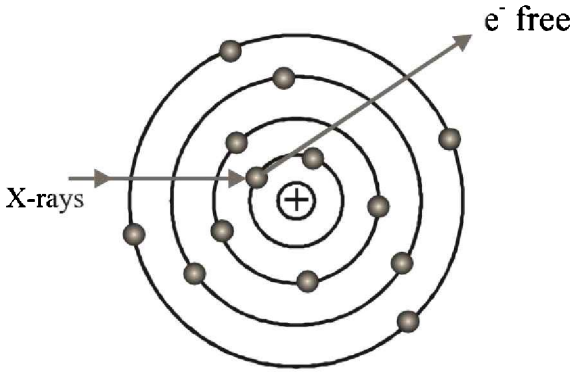


Sol. Intensity for metal B is decreasing at faster rate than that of A. It means, metal B absorbs more X-rays. Hence, $\mu_A < \mu_B$

2. We saw that when photons hit a metal, they penetrate into it. Variation of μ with frequency of X- rays for heavy metals like tungsten is plotted in the figure below. These peaks in graph may surprise you. Let's see why they are there.



When X- ray hits an electron of an atom, an electron may absorb the energy from X- rays and may jump to a higher orbit. But in heavy metals such as tungsten or molybdenum, there are almost no vacant orbits. So, an electron cannot jump to a higher orbit. The only available option for the electron is to jump out of the atom and become free.



We can say that an electron will absorb the photon only if the photon has sufficient energy to take the electron out of the atom.

In the graph shown above, the edges or peaks are obtained at frequencies ν_1, ν_2 and ν_3. At these frequencies, photons have sufficient energy to free electrons from their orbits. Thus these frequencies can be absorbed by metal in larger amount. This increases co-efficient of absorption significantly at these frequencies. So, in graph, we see peaks at these frequencies.

K- edge, as in the figure, represents photons which free an electron from K shell**.** Similarly **L-edge** and **M-edge** represent photons which free electrons from L and M shells respectively.

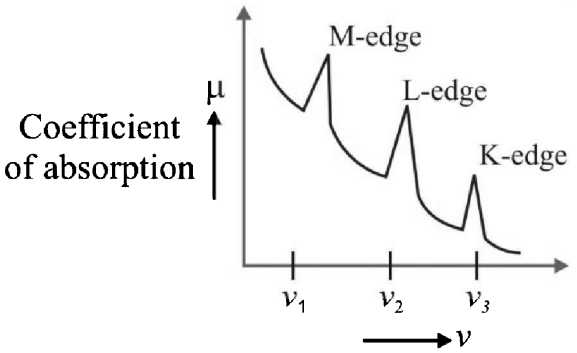
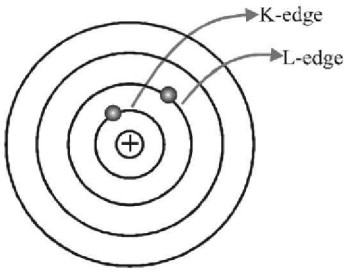


Illustration 6

Find out the frequency of K_α radiation if ν (K-edge) = 5×10^{18}, ν(L-edge) = 3×10^{18}.



Sol.

In case of K-edge,

$E = h\nu$

$= h\,(5 \times 10^{18})$

In case of L-edge

$E = h\nu = h\,(3 \times 10^{18})$

We know,

Energy associated with K_α= Energy gap between n = 2 to n = 1.

$$\Rightarrow \quad h\nu_{K_\alpha} = E_{\text{K-edge}} - E_{\text{L-edge}}$$

$$\Rightarrow \quad h\nu_{K_\alpha} = h\left(5 \times 10^{18} - 3 \times 10^{18}\right)$$

$$\Rightarrow \quad \nu_{K_\alpha} = 2 \times 10^{18}$$

3.7 Soft and Hard X-rays

X- Rays which have photons of lower energy are called **soft X-rays**. While X- rays which have photons of higher energy are called **hard X- rays**. This implies that

$$\nu_{\text{Hard}} > \nu_{\text{Soft}}$$

Harder X-rays have smaller value of coefficient of absorption in comparison to soft X-rays.

Exercise of X-Rays

Multiple choices Question

Only one option correct

Q.1 The potential difference applied to an X-ray tube is 10 kV and the current through it is 6.4 A. Then the number of electrons striking the target per second is

(a) 4×10^{19} (b) 5×10^{6} (c) 1×10^{17} (d) 4×10^{15}.

Ans. **[a]**

Q.2 An X –ray tube operates at 25 kV. Find the maximum speed of the electrons striking the anode, given the charge of electron = 1.6×10^{-19} coulomb and mass of electron = 9×10^{-31} kg.

(a) 7×10^{14} m/s (b) 8.4×10^{8} m/s

(c) 0.712×10^{23} m/s (d) 9.42×10^{7} m/s

Ans. **[d]**

Q.3 The X-rays coming from an X-ray tube will be

(a) Monochromatic

(b) Having all wavelengths smaller than a certain minimum wavelength

(c) Having all wavelengths greater than a certain minimum wavelength

(d) Having all wavelengths between certain minimum and maximum wavelengths

Ans. **[c]**

Q.4 The K_αX-ray emission line of tungsten occurs at $\lambda = 0.042$m. The energy difference between K and L levels in this atom is about

(a) 0.51 MeV (b) 1.2 MeV (c) 29KeV (d) 13.6eV

Ans. **[c]**

Q.5 The minimum wavelength of X-rays produced by an X-ray tube operated upon 1243 kV is (Given h = 6.625×10^{-34} J-sec, e = 16×10^{-19} C and c = 3×10^{8} m/sec)

(a) 1 Å (b) 0.1 Å (c) 10 Å (d) 0.001 Å

Ans. **[d]**

Q.6 Electrons with energy 160 KeV are incident on the tungsten target of an X-ray tube. K-shell electrons of tungsten have –72.5 KeV energy. X-ray emitted by the tube contain only

(a) A continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of ~0.155 Å.

(b) A continuous X-ray spectrum (Bremsstrahlung) with the minimum wavelengths.

(c) The characteristic X-ray spectrum of tungsten

(d) A continuous X-ray spectrum (Bremsstrahlung) with a minimum wavelength of ~0.0775Å and the characteristic X-ray spectrum of tungsten.

Ans. **[d]**

Q.7 The wavelength of K_α X- ray of tungsten is 0.63Å. If the energy of a tungsten atom with an L electron knocked out is 11.3 KeV, what will be the energy of this atom when a K electron is knocked out? (h = 4.14×10^{-15} eVs)

(a) 31.014 KeV (b) 19.714 KeV

(c) 29.013 KeV (d) 14.012 KeV

Ans. **[a]**

Q.8 An X – ray tube is operated at 12.4 kV. In the continuous spectrum of the emitted X – rays, Find maximum speed of X – ray emitted.

(a) 3.5×10^{10} m/s (b) 4×10^{15} m/s

(c) 3×10^{18} m/s (d) 4.58×10^{7} m/s

Ans. **[d]**

Q.9 The element which has a K_α X-rays line of wavelength 1.8 Å is:

($R = 1.1 \times 10^7 m^{-1}$, b = 1 and $\sqrt{5/33 = 0.39}$)

(a) Co, Z=27 (b) Fe, Z = 26 (c) Mn, Z = 25 (d) Ni, Z = 28

Ans. **[a]**

Q.10 The wavelength of the characteristic X-ray K_α line emitted by a hydrogen like atom is 0.32 Å. The wavelength of K_β line emitted by the same element is:

(a) 0.18 Å (b) 0.48 Å (c) 0.27Å (d) 0.38 Å

Ans. **[c]**

Q.11 The wavelength of K_α X-ray produced by an X-ray tube is 1.52 Å. The atomic number of anticathode material is

(a) 82 (b) 29 (c) 20 (d) 10

Ans. **[b]**

Q.12 K_α wavelength of an element with atomic number Z = 23 is λ. For what of Z, the wavelength becomes 4λ?

(a) 44 (b) 11 (c) 12 (d) 4

Ans. **[c]**

Q.13 If λ_1, λ_2 and λ_3 are the wavelength of K_αX-rays emitted by ^{112}Sn, ^{114}Sn and ^{116}Sn tin isotopes, then

(a) $\lambda_1 = \lambda_2 = \lambda_3$ (b) $\lambda_1 > \lambda_2 > \lambda_3$

(c) $\lambda_1 < \lambda_2 < \lambda_3$ (d) $\frac{1}{\lambda_1} + \frac{1}{\lambda_3} = \frac{2}{\lambda_2}$

Ans. **[a]**

Q.14 If the short wavelength limit of the continuous spectrum coming out of a Coolidge tube is 100Å, then the debroglie wavelength of the electrons reaching the target metal in the Coolidge tube is approximately:

(a) .109 Å (b) 0.11Å (c) .05Å (d) .85Å

Ans. **[a]**

More than one correct

Q.15 In an X-ray tube the voltage applied is 20kV. The energy required to remove an electron from L shell is 19.9keV. In the X-ray emitted by the tube

(a) Minimum wavelength will be 62.0pm

(b) Energy of the characteristic X-rays will be equal to or less than 19.9 KeV

(c) K_αX-ray may be emitted.

(d) L_α X-ray will have energy 19.9 KeV.

Ans. **[a, b]**

Q.16 The potential difference across an X-ray tube is 10 MV and the current through it is 5 mA. The maximum speed of cathode rays produced is v_{max} and the rate of production of heat at the anticathode if only 0.1% of energy is converted into X-radiation is U. Then

(a) $v_{max} = 1.88 \times 10^6$ m/s (b) $v_{max} = 5.99 \times 10^9$ m/s

(c) U = 250 J/s (d) U = 500 J/s

Ans. **[b, d]**

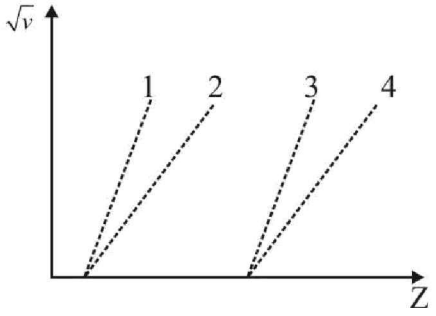
Q.17 The potential difference across an X-ray tube is 50000 volt and the current through it is 2.5mA Then

(a) The number of electrons striking the anode per second is nearly 10^8.
(b) The speed with which they strike is.
(c) Energy of electrons striking the anode is 5×10^4 eV
(d) Cut- off wavelength of x rays is 0.125 nm

Ans. **[b, c]**

Matrix match type

Q.18 **$\sqrt{v}$ versus Z- graph for characteristic X-rays is as shown in figure. Now match the items in Column I with those in Column II.**



Column I	Column II
(a) Line -1	(p) L_β
(b) Line -2	(q) K_α
(c) Line -3	(r) L_α
(d) Line-4	(s) K_β

Ans. **[a → s; b → q; c → p; d → r]**

Assertion and reasoning

Some questions (Assertion–Reason type) are given below. Each question contains STATEMENT – 1 (Assertion) and STATEMENT – 2 (Reason). Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct. So select the correct choice:

Choices are:

(a)Statement – 1 is True, Statement – 2 is True; Statement – 2 **is** a correct explanation for Statement – 1.
(b)Statement – 1 is True, Statement – 2 is True; Statement – 2 is NOT a correct explanation for Statement – 1.
(c)Statement – 1 is True, Statement – 2 is False.
(d)Statement – 1 is False, Statement – 2 is true.

Q.19 **Statement – 1:** Wavelength of continuous X– rays depends of atomic number of the atom of target material.
Because
Statement – 2: It is produced due to collision of electrons with the atoms of target atom.

Ans. **[d]**

Q.20 **Statement – 1:** It is necessary to keep high vacuum in Coolidge tube to produce X- rays.
Because

Statement – 2: High vacuum is kept in Coolidge tube so that the electron emitted from the filament of the tube may not lose their energy in colliding with the atom of the gas in the tube.

Ans. **[a]**

Q.21 **Statement – 1:** Elements with high atomic numbers are suitable targets for producing X–rays.
Because
Statement – 2: Energy of the atom increase with increase in atomic number.

Ans. **[c]**

Q.22 **Statement – 1:** The frequency of K_β X–radiation is greater than K_α for a given target material.
Because
Statement – 2: K_α radiation is produced when an electron from n = 2 jumps into the vacancy in n = 1 orbit; whereas in K_β radiation the transition takes place from n = 3 to n = 1.

Ans. **[a]**

Q.23 **Statement – 1:** In a hydrogen atom, energy of emitted photon corresponding to transition from n = 2 to n = 1 is much greater as compared to transition from n = ∞ to n = 2.
Because
Statement – 2: Wavelength of photon is inversely proportional to the energy of emitted photon.

Ans. **[a]**

Subjective type

Q.24 The battery connected across a Coolidge tube operates at a power level of 100W, when the no. of electrons hitting the target in 1 second is 12.25×10^{10}. Find the minimum wavelength in the resulting X-rays spectrum.

Ans. **[24800 Å]**

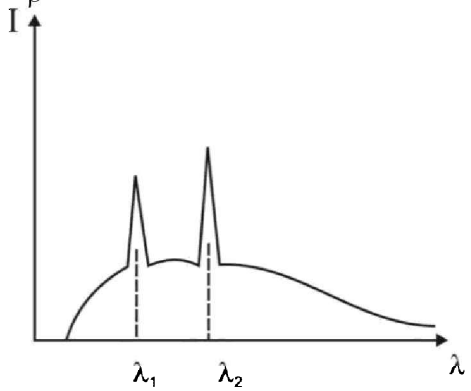
Q.25 Calculate the wavelength of the emitted characteristic X-ray from a manganese (Z = 25) target when an electron drops from an M shell to a vacancy in the K shell.

Ans. **[λ = 1.79Å]**

Q.26 In a Coolidge tube, the tungsten (Z = 74) target is bombarded by electrons. What is the minimum value of the accelerating potential to enable emission of characteristic K_α and K_β lines of tungsten? (The K, L & M levels of tungsten have Binding energies of 69.5, 11.3 & 2.30 KeV respectively.)

Ans. **[69.5kV]**

Q.27 Find which is K_α and K_β.

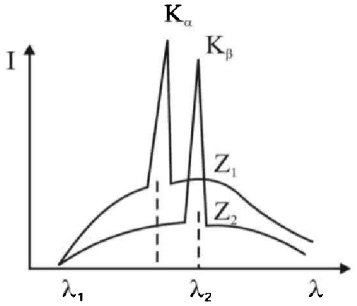


Ans. **[line 1 is K_β and 2 is K_α.]**

Q.28 An X-ray tube operated at 40 kV emits a continuous X – ray spectrum with a short wavelength limit $\lambda_{min} = 0.930\text{Å}$. Calculate the Planck's constant.

Ans. **[79.36×10^{-34}J-s]**

Q.29 Find in Z_1 and Z_2 which one is greater?



Ans. **[$Z_1 > Z_2$]**

Q.30 A cobalt target is bombarded with electrons and the wavelength of its characteristic spectrum are measured. A second, fainter, characteristic spectrum is also found because of an impurity in the target. The wavelength of the K_a lines are 178.9 pm (cobalt) and 143.5 pm (impurity). What is the impurity?

Ans. **[Z_x = 30.0, Hence, the impurity is zinc]**

Q.31 An X-ray tube with a copper target is found to be emitting lines other than those due to copper. The K_α line of copper is known to have a wavelength 2.5405 Å and the other two K_α lines observed have wave lengths 1.709 Å and 2.6578 Å. Identify the impurities.

Ans. **[Z_2 = 30, Hence the impurity is Zinc]**

Q.32 Find the constants a and b in Moseley's equation $\sqrt{v} = a(Z-b)$ from the following data.

Element	**Wavelength of K_α X – ray**	**Z(Atomic number)**
Mo	71 pm	42
Co	178.5 pm	27

Ans. **[$a = 5.0 \times 10^7$, b = 1.137]**

Q.33 When the voltage applied to X- ray tube increased from V_1=10kV to V_2 = 20kV, the wavelength interval between the K_α line and the short wave cut off of the continuous X-ray spectrum increases by a factor of 3. Find the atomic number of the element of which target of the tube is made.

Ans. **[Z = 29]**

04 NUCLEAR PHYSICS

4.1 Introduction

In the beginning of 20th century, it was established that there exists a small particle called **atom**. These atoms were known to be the building blocks of all the materials. People were curious to know that if everything is made up of atoms then what is an atom made up of? This query was resolved later on by various experiments and it was discovered that atoms are made up of tiny particles called **subatomic particles**. Scientists around the world started research on these subatomic particles and they discovered three subatomic particles: electrons, protons and neutrons. Neutrons and protons together constitute a nucleus. Study of nucleus led us to generate electricity in nuclear power plants. On the other hand, its study also led us to develop nuclear weapons. Let's begin the study of nucleus with some important terms.

Some Important Terms

- **Constituents of Nucleus -** Nucleus is made up of neutrons and protons. They are collectively called **nucleons**.

  Nucleons = Neutrons + Protons

- **Mass of Nucleus (M) -** Mass of nucleus is equal to the sum of the masses of protons and neutrons.

  Mass of proton (m_p) = 1,007,276,466 u

  $= 1.672621278 \times 10^{-27}$ kg

  Mass of neutron (m_n) = 1,008,664,915 u

  $= 1.674927351 \times 10^{-27}$ kg

  $\mathbf{M = m_p + m_n}$

We can see that the mass of a neutron is slightly greater than the mass of a proton. But sometimes they can be taken as equal as the difference is very small.

Unit:

Kilogram is not a very convenient unit to measure such small quantities. So, we use another unit called atomic mass unit (amu) represented by symbol 'u'. One amu is defined as–

$$1u = \frac{\text{mass of one } {}^{12}C \text{ atom}}{12} = \frac{1.992647 \times 10^{-26} \text{kg}}{12} = 1.660539 \times 10^{-27} \text{ kg}$$

- **Discovery of Neutron -** In year 1932, James Chadwick conducted an experiment. In his experiment, he bombarded helium on Beryllium. Due to this, some neutral particles were emitted from Beryllium. He named these particles as **Neutrons.** For his work, **James Chadwick** was awarded the **Nobel Prize**.
- **Atomic Number -** For any nucleus, we define a term called atomic number represented by 'Z'. Atomic number is the number of protons present inside the nucleus. If any nucleus has 5 protons then for that nucleus Z = 5.

  Z= No. of protons present inside the nucleus
- **Mass Number -** It is represented by 'A'. Mass number is the sum of the number of protons and the number of neutrons in any nucleus.

  M = No. of protons + No. of neutrons
- For an element (say X), we represent mass number and atomic number as shown below–

$${}^{A}_{Z}X$$

For example, helium atom has mass number 4 and atomic number 2, so we write helium as–

$${}^{4}_{2}He$$

- **Isotopes –** Elements which have same atomic no. but different mass no. are called isotopes. Consider two elements, hydrogen (H) and deuterium (D). They are represented as

$${}^{1}_{1}H \qquad {}^{2}_{1}D$$

Note: atomic numbers of H and D is same but their mass numbers are different. Such pairs of elements are called isotopes.

- **Isobars –** Elements which have same mass number and different atomic number are called isobars. Consider two elements, tritium (^{3}H) and helium (He). Tritium is an isotope of H which has mass number 3. We can represent them symbolically as–

$${}^{3}_{1}H \qquad {}^{3}_{2}He$$

Note: These two elements have different atomic number but they have same mass number. Such pairs of elements are called isobars.

- **Isotones –**Elements having same number of neutrons are called isotones. Consider two elements, mercury (Hg) and gold (Au). Their symbolic representation will be –

$${}^{198}_{80}Hg \qquad {}^{197}_{79}Au$$

Note: atomic number and mass number of these two elements are different. Let's check the number of neutrons in them –

n_n in Hg = 198 – 80 = 118

n_n in Au = 197 – 79 = 118

As they have same number of nucleons, they are isotones.

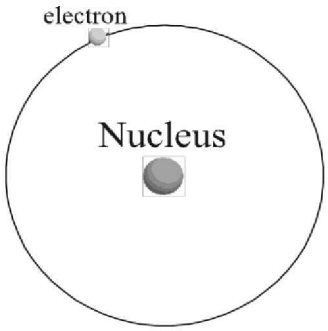
- **Stability of neutrons and protons -** Consider an isolated proton i.e. left free in space. Even in this state a proton can exist and remain stable. But neutrons cannot exist without a nucleus. They become unstable and break down according to this reaction –

$$n_{(free)} \longrightarrow p^{+} + e^{-} + \bar{\nu}$$

We will study and understand this equation later in this chapter.

4.2 Size of Nucleus

Size of nucleus is very small in comparison to the size of an atom. If we assume atom to be of the size of a cricket ground, then nucleus is even smaller than a cricket ball. Nucleus was discovered by Rutherford in his gold foil experiment. During his experiment, Rutherford guessed that at centre of an atom there is a very tiny heavy particle and named it nucleus.



Atom

From the data collected from gold foil experiment, Rutherford calculated the size of the nucleus but it was not very accurate. Later scientists performed different experiments to calculate its size. In their experiments, scientists bombarded fast moving electrons on various elements and the data collected from these experiments helped to find the size of nucleus more accurately.
It was found out that radius of nucleus is given as–

$$R = R_0 A^{\frac{1}{3}}$$

Here, R_0 is a constant whose value is 1.2 fm and A is mass number.

fm stands for femtometre (1 m = 10^{-15} fm).

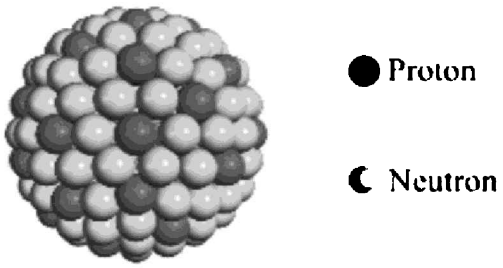
Density of nucleus (ρ) $= \dfrac{\text{mass}}{\text{volume}} = \dfrac{A}{\frac{4}{3}\pi R^3} = \dfrac{A}{\frac{4}{3}\pi R_0^3 A} = \dfrac{3}{4\pi R_0^3} = 2.4\times10^{17}\ \text{kg/m}^3$

Key Points –

- ρ is a constant, independent of A. For nucleus of any element, ρ is the same.
- Value of ρ is very high. It means protons and neutrons inside a nucleus are very tightly packed.

4.3 Nuclear Force

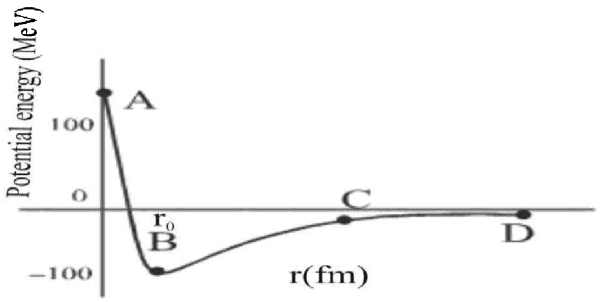
We have seen that inside the nucleus, there are some protons and some neutrons.



Protons in a nucleus will exert electrostatic repulsive force on other protons present inside the nucleus. If they all exert repelling force on each other, it means they should move away and nucleus should break apart. But this does not happen. Can you guess the reason?
Actually, there is another force that binds the nucleus. This force is called **nuclear force**. Consider two nucleons, separated by distance 'r'. Nuclear force will act between them. In mechanics, we have studied that conservative force is associated with some potential energy which is given by relation –

$$F = -\frac{dU}{dr} \qquad \text{... (1)}$$

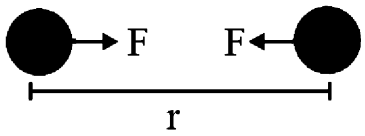
Where F is the force and U is the potential energy.
The following graph shows potential energy due to force between two nucleons.



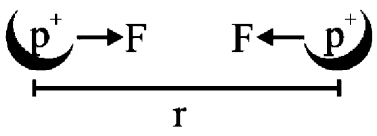
In the above graph, three points A, B, C and D are marked. We can see that from point A to B, potential energy is decreasing and from point B to C it is increasing. When PE is decreasing, it means force between two nucleons is repulsive and when PE is increasing it means force between them is attractive.

From the graph we can make some observations.

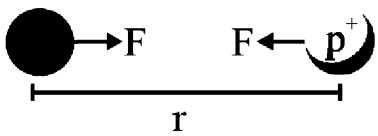
1. When $r < r_0$, nuclear forces are **repulsive**.
   When $r > r_0$, nuclear forces are **attractive**.
2. Magnitude of nuclear force is given by magnitude of slope of this graph. In the graph, slope between point B and C is more, it means attractive forces are more in this region and slope between points C and D is close to zero or very less, meaning attractive forces are very less in this reason.
   We can see that nuclear forces come into existence only for distance r_A to r_C. For distances beyond r_C, nucleus forces are negligible. From this we can conclude that nuclear forces are short-ranged forces.
3. Nuclear forces do not distinguish neutrons and protons. Suppose two neutrons are placed apart by distance 'r'. At this distance they experience an attractive force (Say F).



Now suppose we replace neutrons with protons. Then also they will experience the same attractive force F.



Now suppose we replace one proton with one neutron, then again, there is the same attractive force F.



Inside the nucleus, there is electrostatic repulsive force acting among protons but at the same time, attractive nuclear force is also acting among all neutrons and protons. The effect of attractive nuclear

force is much greater than electrostatic repulsive force. Hence there exists net attractive force inside the nucleus so nucleons remain bound inside it.

4.4 Binding Energy

Binding energy is the energy required to move all the constituents of the nucleus far apart. Here we define a term called 'binding energy per nucleon'.

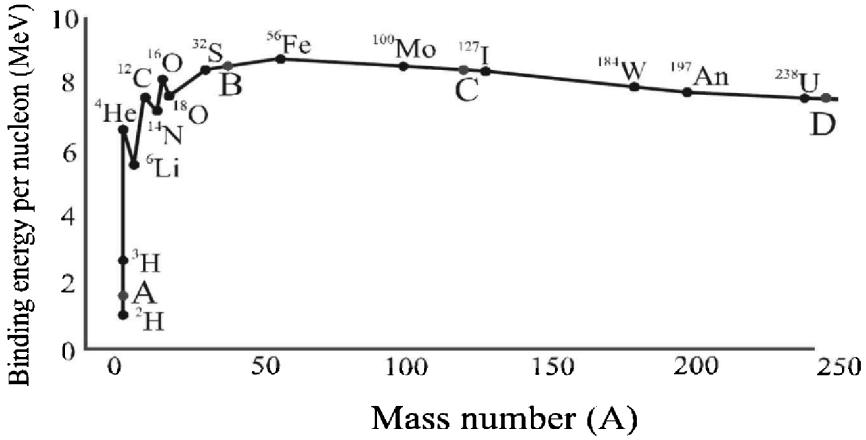
$$\textbf{BE per nucleon} = \frac{\text{Binding energy}}{\text{mass number}} = \frac{\text{BE}}{\text{A}}$$

Significance of Binding Energy per Nucleon

Binding energy is the energy required to separate all the nucleons apart from each other. Binding energy per nucleon is the amount of energy required to separate one nucleon from the nucleus.

Graph

Consider the following graph of binding energy per nucleon versus mass number.



As we move along the x-axis, mass number increases which means that size of an atom is increasing. We can divide this graph into **three regions:**

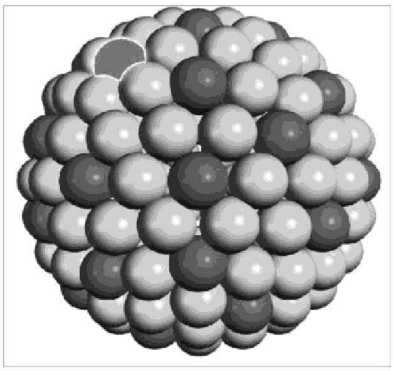
- **From point A to B,**
- **From point B to C**
- **From point C to D.**

Let's analyse each region separately.

Region A to B – The curve in this region is very steep. To understand this nature of the curve, consider a small nucleus which has two nucleons. If we want to extract any one nucleon, it is easy to extract it as there is only one nucleon which is applying attractive force on it. Now suppose we have a slightly bigger nucleus, say nucleus of five nucleons. Now, if we try to extract any one nucleon, it is tough as it is experiencing attractive force from other four nucleons and they will try to stop it. It means we have to apply more energy to extract this nucleon. Similarly, if we have a much bigger nucleus then more energy will be required to extract the nucleon out of it. So we can see that as the mass number increases, binding energy per nucleon also increases. In the graph, you can see some discrepancies but there is a general increasing trend in region A to B. The explanation of these irregularities is beyond the scope of this book.

Region B to C - In this region we can see that the curve is almost parallel to the x-axis or we can say that the curve is flat. It means even though mass number of atoms are increasing but still binding energy per nucleon is almost constant. Let's understand why?

Consider the nucleus as shown in the figure.



When we try to extract the highlighted nucleon as shown in the figure in blue all the other nucleons present inside the nucleus attract the nucleon and thus we have to supply a lot of energy in order to extract this nucleon. Now we have studied that nuclear forces are short-ranged forces. If two nucleons are far there is no attractive force between them. As size of nucleus increases, distance between the nucleon that we want to extract and the nucleon present at the other end of the nucleus increases. This distance becomes so large that nuclear forces due to this nucleon cease to exist. Thereafter, no matter what the size of the nucleus is, attraction force remains same.

Region C to D- In this region, there is slow decrease in the binding energy per nucleon as mass number increases.

We have seen that as the size of nucleus increases beyond a certain limit then attractive forces for any nucleon becomes constant. We know that nuclear forces are short-ranged forces, but electrostatic forces are long-ranged forces, i.e. they can act even at longer distances. Since the mass number of nucleus is increasing, therefore numbers of protons in the nucleus are also increasing. Suppose we want to extract a proton out of this nucleus, it will experience a constant attractive force due to nucleons present close to it, but this proton will also experience a repulsive electrostatic force due to protons present at the other end of the nucleus. Thus net attractive force decreases.

4.5 Mass Defect

Suppose there are four nucleons separated by large distances. Now we bring all the four nucleons close to each other and combine them to form a nucleus. Initially, the combined mass of all the four nucleons was m_1. When we combined them, then the mass of the nucleus was found to be m_2. Now what do you think, what is the relation between m_1 and m_2? Is m_1 greater than m_2 or m_2 greater than m_1 or both of them are equal?

Logically, it seems that m_1 should be equal to m_2. But actual calculations based on various experiments show that m_1 is greater than m_2.Surprised!!! Let's see how?

Reason

The reason behind this mass defect was given by Albert Einstein with his popular equation –

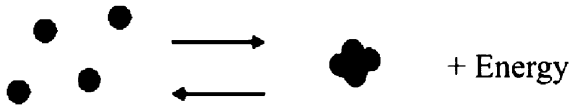
$$E = mc^2$$

Einstein said that there is defect in mass as some of the mass gets converted into energy. When we combine these four nucleons to create a nucleus some energy is released. This energy has come from the mass which has converted into energy. This is the reason why the mass of combined nucleons decreases.

The total energy released in this case will be –

$$E = (m_1 - m_2)\, c^2$$

If we are saying that mass is converted into energy then from the similar concept we can also say that energy can also be converted into mass. According to Einstein, mass and energy can be convertible into each other. It means that if we break a nucleus into four nucleons then it can be done by supplying some energy to the nucleus. This energy will be converted into mass.



Isolated Nuclei Nucleus

This difference in masses of isolated nucleons and nucleus is called **mass defect**. In this case, mass defect is $(m_1 - m_2)$.



Quiz-1

1kg water is given heat of 100 J. Calculate the increase in mass of water.

Sol. According to Einstein, as we are supplying energy to water, some energy will be converted into .mass. Let's calculate

$E = \Delta mc^2$

$\Rightarrow \quad 100 = \Delta m\,(3\times10^8)^2$

$\Rightarrow \quad \Delta m = \dfrac{100}{9\times10^{16}}\,kg = 0.1\times10^{-14}kg.$

From this problem we can see that even in our day-to-day activities, such as heating water, we convert energy to mass and vice-versa. But the amount of mass that we convert or we obtain is so small that it is very difficult to calculate or measure. So usually we neglect these changes in mass. But in case of nucleons, we have to consider the mass defect in our calculations because the mass of nucleons is already so small that we cannot neglect these small changes.



Quiz-2

The equation given below shows that two protons and two neutrons combine to give a helium atom.

$$2p^+ + 2n \longrightarrow {}^4_2He$$

In the above process, is energy released or supplied?

Sol. We can see that initially system of $2p^+$ and 2n was not bound. Finally, they are bound to make a He atom. We have seen that if the system is bound then its mass reduces. As mass has reduced, it means reduced mass has converted into energy. So, energy will be released in this process.
This is a very important concept. ***If a system is more bound, mass is lesser.***



Quiz-3

The equation given below shows that a nucleus is broken into two smaller nuclei.

$$\mathbf{{}^{120}_{50}N \longrightarrow 50p^+ + 70n}$$

Is energy released or supplied in above process?

Sol. We can see that initial system was bound, but after breaking, it is no more bound so some

mass must have increased. This extra mass has come from the energy that we supplied to the nucleus. So in this process energy will supplied.

Illustration 1

Consider the following equation

$$2p^+ + 1n \longrightarrow {}^3_2He$$

$$m_{He} = 2.017 \text{ amu}$$

$$m_p = 1.008 \text{ amu}$$

$$m_n = 1.009 \text{ amu}$$

Find the energy released.

Sol. The system was initially not bound, but finally it becomes bound. It means, its mass has reduced and this reduced mass has converted into energy.

To calculate this energy, first we will find the mass defect.

$\Delta m = (2m_p + m_n) - (m_{He})$

$$= 2.016 + 1.009 - 2.017 = 1.008 \text{ amu} = 1.008 \text{ x } 1.67 \times 10^{-27} \text{ kg}$$

So, energy released will be –

$$E = \Delta mc^2 = (1.008 \times 1.67 \times 10^{-27} \text{ kg})\, c^2$$

$$\Rightarrow E = (1.008) \times 1.67 \times 10^{-27} \times 9 \times 10^{16} = 15.15 \times 10^{-11} \text{ J}$$

But in nuclear physics, we usually represent energy in terms of electron volt. To convert joules into eV, we will divide our result with 1.6×10^{-19}.

So we get,

$$E = 9.46 \times 10^8 \text{ eV}$$

Key Points –

Note that there was too much calculation in the last example. Mass was given in amu so we had to convert it to kg. Then we did multiplication by c^2. Then to give answer in eV, we divided by 1.6×10^{-19}. Such questions are common so we would like to avoid these calculations. Instead of doing this long calculation, we can just find Δm in amu and multiply it by 931.5×10^6 eV/u to find energy in eV.

$E = \Delta m\, (931.5 \times 10^6)$ eV where Δm is in amu.

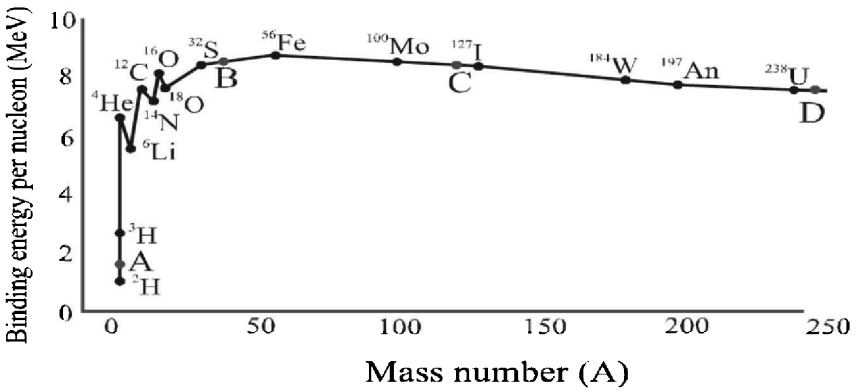
Illustration 2

A nucleus is made up of 10 nucleons. Binding energy per nucleon is given 5eV. Now this nucleus is broken into two different nuclei each with five nucleons. The binding energy per nucleon for both the nucleus is given 6eV. Is energy liberated or supplied in this entire process?

Sol. We can see that the nuclei that have been made after breaking up of a bigger nucleus have more binding energy per nucleon. It means, the final smaller nuclei are more bound. It means, mass has decreased. This decreased mass is converted into energy. Thus we can say that energy is liberated in this process

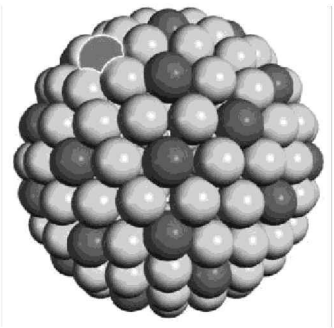
4.6 Radioactive Decay

Again consider the graph of binding energy per nucleon and mass number.



In this graph, we have seen that in region C to D, binding energy reduces slowly. We all have seen the reason behind this.

Consider the nucleus as shown in the figure below.



In this figure, the nucleon which has been highlighted in blue will experience an attractive force due to nearby nucleons. Now suppose that this nucleon is a proton. Then it will also experience electrostatic repulsive force due to nucleons present throughout the nucleus. The net force acting on this nucleon is *weak* attractive force. That's why though the nucleon is bound to the nucleus, it has tendency to leave the nucleus easily. That's the case with many nucleons on periphery of nucleus. Therefore nuclei having large mass number are unstable.

Out of all the known elements, nuclei of **82 elements** have been labelled as **unstable.** For example, uranium, thorium, radium etc. These big nuclei have tendency to break down into smaller stable nuclei.

This process of breaking down of a heavy, unstable nucleus into smaller stable nuclei is called **radioactive decay**.

Radioactive decay can take place in various forms. Let's see some important forms of radioactive decay.

1. α- decay

Positively charged helium ion is called α-particle.

Consider the two equations.

$$^{238}_{92}\mathrm{U} \longrightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$$

$$^{210}_{84}\mathrm{P_0} \longrightarrow {}^{206}_{82}\mathrm{Pb} + {}^{4}_{2}\mathrm{He}$$

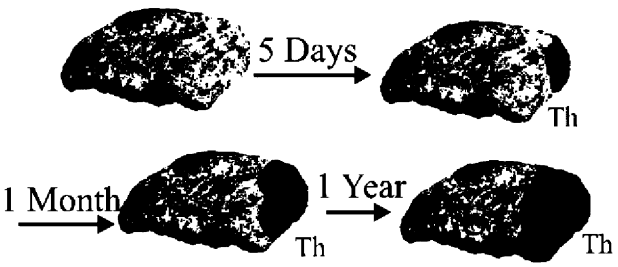
In the above equation 'He' particles are called **α-particles.**

These types of radioactive decays in which α-particles are one of the products are **called α-decay.**

General α-decay equation for an element 'X' is–

$$^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$$

In the above example we can see that uranium nucleus decays and converts into thorium and α-particle. But this process of conversion is extremely slow. If 1kg of uranium is kept, it will not convert into thorium and α- particles within few seconds, it will take hundreds of years.



Q-value /Q- factor - Nuclear reactions are those reactions which involve nucleus. α-decay is an example of nuclear reaction. In a nuclear reaction, we talk about Q-value /Q-factor.
Q-value is defined as –

$$Q = (m_i - m_f)\, c^2$$

Here, m_i is initial mass of nuclei

m_f is final mass of decayed nuclei

c is the speed of light

For decay equation of uranium we can write Q factor as –

$$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

$$Q = [m_U - (m_{Th} + m_{He})]\, c^2$$

Q factor tells us the amount of mass that has been converted into energy or amount of energy converted into mass.

Key Points –

- If the value of Q factor is positive, it means $m_i > m_f$. Thus mass has been converted into energy. So, we can call these reactions **as exothermic reactions**. These reactions are spontaneous, i.e. they are initiated by themselves.
- If the value of Q is negative, it means $m_i < m_f$. Mass has increased in the final product. It means we have given some energy, so we call it an **endothermic reaction**. These reactions are non-spontaneous, i.e. they cannot be initiated by themselves. Some energy has to be provided in order to start the reaction.

Illustration 3

You are given following atomic masses–

$^{238}_{92}$**U = 238.050 u,** $^{4}_{2}$**He = 4.0026 u**

$^{234}_{90}$**Th = 234.0436 u,** $^{1}_{1}$**H = 1.00783 u**

$^{237}_{91}$**Pa = 237.05121 u**

(i) Calculate the energy released during the alpha decay of $^{238}_{92}$**U .**

Sol. We know the equation of α-decay of Uranium–

$$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

Energy released will be equal to Q factor. So

$$Q = (m_i - m_f)\, c^2$$
$$= [m_U - (m_{Th} + m_{He})]\, c^2$$
$$= (0.00456\ u)\ (931.5 \times 10^6\ eV/u)$$
$$= 4.25\ MeV$$

So, energy released will be 4.25 MeV

(ii) Show that uranium cannot emit a proton spontaneously.

Sol .First we will write an equation for which a photon is emitted.

$$^{238}_{92}U \longrightarrow {}^{237}_{91}Pa + {}^{1}_{1}H$$

Now calculate m_i and m_f for this equation.

$m_i = m_U = 238.050$ u

$m_f = m_H + m_{Pa} = 238.0590$ u

We can see that $m_f > m_i$ which means Q factor is negative. Thus the process is non-spontaneous.

Illustration 4

A nucleus X of mass m is at rest. Another nucleus Y of mass m travelling with some kinetic energy collides with X. After collision a new nucleus Z is formed.
This reaction is written as –

$$X + Y \longrightarrow Z$$

It is given that this reaction is endothermic and Q = 10eV. Calculate minimum KE of Y for the reaction to proceed.

Sol. This reaction is endothermic so for this reaction to proceed, 10 eV of energy must be supplied. In this collision, there will be some loss in KE and this loss in KE will fuel this reaction. Maximum KE that can be lost in a collision is given by $\frac{1}{2}\mu v_{relative}^2$

In this case, $\mu = \frac{m \times m}{m + m} = \frac{m}{2}$

Suppose nucleus Y is coming with velocity v and as X is at rest so $v_{relative} = v$.

$$\Rightarrow \quad KE_{max} = \frac{1}{2}\,\frac{m}{2}\,v^2 = \frac{1}{2}KE_Y$$

It means if Y is coming with kinetic energy KE_Y, only half of it will be lost in the collision.

To process this reaction there must be at least 10 eV of energy.

$$\Rightarrow \quad \frac{1}{2}KE_Y \geq 10\,eV$$

$$\Rightarrow \quad KE_Y \geq 20\ eV$$

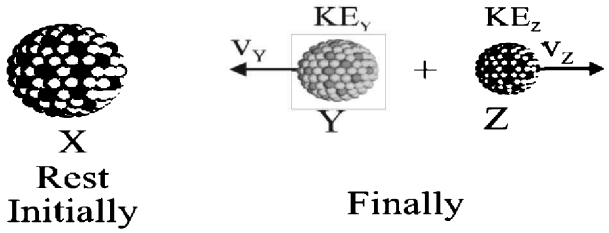
Illustration 5

Consider the following reaction–

$$X \rightarrow Y + Z$$
$$m_X \quad m_Y \quad m_Z$$

Q factor for this reaction is positive. X was initially at rest. When it breaks into Y and Z, the energy produced becomes kinetic energy of Y and Z. Calculate this KE of Y and Z.

Sol.



In this entire process momentum will be conserved. So, Y and Z will move opposite to each other. Let speed of Y and Z be v_Y and v_Z.

From conservation of momentum:

$$m_Y v_Y = m_Z v_Z \quad \text{... (1)}$$

It is given that all the energy released in this reaction has become KE of X and Y, so

$$\frac{1}{2}m_Y v_Y^2 + \frac{1}{2}m_Z v_Z^2 = Q \quad \text{... (2)}$$

On solving equation (1) and (2), we will get the value of KE_Z and KE_Y as–

$$KE_Z = \frac{Q}{1+\frac{m_Z}{m_Y}}$$

$$KE_Y = \frac{Q}{1+\frac{m_Y}{m_Z}}$$

Key Points –

If nucleus X is at rest and it breaks into two different nuclei Y and Z then kinetic energy of Y and Z can be calculated using these formulas.

$$KE_Z = \frac{Q}{1+\frac{m_Z}{m_Y}} \quad \text{and} \quad KE_Y = \frac{Q}{1+\frac{m_Y}{m_Z}}$$

2. β **decay**

It is of two types, β^+ and β^-. First we will see β^- decay.

- β^- **decay** – In β^- decay, a neutron breaks into a proton, an electron and one more particle called antineutrino $(\bar{\nu})$

$$n \rightarrow p^+ + e^- + \bar{\nu}$$

General equation of β^- decay can be written as

$$^{A}_{Z}X \longrightarrow ^{A}_{Z+1}Y + e^- + \bar{\nu}$$

Here, nucleus X has many neutrons. One of these neutrons breaks down and one proton is created. Therefore, atomic number of this new element (Y) so formed will increase by one but mass number (A) remains same.

For example,

$$^{29}_{13}Al \longrightarrow ^{29}_{14}Si + e^- + \bar{\nu}$$

Let's talk a little bit about antineutrinos ($\bar{\nu}$). They are very complicated particles and still scientists are trying to understand them in a better way. They can penetrate through any substance and hence cannot be captured. Millions of antineutrinos come to earth from sun every day but they can penetrate

the entire earth and escape. There must be so many antineutrinos present in the room you are sitting and penetrating your body. As they cannot be captured, it becomes very difficult to study them. One thing which is known about them is that ***there mass is very-very small***. It can be taken as zero.

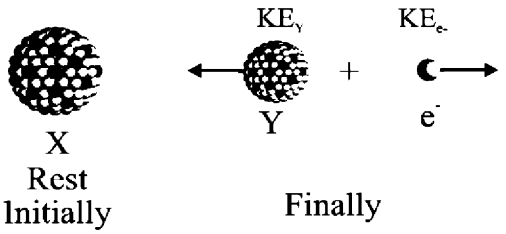
4.7 Discovery of antineutrino

Suppose an element X breaks down according to β^- decay. Its equation will be –

$$X \longrightarrow Y + e^-$$

Here we have skipped writing antineutrino as earlier scientists used to think that β^- decay of any matter X will give Y and e^- only.

We can calculate Q-factor for this equation. Initially, X was at rest. It breaks down and creates Y and e^-. Energy that is released during this reaction will become KE of Y and e^-.



Kinetic energy of this e^- and Y can be calculated using formula that we derived in previous part.

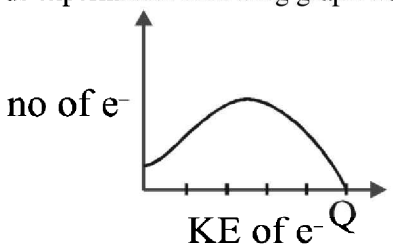
$$KE_{e^-} = \frac{Q}{\left(1+\frac{m_e}{m_Y}\right)}$$

Now $m_Y >> m_e$, so $\frac{m_e}{m_y} \cong 0$

$$\Rightarrow \qquad KE_{e^-} = Q$$

So, almost all the energy that is released in this reaction goes to e^- in the form of KE. Energy obtained by Y is approximately zero. So we can conclude that e^- emitted during β^- decay of any particle has an energy equivalent to Q of the reaction.

This is just a theory. When actual experiments were conducted, then result was something different. With the data collected from various experiments following graph was drawn.



From the above graph, we can see that number of electrons having KE equal to Q were very less. Most of the e^- had KE less than Q. For decades, scientists were not able to find where the remaining energy was going. But later after many more experiments, it was discovered that there is another particle emitted in β^- decay i.e. $\bar{\nu}$. It is this $\bar{\nu}$, which takes away some energy and thus electrons are left with lesser amount of energy.

- **β^+ decay** – This is just opposite of β^- decay. Here a proton decays to produce a neutron, a positron (or anti-electron) and neutrino.

$$p^+ \rightarrow n + e^+ + \nu$$

Mass and charge of positron are equal to that of electron. But charges on electron and positron are opposite in sign. General equation of β^+ decay is given as –

$$^{A}_{Z}X \longrightarrow ^{A}_{Z-1}Y + e^+ + \nu$$

For example,

$$^{64}Cu \longrightarrow ^{64}Ni + e^+ + \nu$$

We have seen that in β^+ decay, a positron is emitted and in β^- decay, an electron is emitted. Other than that they are same. All the things that we studied about β^- decay are also valid for β^+ decay.

Q-Factor for β^- decay –

General equation of β^- decay is –

$$\underset{m_X}{X} \longrightarrow \underset{m_Y}{Y} + \underset{m_{e^-}}{e^-} + \bar{\nu}$$

We have seen formula to calculate Q–factor.

So,

$$Q = \left[m_X - \left(m_Y + m_{e^-}\right)\right]c^2$$

But in case of β^- decay, it is wrong. Actually, Q factor for β^- decay is given as –

$$Q = [m_X - m_Y]\, c^2$$

Let's see why. We know that in β^- decay, one neutron in nucleus of element X breaks down into a proton and an electron.

$$n \longrightarrow p^+ + e^- + \bar{\nu}$$

This e^- so created has high KE, so it moves out of the system. But protons will remain inside the nucleus. As X has got one additional proton, it will change into Y but this Y is not neutral. As electron has escaped, Y has one electron less and thus having positive charge. So the actual reaction of β^- decay can be written as–

$$X \longrightarrow Y^+ + e^- + \bar{\nu}$$

We can write Q factor for this equation –

$$Q = \left[m_X - \left(m_{Y^+} + m_{e^-}\right)\right]c^2$$

But in the general equation of β^- decay we have mentioned that Y is neutral. Actually, this Y^+ ion gets an e^- from external sources.

$$\underset{\text{Y-ion}}{Y^+} + \underset{\text{electron from external source}}{e^-} = Y$$

As, $\quad m_{Y^+} + m_{e^-} = m_Y$

$\Rightarrow \quad \boldsymbol{Q = [m_X - m_Y]c^2}$

Q factor for β^+ decay –

General equation for β^+ decay is –

$$X \longrightarrow Y + e^+ + \nu$$

Q factor for this equation will be –

$$Q = \left[m_X - \left(m_Y + 2m_e \right) \right] c^2$$

We know that in β^+ decay, a proton breaks into neutron, positron and neutrino.

$$p^+ \longrightarrow n + e^+ + \nu$$

The element Y has one proton less than X but has same number of electrons in orbits. It means, the new element so created is in form of Y^-. So, exact equation of β^+ decay should be –

$$X \longrightarrow Y^- + e^+ + \nu$$

So, Q-factor for β^+ decay will be –

$$Q = \left[m_X - \left(m_{Y^-} + m_e \right) \right] c^2$$

As, $m_{Y^-} = m_Y + m_e$

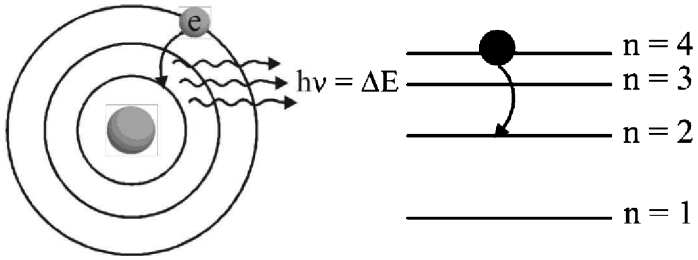
$$\Rightarrow \quad Q = \left[m_X - \left(m_Y + m_e + m_e \right) \right] c^2$$

$$\Rightarrow \quad \boldsymbol{Q = \left[m_X - \left(m_Y + 2m_e \right) \right] c^2}$$

3. Gamma decay (γ- decay)

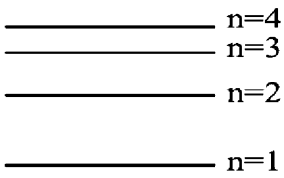
Before we start with γ decay, first we will recall one concept of atomic structure. We studied that when e^- jumps from a higher orbit to lower orbit than its energy reduces and photon is released. The energy of this photon (hν) is equal to energy gap between the two orbits.

$$\Delta E = h\nu$$



Atom and its orbit Energy Diagram

In nucleus, something similar happens. There are different energy levels inside nucleus.



Energy levels of nucleus (inside nucleus)

Detailed description about these energy levels inside the nucleus is beyond the scope of this book. When a nucleon jumps from higher to a lower energy level, its energy reduces and this reduced energy is released in the form of photons. Here also the energies of photons are equal to energy gaps between the two states. This phenomenon is called **gamma decay**.

Equation of the photon that is emitted from nucleus will be –

$$\Delta E = h\nu$$

$$\Rightarrow \quad \nu = \frac{\Delta E}{h}$$

In the nucleus, energy gaps are very large. Due to this, frequencies of photons emitted are also large. These frequencies lie in the range of γ-ray spectrum.

General equation of γ-radiation is –

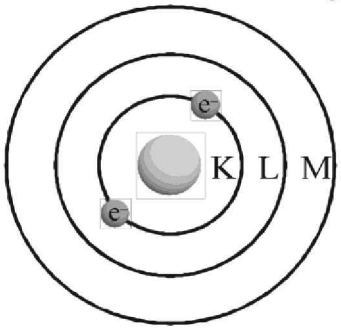
$$X^* \longrightarrow X$$

Here '*' sign represents that nucleons of element X are in excited state.

Note: *In case of α and β decay, elements are converted into other elements. This process is called* ***transmutation****. But in case of γ-decay there is no transmutation.*

4. K- Capture

Consider an atom and shells around its nucleus as shown in the figure.



In this atom, K shell is closest to the nucleus of an atom. Sometimes electrons present in K-shell are captured by protons in the nucleus. This proton and electron combine to produce a neutron.

General equation of this type of decay is –

$$_{Z}^{A}X \longrightarrow {}_{Z-1}^{A}Y$$

Here Z has reduced by one as one proton was converted into neutron.

4.8 Law of Radioactive Decay

In radioactive decay, we have seen that an unstable nucleus decays into different particles which have stable nuclei. But this process occurs over a period of time.

Now suppose that in any small time dt, total dN number of nuclei are decaying.

So, here we define $\frac{dN}{dt}$ as decay rate.

Decay rate is directly proportional to the number of nuclei (N). It means if there are more nuclei of uranium, their decay rate will be more. This seems logical.

So, $$\frac{dN}{dt} \propto N$$

$$\Rightarrow \quad \frac{dN}{dt} = -\lambda N$$

Here, λ is decay constant or disintegration constant. Negative sign shows that N is decreasing.

We can understand the physical significance of λ. If λ is more, decay will be fast. If λ is less, decay will be slow.

We can further solve this equation –

$$\frac{dN}{dt} = -\lambda N$$

$$\Rightarrow \quad \frac{dN}{N} = -\lambda dt$$

$$\Rightarrow \quad \int \frac{dN}{N} = -\lambda \int dt$$

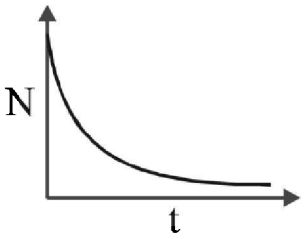
At time t = 0, suppose there were N_0 nuclei and at time t, N nuclei are left.

$$\Rightarrow \quad \int_{N_0}^{N} \frac{dN}{N} = -\lambda \int_{0}^{t} dt$$

$$\Rightarrow \quad \ell n \frac{N}{N_0} = -\lambda t$$

$$\Rightarrow \quad N = N_0 e^{-\lambda t}$$

This is the equation of law of radioactive decay. This equation tells us the number of nuclei of the radioactive unstable element (say of Uranium) left at any time t. We can plot a graph between N and t.





Quiz-4

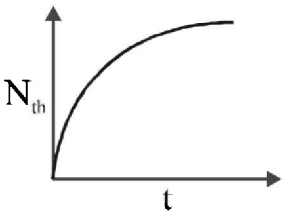
At any time t, calculate the number of nuclei of thorium produced in a decay process of uranium.

Sol. Number of Thorium nuclei produced = number of Uranium nuclei decayed

$N_{Th} = N_0 - N$

$N_{Th} = N_0 - N_0 e^{-\lambda t} = N_0 (1 - e^{-\lambda t})$

We can plot N_{Th} and t –



As time passes, number of thorium nuclei increases.

There are some important terms related to time and decay of nuclei. Let's see –

Half Life - Suppose, uranium is decaying and its nuclei are decreasing with time. Initially, number of nuclei of uranium was N_0. A time will come when they will reduce to half i.e. $N = \frac{N_0}{2}$. The time at which this happens is called half-life.

Let's calculate general equation for half-life –

$$N = N_0 e^{-\lambda t}$$

At time t_{HL}, $N = \frac{N_0}{2}$

$$\Rightarrow \quad \frac{N_0}{2} = N_0 e^{-\lambda t_{HL}}$$

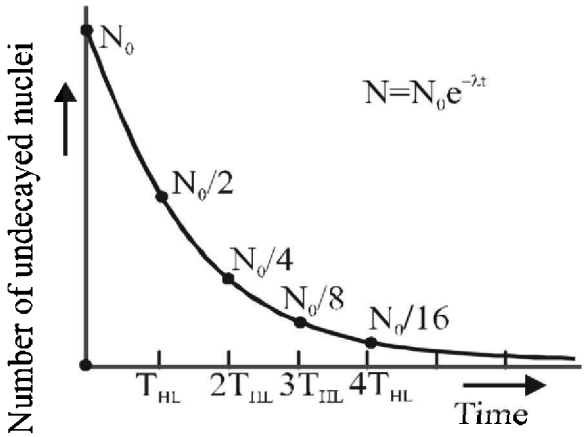
$$\Rightarrow \quad t_{HL} = \frac{\ell n 2}{\lambda}$$

We can understand this process in the following manner.

Suppose initially there were N_0 nuclei present. After $\frac{\ell n 2}{\lambda}$ time there will be $\frac{N_0}{2}$ nuclei. After next $\frac{\ell n 2}{\lambda}$ time there will $\frac{N_0}{4}$ nuclei. After next $\frac{\ell n 2}{\lambda}$ time there will $\frac{N_0}{8}$ nuclei. So on.

$$N_0 \xrightarrow[t_{HL}]{\frac{\ell n 2}{\lambda}} \frac{N_0}{2} \xrightarrow[2t_{HL}]{\frac{\ell n 2}{\lambda}} \frac{N_0}{4} \xrightarrow[3t_{HL}]{\frac{\ell n 2}{\lambda}} \frac{N_0}{8}$$

We can plot graph between the number of un-decayed nuclei and time as shown in the figure below.



Mean Life or Average Life - Suppose we have some uranium which is producing thorium by α-decay. The nucleons inside nuclei decay at different times. Say one nucleon decays in 5 sec, so its lifetime is 5 sec. Say, another nucleon decays in 10 days, so its lifetime is 10 days and so on. If we take the average of all these lifetimes of all nuclei, then it is called mean/average life. It is calculated as –

$$\textit{Mean /average life} = \frac{1}{\lambda}$$

Illustration 6

In a sample of old rock, ratio of thorium to uranium is 0.65. What is the age of rock? T_{HL} of U is 4.5×10^9 years. Assume that initially rock had only uranium.

Sol. It is given that –

$$\frac{N_{TH}}{N_U} = 0.65$$

Initially, rock had only uranium. Over a period of time, this uranium must have decayed to thorium. Let's assume that age of rock is t_0. Initially, when there was only U in rock, number of nuclei was N_0. After time t_0, number of nuclei of uranium left are–

$$N_U = N_0 e^{-\lambda t_0} \qquad \text{... (1)}$$

Numbers of Thorium nuclei created in rock by time t_0

$$N_{Th} = N_0\left(1-e^{-\lambda t_0}\right) \quad \text{... (2)}$$

On dividing equation (2) by (1) –

$$\frac{N_{Th}}{N_U} = \frac{N_0\left(1-e^{-\lambda t_0}\right)}{N_0 e^{-\lambda t_0}}$$

$$\Rightarrow \quad 0.65 = \frac{1-e^{-\lambda t_0}}{e^{-\lambda t_0}}$$

$$\Rightarrow \quad 1.65 = \frac{1}{e^{-\lambda t_0}} \quad \text{... (3)}$$

Here $\lambda = \frac{\ell n 2}{t_{HL}} = \frac{\ell n 2}{4.5\times 10^9}$

On substituting value of λ in equation (3), we get

$t_0 = 3.3 \times 10^9$ years

Illustration 7

A substance X undergoes α-decay and β-decay simultaneously by following equations.

$$X \xrightarrow{\alpha\text{-decay}} Y$$

$$X \xrightarrow{\beta\text{-decay}} Z$$

Decay constants for α and β decays are λ_1 and λ_2 respectively. Find the number of nuclei of X as a function of time. At t = 0, number of nuclei of X are N_0.

Sol. If there were only α-decay, we could have written –

$$\frac{dN}{dt} = -\lambda_1 N_0$$

If there were only β-decay, we could have written –

$$\frac{dN}{dt} = -\lambda_2 N_0$$

As both the decays are going on simultaneously so we will write –

$$\frac{dN}{dt} = \left(-\lambda_1 N_0\right) + \left(-\lambda_2 N_0\right) = -(\lambda_1 + \lambda_2) N_0$$

$$\Rightarrow \quad \int \frac{dN}{N_0} = \int -\left(\lambda_1 + \lambda_2\right) dt$$

At t = 0, there were N_0 nuclei and at time t, there are N nuclei.

$$\Rightarrow \quad \int_{N_0}^{N} \frac{dN}{N_0} = -\int_0^t \left(\lambda_1 + \lambda_2\right) dt$$

On integrating, we get –

$$\ell n \frac{N}{N_0} = -\left(\lambda_1 + \lambda_2\right) t$$

$$\Rightarrow \quad N = N_0 e^{-(\lambda_1+\lambda_2)t}$$

Illustration 8

A nuclear reactor produces radioactive substance X at a constant rate P. The decay constant of X is λ. Find the number of nuclei of X as a function of time.

Sol. Here, first some radioactive substance is decaying and producing X. Now this X is again decaying and producing some other substance say Y.

$$\text{Substance} \xrightarrow{P} X \longrightarrow Y$$

So the processes of production and decay of X are occurring simultaneously.

The rate at which X is being produced is P and the rate at which X is decaying is λN.

We can write –

$$\frac{dN}{dt} = P - \lambda N$$

$$\Rightarrow \quad \frac{dN}{P - \lambda N} = dt$$

$$\Rightarrow \quad \int \frac{dN}{\lambda N - P} = -\int dt$$

At t = 0, as no amount of X was produced, number of nuclei of X was zero.

At time t, there is N number of nuclei of X.

$$\Rightarrow \quad \int_0^N \frac{dN}{\lambda N - P} = -\int_0^t dt$$

On integrating we get –

$$\frac{1}{\lambda}\left[\ell n(\lambda N - P)\right]_0^N = -t$$

$$\Rightarrow \quad \ell n\left(\frac{\lambda N - P}{-P}\right) = -\lambda t$$

$$\Rightarrow \quad \frac{\lambda N - P}{-P} = e^{-\lambda t}$$

$$\Rightarrow \quad N = \frac{P - Pe^{-\lambda t}}{\lambda} = \frac{P}{\lambda}\left(1 - e^{-\lambda t}\right)$$

Illustration 9

Consider the following equation. All the required details are given.

$$\mathbf{X \xrightarrow{\lambda_1} Y \xrightarrow{\lambda_2} Z}$$

at t = 0	$\mathbf{N_0}$	**0**	**0**

Express the number of nuclei of X and Y present at any time t.

Sol. Suppose number of nuclei of X, Y and Z at any time t is N_X, N_Y and N_Z respectively.

For X, we can write –

$$\frac{dN_X}{dt} = -\lambda_1 N_X \qquad \text{... (1)}$$

We can see that element Y is being produced and decaying simultaneously. First we will write equation of rate at which it is producing.

$$\frac{dN_Y}{dt} = \lambda_1 N_X$$

Equation of rate at which it is decaying is –

$$\frac{dN_Y}{dt} = -\lambda_2 N_Y$$

So, overall equation of Y will be –

$$\frac{dN_Y}{dt} = \lambda_1 N_X - \lambda_2 N_Y \quad \text{... (2)}$$

For Z, we can write –

$$\frac{dN_Z}{dt} = \lambda_2 N_Y$$

Now we will solve each equation one by one. First solve equation (1).

$$\frac{dN_X}{dt} = -\lambda_1 N_X$$

$$\Rightarrow \quad \frac{dN_X}{N_X} = -\lambda_1 dt$$

$$\Rightarrow \quad \int \frac{dN_X}{N_X} = -\lambda_1 \int dt$$

At t = 0, there were N_0 nuclei of X and at time t, there are N_X nuclei of X.

$$\Rightarrow \quad \int_{N_0}^{N_x} \frac{dN_X}{N_X} = -\lambda_1 \int_0^t dt$$

$$\Rightarrow \quad \left[\ell n\, N_X\right]_{N_0}^{N_X} = -\lambda_1 t$$

$$\Rightarrow \quad \ell n \frac{N_X}{N_0} = -\lambda_1 t$$

$$\Rightarrow \quad N_X = N_0 e^{-\lambda_1 t} \quad \text{... (3)}$$

Equation (2) –

$$\frac{dN_Y}{dt} = \lambda_1 N_X - \lambda_2 N_Y$$

We can substitute value of N_X from equation (3)

$$\frac{dN_Y}{dt} = \lambda_1 N_0 e^{-\lambda t} - \lambda_2 N_Y$$

$$\Rightarrow \quad \frac{dN_Y}{dt} + \lambda_2 N_Y = \lambda_1 N_0\, e^{-\lambda_1 t}$$

Multiplying this equation with $e^{-\lambda_2 t}$

$$\Rightarrow \quad \frac{dN_Y}{dt}.e^{\lambda_2 t} + \lambda_2 N_Y.e^{\lambda_2 t} = \lambda_1 N_0 e^{-\lambda_1 t}.e^{\lambda_2 t}$$

$$\Rightarrow \quad \left(\frac{d}{dt} N_Y\, e^{\lambda_2 t}\right) = \lambda_1 N_0\, e^{(\lambda_2 - \lambda_1)t}$$

$$\Rightarrow \quad d\left(N_Y e^{\lambda_2 t}\right) = \lambda_1 N_0 e^{(\lambda_2 - \lambda_1)t} dt$$

$$\Rightarrow \quad \int d\left(N_Y e^{\lambda_2 t}\right) = \int \lambda_1 N_0 e^{(\lambda_2 - \lambda_1)t} dt$$

On solving, we get –

$$N_Y = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1}\left[\frac{e^{(\lambda_2-\lambda_1)t}-1}{e^{\lambda_2 t}}\right] = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} - e^{-\lambda_2 t}$$

4.9 Activity

Activity of a radioactive substance is defined as the rate at which it decays. Consider the following nuclear reaction.

$$X \rightarrow Y$$

With the passage of time, value of X must be decreasing. For this we write equation–

$$\frac{dN}{dt} = -\lambda N$$

Here, λN is the rate of decay of X so it can be called activity of X. Let's analyse the physical meaning of activity. When X decays at high rate then we say that there is more activity and when X decays at slow rate then we say that there is less activity. .

For example, if we say that activity is 100dps, it means every second 100 nuclei are disintegrating.

Unit of activity is dps which stands for 'disintegration per second'. Other units of activity are Becquerel (Bq) and curie (Ci). Relation between Ci, Bq and dps –

$$1Ci = 3.7 \times 10^{10} Bq = 3.7 \times 10^{10} dps$$

Note: Bq and dps are SI units.



Quiz-5

Consider the following equation

$$X \xrightarrow{\lambda_1} Y \xrightarrow{\lambda_2} Z$$

(i) What will be the activity for X?

Sol .Equation for decay of X will be –

$$\frac{dN_X}{dt} = -\lambda_1 N_X$$

Here, activity of X is $-\lambda_1 N_X$.

(ii) What will be the activity for Z?

Sol. Activity for Z will be the rate at which Z is decaying. But Z is a stable substance and is not decaying. So its activity will be zero.

(iii) What will be the activity for Y?

Sol. In this problem, X is disintegrating and producing Y and this Y is further disintegrating and producing Z. We have already seen equations for such situation in a previous illustration.

$$\frac{dN_Y}{dt} = \lambda_1 N_X - \lambda_2 N_Y$$

Here activity of Y is the rate at which Y is decaying. So activity for Y is $\lambda_2 N_Y$.

Remember, here $\frac{dN_y}{dt}$ represents rate of change of Y, it is not activity. Activity stands for activity of decay.

Activity $\neq \frac{dN_y}{dt}$

Activity at time t

Consider the following equation –

$$X \longrightarrow Y$$

At $t = 0$, N_0

At t, N

We can write activity of X at t = 0 as–

$A_0 = -\lambda N_0$

At any time t, activity of X will be –

$A_t = -\lambda N$

We know that –

$N = N_0 e^{-\lambda t}$

Multiply this equation by λ –

$\Rightarrow \lambda N = \lambda N_0 e^{-\lambda t}$

$\Rightarrow A_t = A_0 e^{-\lambda t}$

Graph of time versus activity will be as shown below –

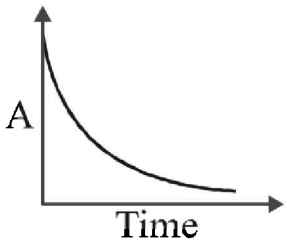


Illustration 10

For a radioactive substance A = 500 dps. After 5 min, A reduced to 400 dps. Find λ.

Sol. Here, $A_t = 400$

$A_0 = 500$

$t = 5 \text{ min} = 5 \times 60 = 300 \text{ s}$

$\Rightarrow A_t = A_0 e^{-\lambda t}$

$\Rightarrow 400 = 500\, e^{-300\lambda}$

$\Rightarrow e^{-300\lambda} = \frac{4}{5}$

$\Rightarrow -300\lambda = \ell n \frac{4}{5}$

$\Rightarrow \lambda = -\frac{1}{300} \ell n \frac{4}{5} = 7.4 \times 10^{-4}$

4.10 Use of radioactivity for Medical purposes

The phenomenon of activity is widely used for medical purposes. Let's see one such case. Imagine that a person met an accident and a lot of blood drained out of his body. Now doctor needs to give blood to the patient. But how does doctor know the amount of blood to be given? Well, here

radioactivity plays its role. With the help of radioactivity, doctors find out the amount of blood present in the patient's body. Now blood to be given is the amount equal to the difference in the amount of blood in the normal person's body and the amount of blood in patient's body. For this purpose, doctors use a radioactive isotope of sodium. Generally, radioactive substances are harmful for the body but this particular element is not harmful. Let's see an example to understand how sodium is used in calculating the amount of blood present in the body.

Illustration 11

^{24}Na is injected into the body of a patient via injection. Activity of this ^{24}Na is 2×10^3 dps. After 5 hrs, 1cm^3 of blood is taken out of the body of the patient. Now this blood is also radioactive. The activity of this blood is measured as 100 dps. Calculate volume of blood in the body (λ = 0.00632).

Sol. Suppose that the volume of blood is V_0.

$A_0 = 2 \times 10^3$ dps

After 5 hours or 18000 seconds, activity of Na in 1cm^3 of blood is 100 dps.
So, activity of Na in V_0 volume of blood will be 100 V_0.

$\Rightarrow$ $A_t = 100\ V_0$ dps

From relation $A_t = A_0 e^{-\lambda t}$, we can write –

$100\ V_0 = (2 \times 10^3)\ e^{-(0.00632)(18000)}$

On solving this equation, we get the amount of blood present inside the body of the patient.

4.11 Nuclear Fission

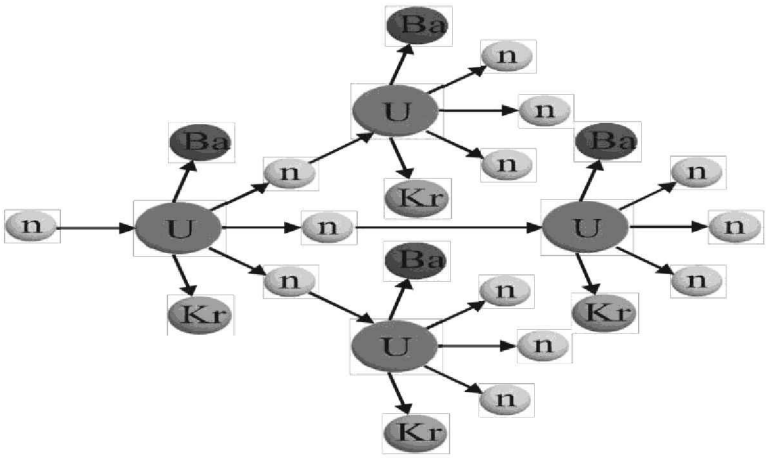
Nuclear fission is a very important type of nuclear reaction. This has impacted mankind in both good and bad ways. In year 1945, atom bombs were dropped over Hiroshima and Nagasaki. The bomb that was dropped on Hiroshima was named 'Little boy' and the bomb which was dropped on Nagasaki was named 'Fat man'. These bombs claimed thousands of lives. In this section, we are going to study and analyse the reactions which were responsible for such mass destruction.

$$^1_0n + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U \longrightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^1_0n$$

In nuclear fission reaction, uranium nucleus is bombarded with slow moving neutron. This neutron penetrates uranium nucleus and forms an isotope of uranium $^{236}_{92}U$. But this uranium isotope is so unstable that it immediately disintegrates into barium (^{141}Ba), krypton (^{92}Kr) and three neutrons $\left(3\,{}^1_0n\right)$.

There are some important observations related to this reaction.

- Suppose a single neutron is bombarded on uranium nucleus. This single neutron will react with a single atom of uranium. Once this reaction is complete, it will form Ba, Kr and three neutrons. Now these three neutrons so produced will again strike three more uranium atoms. As a result three uranium nuclei will disintegrate into Ba, Kr and 9 neutrons. These neutrons will further strike more nuclei of uranium and more such neutrons will be released.



This process will continue until all the uranium is disintegrated. This process is called a **chain reaction.** Once the reaction is triggered with one neutron, it carries on of itself.

- In the disintegration of uranium, it is not necessary that it breaks into Ba and Kr. There might be other possible reactions as well.

$$ {}_0^1n + {}_{92}^{235}U \rightarrow {}_{92}^{236}U \rightarrow {}_{51}^{133}Sb + {}_{41}^{99}Nb + 4\,{}_0^1n $$

$$ {}_0^1n + {}_{92}^{235}U \rightarrow {}_{92}^{236}U \rightarrow {}_{54}^{140}Xe + {}_{38}^{94}Sr + 2\,{}_0^1n $$

But yield of Ba and Kr is very high, so generally we talk the most about that reaction. But note that the number of neutrons produced in these other reactions is four and two. On an average, 2.5 neutrons are emitted per reaction.

- These products formed after fission reaction are generally not very stable themselves. Most of them undergo β-decay and the products so formed are more stable.
- Fission reactions are highly exothermic. They release tremendous amount of energy. This energy converts into kinetic energy of products. These products start to collide with each other. Due to these collisions, the kinetic energy is converted into heat energy. We can calculate amount of energy released by this reaction using Q-factor. In case of Hiroshima and Nagasaki bombardments, large amount of energy that was released from fission reaction was responsible for such mass destruction.

4.12 Fission Material

Uranium is the most commonly used element in fission reactions. All reactions that we have seen till now had isotope ${}_{92}^{236}U$ of uranium. But this isotope is scarcely available in nature. In nature, an isotope ${}_{92}^{238}U$ is available in abundance. But this isotope of Uranium is quite stable and thus cannot be used for fission reaction. If this isotope undergoes neutron bombardment then it undergoes following series of reaction –

$$ {}_0^1n + {}_{92}^{238}U \rightarrow {}_{92}^{239}U \xrightarrow{\beta-\text{decay}} {}_{93}^{239}Np + e^- + \nu^- \xrightarrow{\beta-\text{decay}} {}_{94}^{239}Pu + e^- + \nu^- $$

The final product Pu (plutonium) is suitable for fission reaction. Due to scarcity of ${}_{92}^{236}U$ in nature, we have to use alternate materials to carry out fission reactions. Plutonium is one such alternate. There are reactors which are designed to produce plutonium from ${}_{92}^{238}U$. These reactors are called ***breeder reactors.***

4.13 Nuclear Reactors

The purpose behind nuclear reactions is not only to take life. But it is also used for the benefit of mankind. The tremendous amount of energy released from fission reaction can be harnessed and converted into electric energy. This process is carried out in nuclear power plants. To support this statement, consider the following fact. 1kg of coal when burnt in thermal power plant produces electricity that is sufficient to light a bulb for 8 hours. When 1kg of uranium is used in nuclear power plant then the amount of electricity produced is sufficient to light a bulb for 3×10^4 years.
But while harnessing energy from the fission reaction, utmost care has to be taken because if this reaction gets out of control then reactor may convert into an atom bomb. Many such accidents have already happened around the world.
Let us understand working of a nuclear reactor.

1. Moderator: Nuclear fission requires slow moving neutrons (also called thermal neutrons). In fission reaction, 2.5 neutrons so produced have very high speed. These fast moving neutrons are not appropriate to carry out fission reactions. Their speed is so high that they escape from the reaction chamber. Therefore, we need to slow them down. To do that, we use a substance called ***moderator.*** This moderator is combined with uranium. When neutrons are produced, they collide with molecules of moderator and lose their speed. Generally, water (H_2O), heavy water (D_2O) or graphite is used as moderator.

Key Points –

Hydrogen has three naturally occurring isotopes, denoted as ^1H, ^2H and ^3H called protium, deuterium and tritium respectively.

^1H or protium is represented by symbol H.

^2H or deuterium is represented by symbol D.

^3T or tritium is represented by symbol T.

A water molecule is made up of protium and oxygen (H_2O). If we replace protium with deuterium it is called heavy water (D_2O).

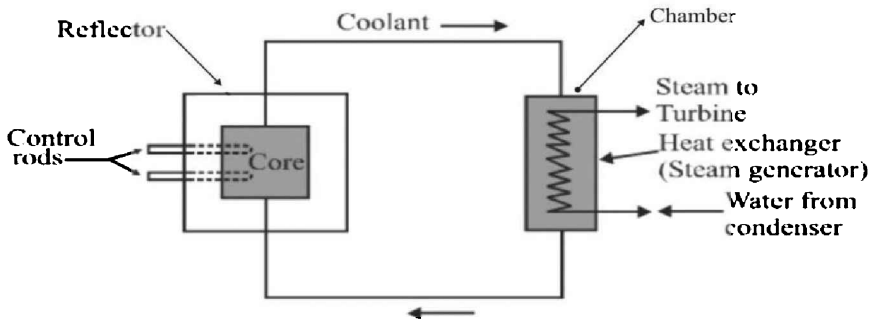
2. Multiplication factor: When uranium is bombarded with a neutron, it produces three neutrons. When all the three neutrons collide with other uranium atoms, then we say that multiplication factor (K) is 3. When only two neutrons out of three neutrons collide with uranium then we say that multiplication factor (K) is 2 and so on.
To perform chain reaction, $K \geq 1$.
If $K < 1$, the reaction will die down.
If $K > 1$, the reaction rate increases and becomes out of control (used in atom bombs).
If $K = 1$, it is called critical operation. At $K = 1$, reaction remains under control and is hence used in power plants.
In nuclear reactors, to control multiplication factor, control rods are used. These rods are made up of cadmium. Cadmium has property to absorb neutrons. These rods are inserted into the reaction chamber. If value of K goes more than 1, rods are inserted deeper into the chamber. If K goes less than 1, rods are pulled out of the chamber.



Other than control rods, there are safety rods also. These safety rods are used to absorb neutrons completely. These rods are inserted in reaction chamber when there is a need to cease the

reaction immediately. At times of emergency, such as when reaction gets out of control or when there is some accident, these rods are inserted completely into reaction chamber.

3. Basic structure of a nuclear reactor: The below diagram shows the basic structure of a nuclear reactor.



Core - It is the reaction chamber. All the reaction takes place in this chamber. It is filled with uranium and moderator.

Reflector - Reflector is a box shaped structure which surrounds the core. A neutron produced in the core tries to escape from the chamber. But this reflector stops neutrons from escaping.

Chamber - All the heat produced in the core is transferred to a chamber. There is supply of water in this chamber. Due to heat, water gets converted into steam and this steam is send to a turbine which generates electricity.

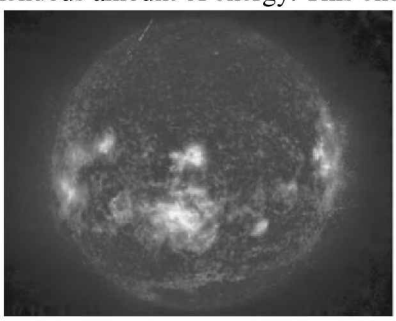
Nuclear Waste

Waste of nuclear reactions is very harmful and needs to be disposed very carefully. It is very difficult to dispose such harmful substances and it has become a worldwide issue of concern.



4.14 Nuclear Fusion

Sun and other stars possess a tremendous amount of energy. This energy comes from fusion reactions.



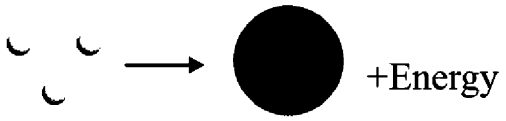
The two terms, fission and fusion may be confusing at times.
In fission reaction, a large nucleus breaks into smaller nuclei.
In fusion reaction, small nuclei merge and produce larger nucleus.

Nuclear Fission



Large Nucleus Smaller Nucleus

Nuclear Fusion

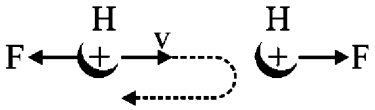


Smaller Nucleus Large Nucleus

Energy released in fusion reaction is more than the energy released in a fission reaction.
Equation of fusion reaction going on in Sun (in highly simplified form) is–

$$_1^2H + _1^2H \longrightarrow _2^4He$$

This reaction can also be used to create a hydrogen bomb which is more destructive than a nuclear bomb. But it is very difficult to perform such reactions. Consider two hydrogen nuclei–



We want to perform a fusion reaction between them. For that we have to fuse one hydrogen nucleus into another. To do that, we impart some velocity (v) to one hydrogen nucleus towards another. But as this hydrogen nucleus reaches close to another nucleus, it experiences force of electrostatic repulsion and turns around without colliding as shown in figure.
Speed of hydrogen nucleus has to be kept such that it overcomes this repulsive force. So, huge amount of kinetic energy is needed for this reaction to start.
In stars, temperature is very high. Due to this, nuclei have very high kinetic energy and thus fusion reaction is possible.
The amount of energy required to start this reaction is about 400 KeV. In the kinetic theory of gases, we studied that for gases –

$$\text{Average kinetic energy} = \frac{3}{2}kT$$

$$\Rightarrow \quad 400\,KeV = \frac{3}{2}kT$$

$$\Rightarrow \quad T = 3 \times 10^9\ K$$

So, to perform a fusion reaction, temperature of hydrogen should be about 3×10^9 K. But the temperature of the Sun is 1.5×10^7 K only but still there is fusion reaction going on in Sun. How?

Actually in Sun, temperature being lesser, average KE < 400 KeV. But this is only the average value. Some of the faster moving atoms of hydrogen in the Sun might be having KE equal to 400 KeV or even more. Such atoms do fusion reaction.
When fission reaction is carried out due to high temperature as in Sun or other starts, it is called **Thermal Nuclear Fusion.**

Examples of Fusion Reactions –

- Reactions inside Sun

The fusion reaction in the Sun is a multi-step process in which hydrogen is burned into helium. These reactions (in detail now) are as follow –

$$_1^1H + _1^1H \longrightarrow _1^2H + e^+ + \nu + 0.42MeV$$

$$e^+ + e^- \longrightarrow \gamma + \gamma + 1.02MeV$$

$$_1^2H + _1^1H \longrightarrow _2^3He + \gamma + 5.49MeV$$

$$_2^3He + _2^3He \longrightarrow _2^4He + _1^1H + _1^1H + 12.86MeV$$

$$4_1^1H + 2e^- \longrightarrow _2^4He + 2\nu + 6\gamma + 26.7MeV$$

Combination of above 4 equations gives fifth equation. Overall, 4 hydrogen are combining to give one helium atom.

- Scientists believe that in the beginning of the universe, there must have been only hydrogen present. This hydrogen must have fused and produced different elements that we see today. See how in the following reactions, oxygen and nitrogen are being formed.

$$_1^1H + {}^{14}N \longrightarrow {}^{15}O$$

$$_1^1H + {}^{12}C \longrightarrow {}^{13}N$$

- **Controlled Fusion Reactions**

These days, scientists are trying to conduct controlled fusion reaction.



If this reaction is possible, then it will resolve our entire energy crisis. But if this reaction gets out of control then it will become a disaster.

Exercise of Nuclear Physics

Multiple choice types

Only one option correct

Q.1 In which of the following processes, the number of protons increases in the nucleus?

(a) α -decay (b) β^- - decay (c) β^+ - decay (d) k – capture

Ans. **[b]**

Q.2 A radioactive element X converts into another stable element Y. half-life of X is 2 h. Initially, only X is present. After time t, the ratio of atoms of X and Y is found to be 1: 4. Then t in hours is

(a) 2 (b) Between 4 and 6 (c) 4 (d) 6

Ans. **[b]**

Q.3 If a nucleus such as ^{226}Ra that is initially at rest undergoes alpha decay, then which of the following statements is true?

(a) The alpha particle has less kinetic energy than the daughter nucleus.
(b) The alpha particle has less Momentum than the daughter nucleus.
(c) The alpha particle and daughter nucleus both have same kinetic energy
(d) We cannot say anything about kinetic energy of alpha particle and daughter nucleus.

Ans. **[b]**

Q.4 A radioactive nucleus is being produced at a constant rate 5 per second. Its decay constant is 10. If N_0 are the number of nuclei at time t = 0, then maximum number of nuclei possible are

(a) 0.2 (b) 0.5 N_0 (c) N_0 (d) 0.75 + N_0

Ans. **[a]**

Q.5 A helium atom, a hydrogen atom and a neutron have masses of 4.003 u, 1.008 u and 1.009 u (unified atomic mass units), respectively. Assuming that hydrogen atoms and neutrons can fuse to form helium, what is the binding energy of a helium nucleus?

(a) 2.01 u (b) 3.031 u (c) 1.017 u (d) 0.031 u

Ans. **[d]**

Q.6 The activity of a radioactive element decreases to two- third of the original activity I_0 in a period of nine years. After a further lapse of nine years, its activity will be

(a) $\frac{4}{9} I_0$ (b) $\frac{2}{3} I_0$ (c) $\frac{I_0}{9}$ (d) $\frac{I_0}{6}$

Ans. **[a]**

Q.7 A 15×10^{-4} Å photon produces an electron – positron pair in the vicinity of a heavy nucleus. Rest energy of electron is 1.012 MeV. If they have the same kinetic energies, the energy of each particle is nearly.

(a) 3.098 MeV (b) 4.121 MeV (c) 5.121 MeV (d) 6.121 MeV

Ans. **[a]**

Q.8 Assuming that about 20 MeV of energy is released per fusion reaction.

$$_1H^2 +_1 H^2 \rightarrow 2He^4 + E + other\ \ particles$$

Then the mass of $_1H^2$ consumed three day in a fusion reactor of power 2×10^6 watt will approximately be

(a) 180 g (b) .18 g (c) 18 g (d) 1.8 g

Ans. **[b]**

Q.9 In a fusion reaction ${}_1^2He + {}_1^2H \rightarrow {}_2^3He + {}_0^1n$, the masses of deuteron, helium and neutron expressed in a.m.u are 4.030, 5.340 and 1.009, respectively. If 1 kg of deuterium undergoes complete fusion, find the amount of total energy released. (1 a.m.u. = 931.5 meV/c^2)

(a) 6.02×10^{13} J (b) 5.6×10^{13} J

(c) 9.0×10^{13} J (d) 0.9×10^{13} J

Ans. **[c]**

Q.10 The compound unstable nucleus ${}_{92}^{236}U$ often decays in accordance with the following reaction

${}_{92}^{236}U \rightarrow {}_{54}^{140}Xe + {}_{38}^{94}Sr + other\ particles$

In the nuclear reaction presented above, the 'other particle' might be

(a) An alpha particle , which consists of two protons and two neutrons

(b) Two protons

(c) One proton and one neutron

(d) Two neutron

Ans. **[d]**

Q.11 The luminous dials of watches are usually made by mixing a zinc sulphide phosphor with a α particle emitter. The mass of radium (mass number 226, half – life 1620 years) that is needed to produce an average of 10 α - particles per second for this purpose is

(a) 2.77 mg (b) 2.77 g

(c) 2.77×10^{-23} g (d) 2.77×10^{-13} kg

Ans. **[d]**

Q.12 A neutron of energy 1 MeV and mass 3.2×10^{-27} kg passes a proton at such a distance that the angular momentum of the neutron relative to the proton approximately equals 10^{-23}Js. The distance of closest approach neglecting the interaction between particles is

(a) 3.125×10^{-2} m (b) 3125 m

(c) 31.25 m (d) 3.125×10^{-4} m

Ans. **[d]**

Q.13 The half-life of radioactive radon is 7.6 days. The time at the end of which $1/40^{th}$ of the radon sample will remain undecayed is (given $\log_{10} e = 0.4343$)

(a) 40.49 day (b) 42 day (c) 39.99 day (d) 38.92 day

Ans. **(a)**

Q.14 In the options given below let E denote the mass energy of a nucleus and E (n) that of a neutron. The correct option is

(a) $E\left({}_{92}^{236}U\right) > E\left({}_{53}^{137}I\right) + E\left({}_{39}^{97}Y\right) + 2E(n)$

(b) $E\left({}_{92}^{236}U\right) < E\left({}_{53}^{137}I\right) + E\left({}_{39}^{97}Y\right) + 2E(n)$

(c) $E\left({}_{92}^{236}U\right) < E\left({}_{56}^{140}I\right) + E\left({}_{36}^{97}Y\right) + 2E(n)$

(d) $E\left({}_{92}^{236}U\right) > E\left({}_{56}^{140}Ba\right) + E\left({}_{36}^{94}Kr\right) + 2E(n)$

Ans. **(a)**

Q.15 The half – life of ^{131}I is 8 days .Given a sample of ^{131}I at time t = 0, we can assert that

(a) No nucleus will decay before t = 4 days

(b) No nucleus will decay t = 8 days

(c) All nuclei will decay before t = 16 days
(d) A given nucleus may decay at any time after t = 0

Ans. **(d)**

Q.16 After an interval of one day. $1/32^{th}$ initial amount of a radioactive material remains in a sample. Then, its half-life is

(a) 4.8 h (b) 5 h (c) 4 h (d) 6 MeV

Ans. **(a)**

Q.17 Two radioactive materials X_1 and X_2 have decay constants 11λ and λ, respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $1/e$ after a time

(a) $\frac{1}{10\lambda}$ (b) $\frac{1}{11\lambda}$ (c) $\frac{11}{10\lambda}$ (d) $\frac{1}{9\lambda}$

Ans. **(a)**

Q.18 The probability of survival of a radioactive nucleus for one mean life is

(a) $\frac{1}{e}$ (b) $1-\frac{1}{e}$ (c) $\frac{\ln 2}{e}$ (d) $1-\frac{\ln 2}{e}$

Ans. **(a)**

Q.19 The total binding energy of a α particle (^4_2He) is 24.4 MeV whereas the total binding energy of a deuteron (^2_1H) is merely 2.2 MeV. When two deuterons are made to combine.

(a) An α particle will not be obtained
(b) An amount of 22.2 MeV energy will be released
(c) An amount of 20.0 MeV energy will be released
(d) An amount of 22.2 MeV energy will be consumed

Ans. **(c)**

More Than One Correct

Q.20 Mark out the correct statement(s).

(a) Higher binding energy per nucleon means the nucleus is less stable.
(b) If the binding energy of nucleus were zero, then it would spontaneously break apart.
(c) Binding energy of a nucleus can be negative
(d) Binding energy of a nucleus is always positive.

Ans. **[b, d]**

Q.21 Mark the wrong statement(s).

(a) For an exothermic reaction, if Q value is +12.56 MeV and the KE of incident particle is 2.44 Me V, then the total KE of product of reaction is 15.00 MeV.
(b) For an exothermic reaction. If Q value is +12.56 MeV and the KE of incident particle is 2.44 MeV, the total KE of product of reaction is 12.56 Me V
(c) For an endothermic reaction, if we give the energy equal to $|Q|$ value of reaction, then the reaction will be carried out.
(d) For an exothermic reaction, the BE per nucleon of products should be greater than the BE per nucleon of reactants.

Ans. **[b, c]**

Q.22 During β - decay (beta minus), the emission of antineutrino particle is supported by which of the following statement(s)?
(a) Angular momentum conservation holds well in any nuclear reaction.
(b) Linear momentum conservation holds good in any nuclear reaction
(c) The KE of emitted β - particle is varying continuously to a minimum value.
(d) None of the above.

Ans. **[a, b]**

Q.23 Two samples A and B of same radioactive nuclide are prepared. Sample A has twice the initial activity of sample B. For this situation. Mark out the correct statement(s).
(a) The half – lives of both the samples would be same.
(b) The half – lives of the samples are different
(c) After each has passed through 5 half – lives, the ratio of activity of A to B is 2: 1.
(d) After each has passed through 5 half – lives, ratio of activities of A to B is 64: 1.

Ans. **[a, c]**

Q.24 The decay constant of a radioactive substance is 0.693 $year^{-1}$. Therefore,
(a) Nearly 63% of the radioactive substance will decay in (1/0.173) year
(b) Half –life of the radioactive substance is (1/0.173) year
(c) One of the radioactive substances will be left after one year.
(d) all of the above

Ans. **[a, c]**

Q.25 If A, Z and N denote the mass number, the atomic number, and the neutron number for a given nucleus, we can say that
(a) A = Z + N
(b) Isobars have the same A but different Z and N.
(c) Isotopes have the same Z but different N and A.
(d) Isotones have the same N but different A and Z.

Ans. **[a, b, c, d]**

Q.26 The phenomenon of nuclear fission can be carried out both in a controlled and in an uncontrolled way. Out of the following the correct statements vis-à-vis these phenomena are:
(a) The fission energy released per reaction is much more than conventional nuclear reactions and one of the products of the reaction is that very particle which initiates the reaction.
(b) It is the 'surface to volume' ratio of the sample of nuclear fuel used which determines whether or not the reaction would sustain itself as a 'chain reaction'.
(c) The 'control rods' in a nuclear reactor must be made of material that release neutrons effectively.
(d) The energy released per fission as well as energy released per unit mass of the fuel in nuclear fission are both greater than the corresponding quantities for nuclear fusion.

Ans. **[a, b]**

Q.27 An S^{32} nucleus is spherical and has a radius R and a volume $V = \frac{4}{3}\pi R^3$. According to the empirical observation, the volume of the $_{54}X^{128}$ nucleus assumed to be spherical is V' and radius is R'. Then
(a) V' =8V (b) V' = 4V (c) R' = 1.58 R (d) R' = 8R

Ans. **[b, c]**

Q.28 In a nuclear reactor
(a) The chain reaction is kept under control by rods of cadmium, which reduces the rate
(b) The thick concrete shield is used to slow down the speed of fast neutrons.
(c) Heavy water (or graphite) moderates the activity of the reactor
(d) Out of U^{238} and U^{235} natural uranium has less than 1% of U^{235}

Ans. **[a, d]**

Q.29 From the following equations, pick out the possible nuclear fusion reaction:
(a) ${}_6C^{13} + {}_1H^1 \rightarrow {}_6C^{14} + 4.3\ MeV$
(b) ${}_6C^{12} + {}_1H^1 \rightarrow {}_7C^{13} + 2\ MeV$
(c) ${}_6C^{13} + {}_1H^1 \rightarrow {}_8C^{17} + 8.3\ MeV$
(d) ${}_{92}C^{235} + {}_0n^1 \rightarrow {}_{54}Xe^{104} + {}_{38}Sr^{94} + {}_0n^1 + {}_0n^1 + y + 200\,MeV$

Ans. **[a, b]**

Matrix type

Q.30 For physical quantities are given in column I and their order of values in column II. Match appropriately.

Column I	Column II
a. Thermal energy of air molecules at room temperature	p. 10 KeV
b. Binding energy of heavy nuclei for 3 nucleon	q. 2 eV
c. X – ray photon energy	r. 0.02 eV
d. Photon energy of visible light	s. 21 MeV

Ans. **[a → r, b → s, c → p, d → q]**

Q.31

Column I	Column II
a. Photoelectric effect	p. Photon
b. Wave	q. Frequency
c. X – rays	r. K capture
d. Nucleus	s. γ-rays

Ans. **[a → p, b → q, c → r, d → s.]**

Assertion–Reasoning type

In the following set of questions, a statement 1 is given and a corresponding statement 2 is given just below it. Mark the correct answer as:
(a) If both statement 1 and statement 2 are true and statement 2 is the correct explanation of statement 1.
(b) If both statement 1 and statement 2 are true but statement 2 is not the correct explanation of statement 1.
(c) If statement 1 is true but statement 2 is false.
(d) If statement 1 is false but statement 2 is true.

Q.32 **Statement 1:** Two protons in the same nucleus do not fly apart under the influence of electrostatic repulsion.
Statement 2: Coulomb's law is valid only for large distance.

Ans. **[c]**

Q.33 **Statement 1:** On earth, at present, the amount of unstable nuclides is estimated to be more than the amount of unstable nuclides.
Statement 2: The half-life of stable nuclides is infinite.

Ans. **[d]**

Q.34 **Statement 1:** In any radioactive sample, after one half – life exactly half of the nuclei would have decayed.

Statement 2: According to the law of radioactivity, $\frac{N}{N_0} = E^{-\lambda t}$

Ans. **[d]**

Q.35 **Statement 1:** Nuclear fusion occurs only at very low temperatures.
Statement 2: The average kinetic energies of the nuclei undergoing fusion should be sufficient to overcome coulomb barrier.

Ans. **[d]**

Q.36 **Statement 1:** Neither fission nor fusion of ^{62}Ni would yield energy.

Statement 2: The binding energy per nucleon of ^{62}Ni is minimum.

Ans. **[c]**

Subjective type

Q.37 The half-life of ^{198}Au is 5.2 days. Find the decay constant,

Ans. **[1.5×10^{-6} s^{-1}]**

Q.38 In a nuclear reactor, ^{235}U undergoes fission liberating 200 MeV of energy. The reactor has a 10 % efficiency and produces 1000 MW power. If the reactor is to function for 10 years, find the total mass of uranium required.

Ans. **[38.2×10^3kg]**

Q.39 A radioactive element decays by β - emission a detector records n beta particles in 2 s and in next 2 s it records 1.50n beta particles. Find mean life correct to nearest whole number. [Given: - ln 2 = 0.6931, ln 3 = 1.0986.]

Ans. **[5s]**

Q.40 A 50.0 g sample of carbon is taken from the pelvis bone of a skeleton and is found to have a ^{14}C decay rate of 200.0 decays min^{-1}. It is known that carbon from a living organism has a decay rate of decays /min^{-1} g^{-1} and that ^{14}C has a half-life of 5730 year = 3.01×10^9 min. Find the age of the skeleton.

Ans. **[2.30×10^{-10} min^{-1}]**

Q.41 In the process of nuclear fission of 5g uranium, the mass lost is 3.68 mg. The efficiency of power house run by the fission reactor is 30%. To obtain 800 megawatt power from the power house, how much uranium will be required per hour? ($c = 3\times10^8$ m s^{-1})

Ans. **[14.49×10^{-5} kg]**

Q.42 What is the power output of a $_{92}U^{235}$ reactor if it takes 60 days to use up 5 kg of fuel, and if each fission gives 94 MeV of usable energy?

Ans. **[3.71×10^7 W]**

Q.43 One gram of a radioactive material having a half – life period of 4 years is kept in store for duration of 16 years. Calculate how much of the material remains unchanged?

Ans. **[0.0625 gm.]**

Q.44 A certain radioactive material can undergo three different types of decay. Each with a different decay constant 2λ, 4λ and 6λ. Then, the effective decay constant λ_{eff} is equal to $n\lambda$. What is the value of n?

Ans. **[n = 12]**

Q.45 A count rate meter is used to measure the activity of a given sample. At one instant, the meter shows 8250 counts per minute. 10 minutes later, it shows 5400 counts per minute. Find (a) the decay constant and (b) the half-life of the sample. ($\log_{10} 1.527 = 0.184$)

Ans. **[(a) $\lambda = 0.042\ min^{-1}$ (b) $T = 16.50\ min^{-1}$]**

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